



Generic Commissioning Instructions for Vigilon Plus panel, Vigilon Compact Plus panel & Vigilon Plus Nodes based fire Detection and Alarm system



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Preface

This is the ninth issue covering system Commissioning instructions for the EN fire alarm system based on the **Vigilon Plus 4/6 loop panels, Vigilon Compact Plus panel and Vigilon Plus Network nodes**. Where panels are fitted with Master Control Card / Master Control Board having software at version 4.54 or higher. It also covers information on hybrid network card that allows satellite of networked EN and BS systems to temporarily coexist in multi network.

Associated documents

The Vigilon Plus control panels and network nodes are supplied with respective copies of the operating and installation instruction manuals:

EN54 Vigilon Plus 4/6-loop panel based system - Installation instructions & Operating instructions

EN54 Vigilon Compact Plus panel based system - Installation instruction & Operating instructions

Vigilon Compact Plus Network node - Installation instructions

Vigilon Plus Network Node - Installation instructions

Commissioning tool - User guide

Conventions



This is a note to highlight important text that is normally hidden in the main text.



This is either a caution to prevent damage to the equipment or a warning to inform of dangerous conditions that may result in injury or death.

Symbol Keys



What you will see.



What you will hear.

Abbreviations

ACC	Audio Control Card	Mpeg	Moving picture expert group
ADC	Analogue to digital converter	N/C or NC	Normally closed
C	Common	N/O or NO	Normally open
CH	channel	NVM	Non Volatile Memory (CARD14)
CO	Carbon monoxide	O/C or OC	Open circuit
DAU	Distributed Amplifier Unit	PA	Public Address
DEV	Device	PCB	Printed circuit board
DIL	Dual in line	PIN	Personal identification number (usercode, password, access code)
DKC	Display keyboard card	PSU	Power supply unit
DPCO	Double pole change over (relay contacts)	PTT	Press to Talk
EEPROM	Electrically Erasable Programmable ROM	PVC	Polyvinyl chloride
EOL	End of line	QB	Quick blow (fuse)
FAB	First action byte	S/C or SC	Short circuit
IO or I/O	Input Output channels	SAB	Second action byte
IP	Ingress protection	SAFE	Software addressed firmware encoded
LED	Light emitting diode	SPCO	Single pole change over relay contacts
LRT	Loop Radio Transceiver	SPL	Sound pressure level
LPC	Loop processor card	T	Anti-surge (fuse)
LPCB	Loss prevention council certification board	USB	Universal serial bus
MCB	Master control board (CARD 0)	VA	Voice Alarm
MCC	Main control card or Main controller card (CARD 0)		
MCP	Manual call point		

Preliminary information

Safety information



1. Do not remove or replace printed circuit boards, fuses or attempt to wire the control panel with the panel powered up. Always power-down the mains supply at the fused spur unit and disconnect the battery supply to the panel.

2. When powering-up always power-up the mains supply first before the battery supply. The power-down should be done in reverse order.

3. When installing the cards into the master control board or backplane always use anti-static work procedures.

4. Do not use anti-static procedures on live equipment.

Pre-visit checks

- ☐ Ensure there are accurate *as fitted wiring drawings* available, 2 copies are required
- ☐ Any damaged equipment has been noted for replacement
- ☐ Ensure access will be provided to system equipment installed in the protected premises
- ☐ The installer (*electrical contractor*) will be in attendance until the installation is proved
- ☐ Site contact or representative will be available during the visit
- ☐ Ensure the commissioning tool along with the associated cables, printer and instructions are available
- ☐ Ensure spare parts are available, such as:
 - MCP glasses
 - MCP test key
 - Printer paper roll
 - Equipment door keys.

Pre-commissioning

- ☐ Check the installation of fire alarm equipment with reference to the most recent *as fitted wiring drawings*.
- ☐ Get the feel of the operating condition of areas on the site:
 - action the installer to carry out any rectification work plus
 - report discrepancies for administration purposes.
- ☐ Where the operating condition of an area is not right for the equipment installed, then the appropriate replacement action must be taken.
- ☐ Ensure the fire system equipment is installed in accordance with the appropriate standards and project specification.

Points to remember

Earth leads

- ☐ All *earth leads* supplied with the system equipment must be securely fitted to maintain earth continuity.

Parts for later installation

- ☐ All unused parts should be retained in their respective container for safe keeping until required.

Loop wiring

- ☐ The loop cable should have been connected to the appropriate terminals at each device, as shown in the installation manual in locations identified on the *as fitted wiring drawings*.

Enclosure

- ☐ Access into equipment enclosure is usually by means of opening an outer door/cover. A panel may also have an inner door which will also need opening.

Unattended equipment

- ☐ Where equipment is to be left unattended, then it is important to close the door /cover for safety.

Copper fingers

- ☐ Copper fingers are conductive spring like strips fitted to metal assemblies. They are fitted to shield against electromagnetic and radio frequency interferences. Ensure the copper finger strips are intact and no damage has occurred. Damaged fingers will reintroduce the gap to let in/out interferences.

Static precaution

- ☐ The discharge of static electricity can damage or degrade sensitive electronic components on printed circuit boards. Anti-static procedures should be followed when handling *static sensitive boards*.



It is important that anti-static procedures are NOT carried out on live equipment.

Removal and disconnection

- ☐ Any disconnection of cables or removal of parts from an assembly must be restored or replaced.

Battery

- ☐ To prevent damage to batteries and equipment, the terminals of the battery must not simultaneously touch any conductive part of the equipment enclosure.
- ☐ Sealed lead acid battery can have a useful life of up to 5 years from the date of manufacture, it is strongly recommended that batteries are replaced after 4 years. The batteries must be disposed of correctly by following national or local legislation and battery manufacturers recommendations.

Powering up

- ☐ When equipment is being powered up always connect the mains supply before the battery supply. Power-down should be done in reverse order.



Arcing may occur when the battery circuit is connected to mains powered equipment.

Panel Buzzer

- ☐ It may be necessary during commissioning to switch *Off* the panel buzzer. It is possible to selectively switch the disablement, fault, fire, supervisory and command build buzzer sound to *Off* or *On*. It is important to ensure that the buzzer is switched *On* for normal operation after commissioning.

Fire plan

- ☐ The system should be tested in accordance with the project specification.

Sensor cover

- ☐ Each fire sensor installed in the system should have been fitted with a *dust cover* during installation. The dust covers must be removed from all the fire sensors after the panel loops have been satisfactorily powered-up and with addresses allocated to each device ready for further checks and tests.

Site specific installation

- ☐ Plant equipment interfaced to the system should be tested to recommendations made in the project specification.

Test mode (EN)

- ☐ During commissioning of the system you will need to switch *On* the Test mode. It is important to switch *Off* the Test mode after the work is over, to ensure the system operates normally.

Informing responsible persons

- ☐ **It is important to inform the *person(s) responsible* for the fire alarm system that the system is being commissioned.**

A typical commissioning process

 Always power-down the panel when working on the system, for example when wiring or replacing parts.

Pre visit checks

Ensure you have:

- *As-fitted-drawings* that shows installed system equipment
- Access will be available to protected areas having system equipment
- Installer will be present to rectify wiring faults
- Tools and spare parts are available.

Inform responsible person(s)

Inform responsible person(s) that the fire alarm system is being commissioned and occupants in the protected premises will hear test alarms.

Ensure occupants are made aware of alternative site procedures should there be a fire event while the system is commissioned.

Survey the installation

Refer to the most recent *as-fitted-drawings* and ensure that all the fire system equipment has been installed in accordance with the installation instructions, respective standards and project specification.

Before powering up the Panel

Open the outer cover and inner cover of the panel and:

- Check all the internal cables within the enclosure are securely fitted
- Fit the loop card(s) into the required location(s)
- Ensure no external circuits are connected at this stage, except for the mains supply which must be connected but not switched ON
- Fit the *end of line* resistors to the master alarm circuits.
- Connect the external printer to the RS232 port, if required.
- Ensure the input - output lines of interface units on the loop are not connected.

Power up

- Fit the batteries inside the panel enclosure and connect the battery leads and then switch ON the mains supply.

Initial tests and set ups

- Carry out a display test and ensure DISPLAY and LEDs are working
- Set the system clock time and date
- Configure the RS232 printer port, if an external printer is installed.

Loop Devices

- Configure the links on all LV 4-channel interface units connected to the loop. Ensure links are set for either Input or Output application.

 Ensure all interface Input/Output external wiring remains disconnected at this stage, unless otherwise instructed. This action will prevent inadvertent operation of output/plant while the system is being commissioned.

- Power up the mains powered interface units including ASD interfaces on the loop.

Loop device address allocation

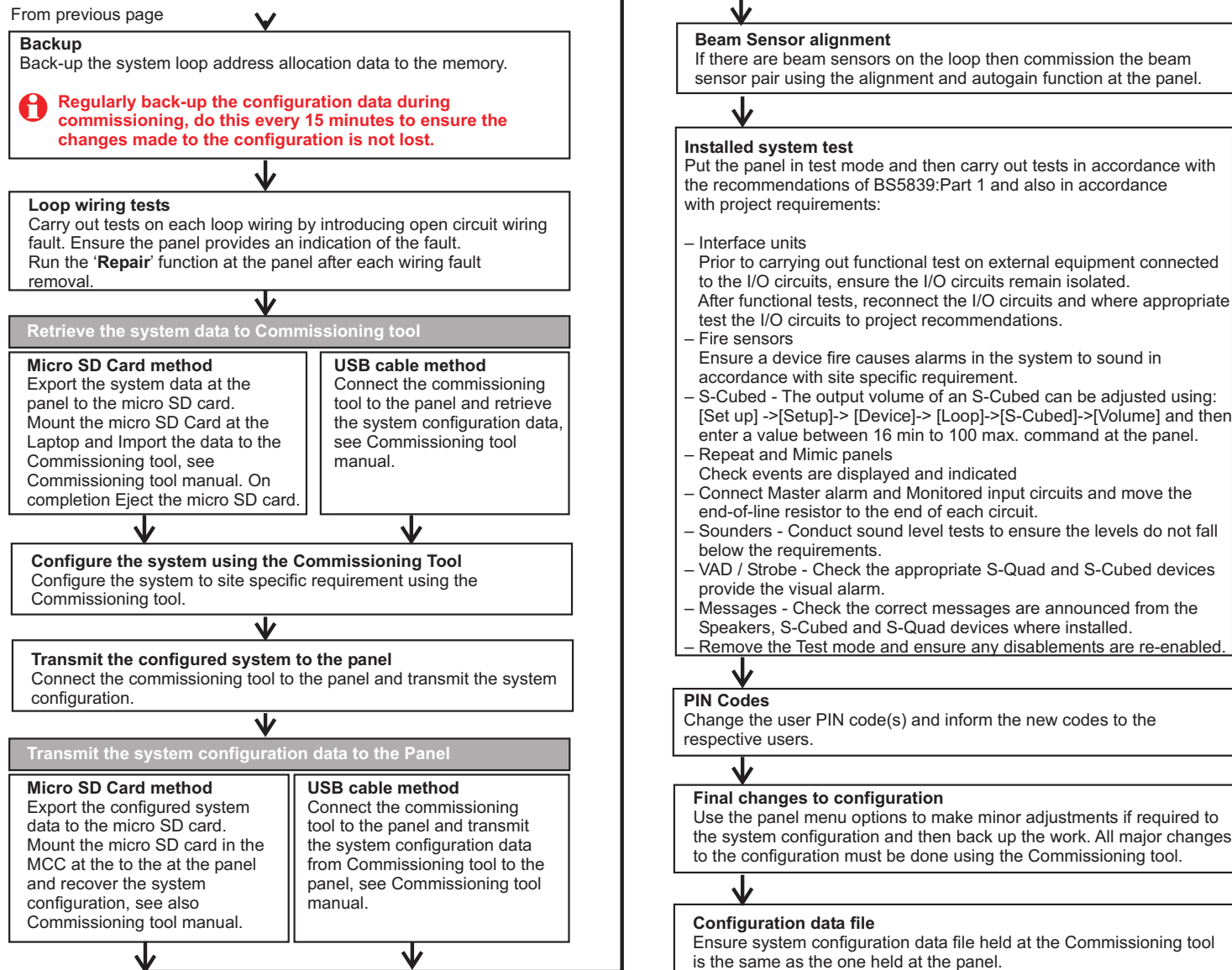
- Connect only End 1 of a loop circuit and power-up the loop
- Allow address allocation to finish. Any fault(s) on the loop during allocation must be rectified. Power-down the loop when correcting loop wiring faults and the power-up to continue address allocation
- Connect only End 2 of the loop circuit and power cycle the loop and ensure address allocation is complete from End 2.
- Connect both End 1 and End 2 of the loop and check the loop allocation is complete with both ends of the loop connected.
- 'Starting Loop n' displayed after successful allocation of addresses.
- Check all the devices on the loop circuit are installed in their correct location using the 'Find Device' function at the panel and by referring to the '*as fitted wiring drawings*'
- Check to ensure correct devices are installed by viewing the 'Device status'
- Repeat the allocation process on other loop circuits.

Pre-allocate the wireless devices

- If the system has wireless devices then pre-allocate the wireless devices and bind them to the respective LRT. Then allocate loop again to allow panel to see the devices.

Radio
Devices

Continue on next page



Product Approval and Standards

Fire detection and alarm control panel

The following fire detection and alarm control panels are NOT YET LPCB approved.

Product number	Description	Standard
VIGPLUS-24	EN Vigilon Plus 4-loop panel	EN 54 Parts 2 & 4
VIGPLUS-72	EN Vigilon Plus 4/6-loop panel	
COMPACT-PLUS	EN Vigilon Compact Plus 2-loop panel	

Interface Units

The following interface units are LPCB approved.

Product number	Description	Standard
S4-34410	1 Input interface module (loop powered)	EN54-17:2005 and EN54-18:2005
S4-34450	4 Input / output Interface module (loop powered)	
S4-34420	1 Output & 1 input (confirmation) interface module (loop powered)	
S4-34415	1 Output Interface	
S4-34411	Single output interface (loop powered)	
S4-34401	Single Channel Mains Switching Interface	
S4-34404	Four Channel mains Switching Interface	EN54-17: 2005 EN54-18: 2005 & EN54-4:1997 + A1: 2002 + A2: 2006
S4-34440-02 and S4-34440-12	Mains Powered Interface Units	

Manual Call points

The following manual call points are LPCB approved.

Product number	Description	Standard
S4-34842	Manual Call Points	EN54 Part 11: 2001
S4-34805		EN54 Part 17: 2005
S4-34485		
S4-34800		

S-Cubed Mark 3

The following S-Cubed Mark 3 devices are LPCB approved.

Product number	Description	Standard
Type A Devices		
S3-S-R	S3 Sounder/Red Body	EN54-3 : 2001, A1:2002, A2:2006 (for Sounder)
S3-V-R	S3 Voice Sounder/Red Body	EN54-23 : 2010 (for Visual Alarm Device VAD)
S3-S-W	S3 Sounder/White Body	EN54-17 : 2005 (for Short circuit isolator)
S3-V-W	S3 Voice Sounder/White Body	
S3-VAD-HPW-R	S3 VAD/High Perf. White VAD/Red body	
S3-VAD-HPR-R	S3 High Perf. Red VAD/Red Body	
S3-S-VAD-HPW-R	S3 Sounder/High Perf. White VAD/Red Body	
S3-S-VAD-HPR-R	S3 Sounder/High Perf. Red VAD/Red Body	
S3-S-VAD-HPW-W	S3 Sounder/High Perf. White VAD/White Body	
S3-S-VAD-HPR-W	S3 Sounder/High Perf. Red VAD/White Body	
S3-S-VAD-LPR-R	S3 Sounder/Standard Perf. Red VAD/Red Body	
S3-S-VAD-LPW-R	S3 Sounder/Standard Perf. White VAD/Red Body	
S3-V-VAD-HPW-R	S3 Voice Sounder/High Perf. White VAD/Red Body	
S3-V-VAD-HPR-R	S3 Voice Sounder/High Perf. Red VAD/Red Body	
Environmentally protected Devices		
S3EP-S-R	S3 Sounder/Red Body/EP	EN54-3 : 2001, A1:2002, A2:2006 (for Sounder)
S3EP-VAD-HPW-R	S3 VAD/High perform. White VAD/Red Body/EP	EN54-23 : 2010 (for Visual Alarm Device VAD)
S3EP-VAD-HPR-R	S3 VAD/High perform. Red VAD/Red Body/EP	EN54-17 : 2005 (for Short circuit isolator)
S3EP-V-VAD-HPW-R	S3 Voice Sounder/High perform. White VAD/Red Body/EP	
S3EP-V-VAD-HPR-R	S3 Voice Sounder/High perform. Red VAD/Red Body/EP	

Commissioning instructions

S-Quad Sensors with Visual alarm device (VAD) - introduced post April 2014

The following **S-Quad with VAD sensors** when operating in the approved STATES are LPCB approved to the respective standard.

Part number	Description	Standard
S4-711-VAD-HPR	S4 Dual Optical & Heat Sensors / Red VAD (HP)	EN54-5 : 2000, A1 : 2002 (heat) EN54-7 : 2000, A1 : 2002, A2 : 2006 (optical) EN54-3 : 2001, A1 : 2002, A2 : 2006 (sounder) EN54-23 : 2010 (visual alarm device VAD) EN54-17:2005 (short circuit isolator) EN54-18:2005 (input output devices)
S4-720-V-VAD-HPR	S4 Heat sensor / Voice Sensor Sounder / Red VAD (HP)	
S4-711-V-VAD-HPR & S4BK-711-V-VAD-HPR	S4 Dual Optical & Heat Sensors / Voice Sounder / Red VAD (HP)	
S4-911-V-VAD-HPR	S4 Dual Optical & Heat & CO Sensors / Voice Sounder / Red VAD (HP)	
S4-711-VAD-LPW	S4 Dual Optical & Heat Sensors / White VAD (Std)	
S4-711-VAD-HPW	S4 Dual Optical & Heat Sensors / White VAD (HP)	
S4-711-V-VAD-LPW	S4 Dual Optical & Heat Sensors / Voice Sounder / White VAD (Std)	
S4-711-V-VAD-HPW	S4 Dual Optical & Heat Sensors / Voice Sounder / White VAD (HP)	
S4-720-V-VAD-HPW	S4 Heat / Voice Sounder / White VAD (HP)	
S4-911-V-VAD-HPW	S4 Dual Optical & Heat & CO Sensors / Voice Sounder / White VAD HP	
Non VAD devices		
S4-720	Heat sensor	EN54 : Part 5 :2002* - (heat)
S4-715	Optical sensor	EN54 : Part 7 : 2000* - (optical smoke)
S4-710	Optical Heat sensors	EN54 : Part 7 : 2000* - (optical smoke)
S4-711 S4BK-711	Dual Optical & Heat Sensors	EN54 : Part 5 : 2000* - (heat) CEA 4021 : 2003-07 Class P heat multisensor detector
S4-711-V & S4BK-711-V	S4 Dual Optical & Heat Sensors / Voice Sounder	EN54-5 : 2000, A1 : 2002 (heat) EN54-7 : 2000, A1 : 2002, A2 : 2006 (optical) EN54-3 : 2001, A1 : 2002, A2 : 2006 (sounder) EN54-17:2005 (short circuit isolator) EN54-18:2005 (input output devices)
S4-770-S	S4 Optical & Heat Sensors / Sounder	
S4-771-S	S4 Dual Optical & Heat Sensors / Sounder	
S4-780-S	S4 Heat Sensor Sounder	
S4-901	S4 Dual Optical & Heat & CO Sensors	

* - these standards are met when the sensor is operating LPCB approved STATES, see STATES table below.



If a S-Quad sensor is configured to operate a non LPCB STATE, then this will contravene the LPCB approval. The required STATE is configured during commissioning and can be also configured using controls at the main panel. The approved Sensor STATES are shown on the following pages.

LPCB approved S-Quad Sensor with VAD device STATES (post March 2016)

The STATE in which an S-Quad sensor operate can be changed from the default factory set STATE to another STATE during commissioning using the Vigilon Commissioning tool. The environment in which an S-Quad device is installed will determine what STATE is suitable for use.

Device	LPCB approved STATE	Meets ~ EN54-7 : 2000, A1:2002, A2:2006 * EN54-5 : 2000, A1:2002
Dual Optical & Heat sensor variant -		
S4-711 and S4BK-711	STATE 0 #	Normal sensitivity optical smoke ~ / Class A1 heat *
	STATE 2	Low sensitivity optical smoke ♦ / Class A1 heat *
(With VAD)	STATE 3	High sensitivity optical smoke ~ / Class A1 heat *
S4-711-VAD-HPR	STATE 4	Normal sensitivity (no spike) optical smoke ~ / Class A1 heat *
S4-711-VAD-HPW	STATE 5	Normal sensitivity optical smoke ~ / Class B heat *
S4-711-V-VAD-LPW	STATE 6	Low sensitivity optical smoke ♦ / Class BS heat *
S4-711-V-VAD-LPR	STATE 7	Normal sensitivity optical smoke ~ / Class A2S heat *
(With Speech)	STATE 8	Delayed normal sensitivity optical smoke ~ / Class A1 heat*
S4-711-V and S4BK-711-V	STATE 11	Low sensitivity optical smoke ♦ / Class B heat *
(With VAD & Speech)	STATE 12	Class A1 heat *
S4-711-V-VAD-HPR and S4BK-711-V-VAD-HPR S4-711-V-VAD-HPW		
(With Sounder)		
S4-771-S		

- factory default settings. ♦ Low sensitivity optical smoke sensing is not approved but may still be useful in certain applications.

Commissioning instructions

Device	LPCB approved STATE	Meets ~ EN54-7 : 2000, A1:2002, A2:2006 * EN54-5 : 2000, A1:2002
Dual Optical, Heat & CO sensor variant- S4-901 (With Speech & VAD) S4-911-V-VAD-HPR S4-911-V-VAD-HPW	STATE 0 #	Normal sensitivity optical smoke ~ / Class A1 heat * Normal sensitivity gas (CO)
	STATE 1	High sensitivity optical smoke ~ / Class A1 heat * High sensitivity gas (CO)
	STATE 2	Medium sensitivity optical smoke ~ / Class A1 heat * Low sensitivity gas (CO)
	STATE 4	Normal sensitivity (no spike) optical smoke ~ / Class A1 heat * Normal sensitivity gas (CO)
	STATE 6	Low sensitivity optical smoke ♦ / Class A1 heat * Normal sensitivity gas (CO)
	STATE 7	Normal sensitivity (no spike) optical smoke ~ / Class A2S heat * Normal sensitivity gas (CO)
	STATE 9	Class A1 heat * Normal sensitivity gas (CO)
	STATE 11	Normal sensitivity (no spike) optical smoke ~ / Class B heat *
	STATE 12	Class A1 heat *
Optical & Heat sensor variant - (With Sounder) S4-710 S4-770-S	STATE 0 #	Normal sensitivity optical smoke ~ / Class A1 heat*
	STATE 2	Low sensitivity optical smoke ♦ / Class A1 heat*
	STATE 3	High sensitivity optical smoke ~ / Class A1 heat*
	STATE 4	Normal sensitivity (no spike) optical smoke ~ / Class A1 heat*
	STATE 5	Normal sensitivity optical smoke ~ / Class B heat*
	STATE 6	Low sensitivity optical smoke ♦ / Class BS heat*
	STATE 7	Normal sensitivity optical smoke ~ / Class A2S heat*
	STATE 8	Normal sensitivity (delayed) optical smoke ~ / Class A1 heat*
	STATE 11	Low sensitivity optical smoke ♦ / Class B heat*
	STATE 12	Class A1 heat*

- factory default settings. ♦ Low sensitivity optical smoke sensing is not approved but may still be useful in certain applications.

Device	LPCB approved STATE	Meets ~ EN54-7 : 2000, A1:2002, A2:2006 * EN54-5 : 2000, A1:2002
Heat sensor variant - S4-720 (With Speech & VAD) S4-720-V-VAD-HPR S4-720-V-VAD-HPW (With Sounder) S4-780-S	STATE 0 #	Class A1 heat *
	STATE 5	Class B heat *
	STATE 6	Class BS heat *
	STATE 7	Class A2S heat *
	STATE 13	Class A2 heat *
Optical sensor variant - S4-715	STATE 0 #	Normal sensitivity optical smoke
	STATE 3	High sensitivity optical smoke
	STATE 4	Normal sensitivity (no spike) optical smoke
	STATE 8	Medium sensitivity delayed optical smoke

Note sensor STATES are not applicable for wireless sensors.

Commissioning instructions

S-Quad Sensors (Legacy product range)

The following **S-Quad sensors** when operating in the STATES shown in table below are LPCB approved to the respective standard.

Product number	Description	Standard
S4-720-ST-VO	Heat sensor Strobe and Speech (Speech and Strobe - no approval)	EN54 : Part 5 : 2002* - (heat)
S4-780	Heat Sensor plus Sounder	EN54 : Part 5 : 2002* - (heat)
S4-780-ST	Heat Sensor plus Sounder & Strobe (Strobe - no approval)	EN54 : Part 3 : 2001 - (sounder)
S4-711-ST	Dual Optical & Heat Sensors Strobe (Strobe - no approval)	EN54 : Part 7 : 2000* (optical smoke)
S4-911	Dual Optical, Heat & CO Sensors (CO - no approval)	EN54 : Part 5 : 2002* (heat) CEA 4021 : 2003-07 Class P heat multisensor detector
S4-711-ST-VO	Dual Optical & Heat Sensors, Speech & strobe (Speech and Strobe - no approval)	EN54 : Part 7 : 2000* (optical smoke)
S4-711-VO	Dual Optical & Heat Sensors, Speech (Speech - no approval)	EN54 : Part 5 : 2002* (heat)
S4-771	Dual Optical & Heat Sensors & Sounder	EN54 : Part 3 2001 - (sounder tone)
S4-911-ST-VO	Dual Optical, Heat & CO Sensors, Speech & Strobe (Speech, Strobe & CO - no approval)	CEA 4021 : 2003-07 Class P heat multisensor detector
S4-770	Optical & Heat sensors and Sounder	

* - these standards are met when the sensor is operating LPCB approved STATES, see STATES table below.

Sounder tone description - see next page



If an S-Quad sensor is configured to operate a non LPCB STATE, then this will contravene the LPCB approval. The required STATE is configured during commissioning and can be also configured using controls at the main panel.

Fire Panels

Vigilon Plus 4/6 loop panels (EN)



Vigilon Compact Plus Panel (EN)



Controls and indications



Vigilon Plus 4/6-loop Panel
Vigilon Compact Plus Panel
Vigilon Plus Node
Vigilon Compact Plus Node



Indications applicable for all Vigilon Plus panels and Nodes

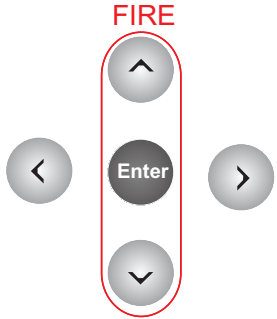
Indicators	Description
Display	The Display provides messages of the system status / events by means of 8 lines by 40 characters per line display.
Zones  (red)	Hidden-until-lit fire zones indicators. When "Zones" text and number(s) are illuminated it indicates that a FIRE has been detected in the specific zone(s). (Zone indicators are not applicable for Network Nodes)
Power  (green)	When illuminated it indicates that a supply to the panel is present.
Fire  (red)	When illuminated it indicates that a FIRE has been detected in the protected premises.
Verify  (amber)	When illuminated it indicates that the Verify button has been pressed and the alarm sounders in the system are delayed from sounding.
Fault  (amber)	When illuminated it indicates that a FAULT has been detected in the fire detection and alarm system or in the audio system.
System Fault  (amber)	When illuminated it indicates that a fault has occurred with the system processor. It is important to investigate this fault because the fire alarm system may not be able to detect fires.
Disablement  (amber)	When illuminated it indicates that a part of the system has been disabled.
Fire routing O/P Active  (red)	Applicable for EN panels only from mid 2015 onwards. When illuminated it indicates the Fire Alarm Routing Equipment is active. Prior to mid 2015 this LED was CB253  (amber) and it operates with Command Build 253.
Fire routing O/P Fault/Dis  (amber)	Applicable for EN panels only from mid 2015 onwards. When illuminated and steady it indicates the Fire Alarm Routing Equipment is faulty. When illuminated and flashing it indicates the Fire Alarm Routing Equipment is disabled. Prior to mid 2015 this LED was CB254  (amber) and it operates with Command Build 254.
Power Fault  (amber)	When illuminated it indicates the battery or mains supply to the panel has failed.
Sounder  (amber)	When illuminated (always with either the FAULT light or the DISABLEMENT light) it indicates that there is a sounder fault (flashing indication) or sounder disablement (steady indication).
Delay  (amber)	When illuminated it indicates that one or more delay blocks are setup on the panel.
Test  (amber)	When illuminated it indicates one or more zones are in Test mode.

Commissioning instructions

Controls applicable for all Vigilon Plus, Compact Plus panels and Nodes

Controls	Description
 MENU ON/OFF	Pressing Menu On/Off will enable/disable the on screen menu facility which gives access to the system menus.
	The 'Fn' buttons are used to select functions and sub-functions of the system menus which appear on the display. Each option available at a menu level has an associated function button and on pressing a function button will select the option described on the display.
 CANCEL BUZZER	The Cancel Buzzer button when pressed will stop the internal panel buzzer from sounding.
 SOUND ALARMS	Pressing the Sound Alarms button will announce 'evacuate message' and sound the evacuate alarms. This button is only pressed in an emergency or at other agreed times, for example when conducting a system test or practice evacuation.
 SILENCE ALARMS	Pressing the Silence Alarms button will stop emergency message announcements and silence the system alarms.
 RESET	Pressing the Reset button will clear any fires and return the panel to the condition seen before the fire event. If a fire condition should reoccur immediately after a 'reset' then the indicated device should be investigated.
 VERIFY	If the Verify facility has been set up, then pressing the 'Verify' button soon after a fire detection will increase the time delay before the alarm sounders are activated. This gives the user time to investigate the cause of the alarm and option to cancel the alarm within the delay time period.
	The 'U' buttons can be configured during commissioning to output user-defined actions, such as to disable devices in areas where smoke may be generated and to activate a plant shutdown, for example during a fire condition. The function of the 'U' buttons should be written on the label that is fitted on the back of the outer door. Note: The Vigilon Plus 4/6 loop panels have four configurable 'U' buttons. At a Vigilon Plus panel to operate a 'Un' key: press the 'Shift' key first then the 'Un' key.
	This is pressed to acknowledge an entry of data or label text.

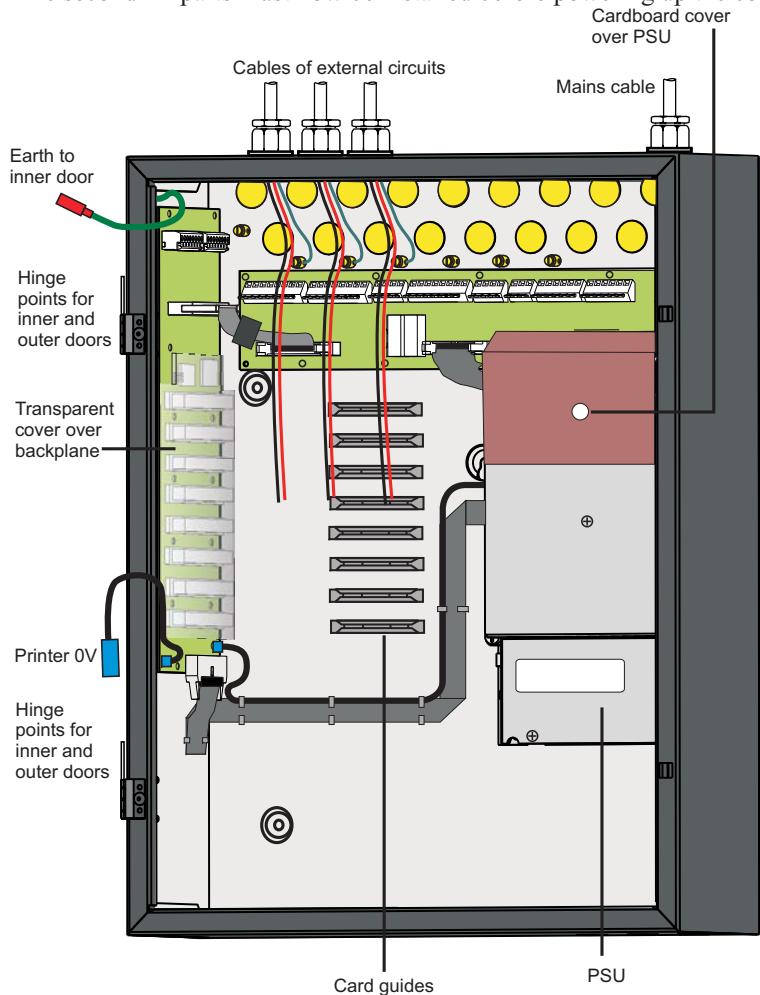
Controls applicable for all Vigilon Plus, Compact Plus panels and Nodes

Controls	Description
	These four buttons are used to scroll the displayed text. The up and down arrows for fire message display are used to scroll fire events.

Vigilon Plus 4/6 loop Panels and Plus Network node

These procedures assume the respective 1st fix assembly of the Vigilon Plus 4-loop (VIGPLUS-24), Vigilon Plus 6-loop (VIGPLUS-72) panel or Plus Network node (VIGPLUS-NODE) is already installed. The 1st fix backbox assembly may be surface or flush mounted.

☐ The second fix parts must now be installed before powering up the control panel.



Second fix parts

☐ Check that you have the second fix parts, which are supplied with the panel:

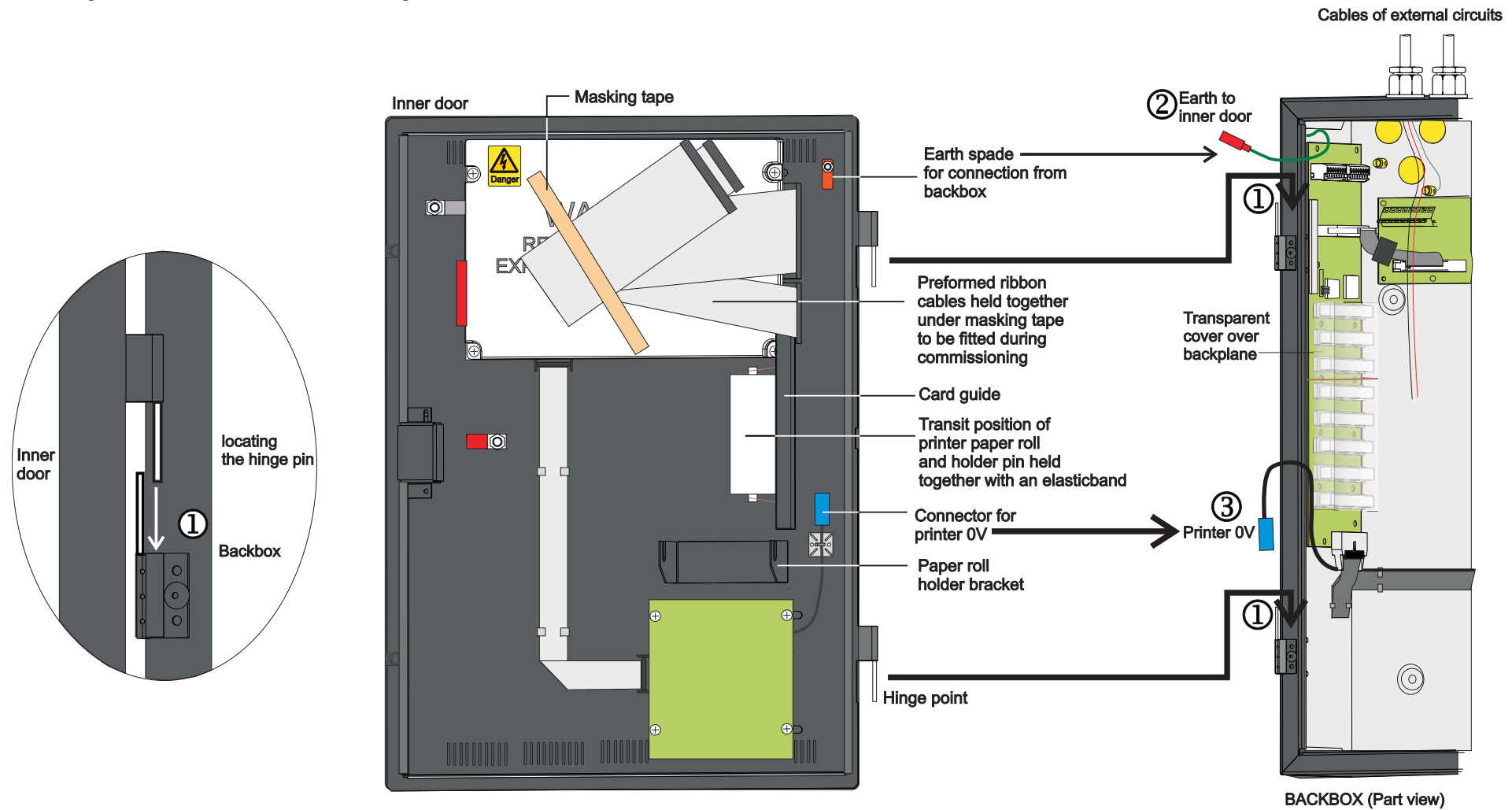
Parts	EN Vigilon Plus 4 loop Control panel (VIGPLUS-24) Vigilon Plus Network Node (VIGPLUS-NODE)	EN Vigilon Plus 6 loop Control panel (VIGPLUS-72)
Inner door assembly	1	1
Battery Pack (2 x 12V 21Ah)	1	2
Battery box		1
Outer door assembly	1	1
Main Control Card	1	1
Loop Card	1 (Option of up to 4 maximum) for VIGPLUS-24 only	1 (option of up to 6 maximum)
Network Card	1 for VIGPLUS-NODE only	-
IO Card	1 for VIGPLUS-NODE only	-
Spares pack (see installation manual for pack content)	1 pack	1 pack

Remove the protective covers

- ☐ Remove the cardboard protection cover fitted over the PSU. The cover is held in by a retaining clip.
- ☐ Remove the transparent protection cover fitted over the backplane held together by an elastic band.

Fitting the inner door

- ☐ Locate the hinge pins on the inner door assembly into the two hinge pin holes ① on the backbox outer face.
- ☐ Fit the earth lead from the backbox to the inner door spade connector ②.
- ☐ Fit together the two blue connectors of the printer 0V leads ③, the leads are located at the inner door and backbox intersection.



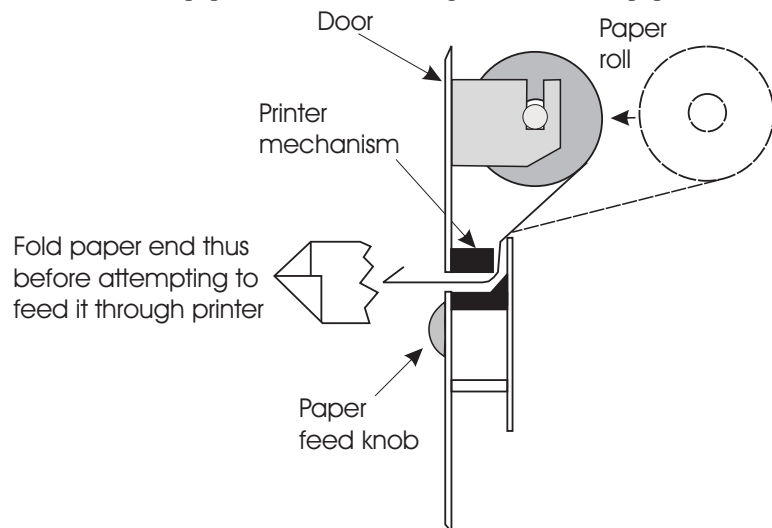
Commissioning instructions

Printer paper roll



The printer paper roll is secured with an elastic band to the card guide on the inner door.

- ☐ Remove the paper roll from the card guide and install paper into the printer mechanism. Ensure the paper roll enters the printer mechanism as shown.

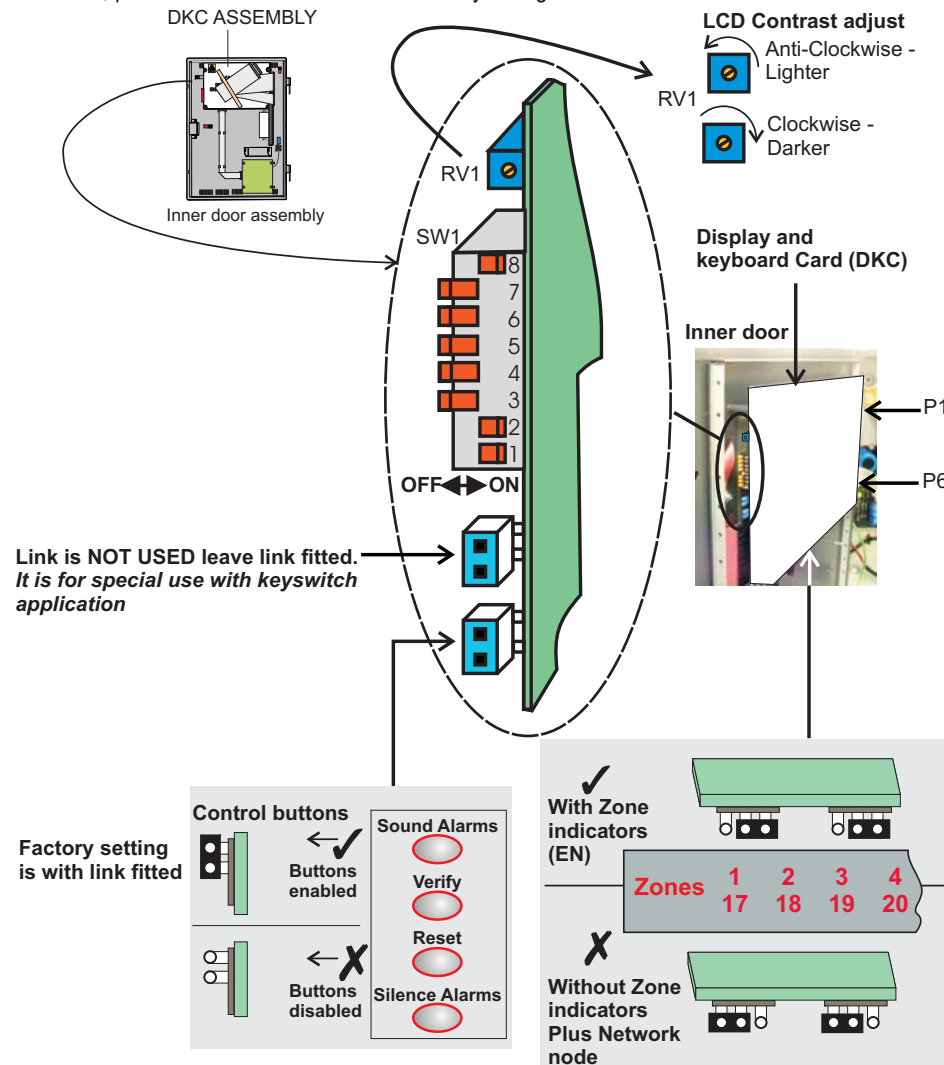


On the outside of the inner door there is a paper feed knob, DO NOT turn the knob in an upward direction as this may damage the integral printer.

- ☐ Upon completion of all commissioning work a new paper roll should be fitted.

Setting the DKC card

The links, pot and switch on the DKC are factory configured as shown below.



Control buttons



The emergency control buttons must always be left in an enabled state, with the 'control buttons' link fitted.

There is no indication given at the panel when the link is set for inactive controls.

Display Contrast

- The display contrast is set using the Pot RV1 on the DKC, which is factory adjusted for optimum contrast. You can also adjust the contrast using the menu option under [Test/Eng], see Appendix A menu maps.



The contrast adjustment function using the menu options may not be applicable for some older builds.

Domain Address and baud rate settings

Typically the Baud rate set using these switches should be the same as the setting made using the [Test/Eng] menu for baud rate setting. A set baud rate using the [Test Eng] menu must be backed up to the NVM.

Baud Rate			Domain address						
1	2	Baud	3	4	5	6	7	8	Address
Off	Off	1200	Off	Off	Off	Off	Off	Off	64
Off	On	2400	Off	Off	Off	Off	Off	On	1
On	Off	9600	Off	Off	Off	Off	On	Off	2
On	On	19200	etc						
			On	On	On	On	On	Off	62
			On	On	On	On	On	On	63

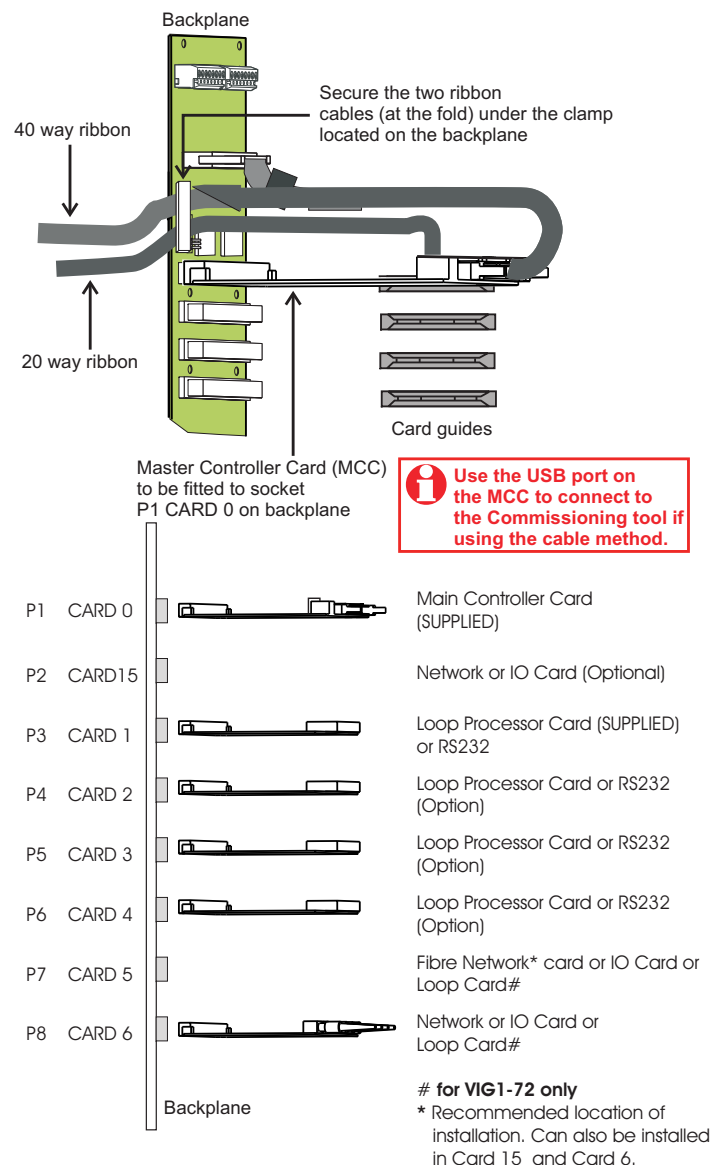
Factory set domain address - 1 with 19200 baud (see SHADED cells)

Zone indicators

The Zone indicators are enabled for normal operation at an EN control panel and must not be disabled.

Commissioning instructions

Card installation



When installing cards into the backplane always use anti-static work procedures. DO NOT use anti-static procedures on live equipment.



An IO Card is not required for connection to DKC and neither is it required to connect to the Commissioning tool. The DKC now connects directly to the MCC card. The Commissioning tool connects directly to the USB port on the MCC for wired method.

The two ribbon cables held together under a masking tape on the DKC assembly must be routed into the backbox to connect to the Main Controller Card.



Ensure the two ribbon cables from DKC are secured under a clamp on the backplane. Fit the Main Controller Card (MCC) into the backplane before the ribbon cables are connected to the MCC.



Always ensure the panel is completely powered down before removal or fitting of cards into the backplane. Power down the battery supply before the mains supply. The power up should be done in reverse order.

- ☐ Ensure all the cards are installed in their correct location into the backplane and are firmly seated into their respective slots.

Terminals on Terminal Card

Backplane

NETWORK CARD IN SLOT P8
0V1 +VE1 -VE1 0V2 N/C +VE2 -VE2 N/C

LOOP CARD IN SLOT P8 (Loop 6)
L1 0V L2 0V

IO CARD IN SLOT P8
A 5V B 0V CTS Rx RTS TX
□ □ □ □ □ □ □ □

PB1

IO (RS232) CARD IN SLOT P7
⊥ N/C ⊥ 0V CTS Rx RTS TX

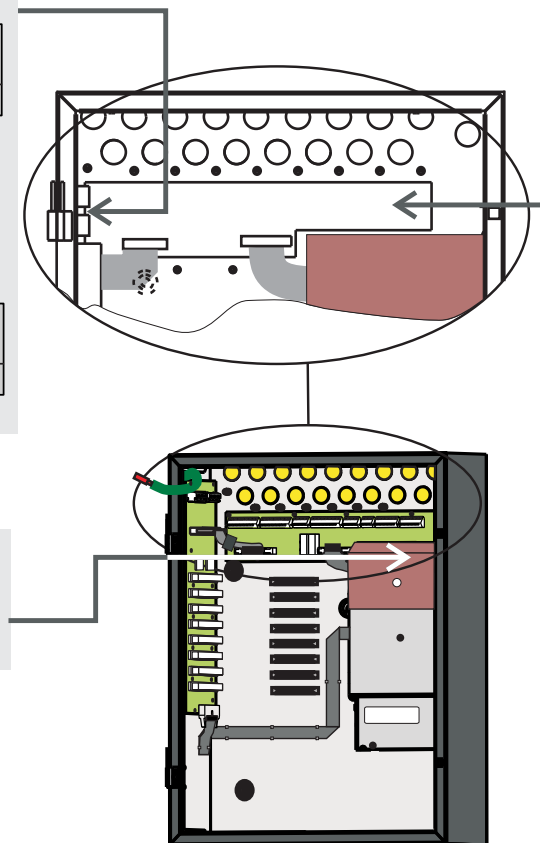
LOOP CARD IN SLOT P7 (Loop 5)
L1 0V L2 0V

□ □ □ □ □ □ □ □

PB2

PSU board (located behind the cardboard cover)

L N P2
□ □ □



Terminal card

Terminals for Network node

CARD 4				CARD 3				CARD 2				CARD 1			
RS232				RS232				RS232				RS232			
CTS	RX	RTS	TX	CTS	RX	RTS	TX	CTS	RX	RTS	TX	CTS	RX	RTS	TX
□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□

Terminals for Control panel

LOOP 4				LOOP 3				LOOP 2				LOOP 1			
L1	0V	L2	0V	L1	0V	L2	0V	L1	0V	L2	0V	L1	0V	L2	0V
□	□	□	□	□	□	□	□	□	□	□	□	□	□	□	□

P2 P3

Master alarm
MA1+ MA1- MA2+ MA2-

Clean C
NC C NO

Terminals for Control panel and Network node

□ □ □ □ □ □ □ □

□ □ □ □ □ □ □ □

P5

P6

0V1 +VE1 -VE1 0V2 N/C N/C +VE2 -VE2 (WITH NETWORK CARD IN SLOT P2 OF BACKPLANE)

RS485
A 5V B 0V Tx CTS Rx RTS

RS232
Tx CTS Rx RTS

□ □ □ □ □ □ □ □

P4

Auxiliary Relay 1
NC C NO NC C NO

Auxiliary Relay 2
NC C NO NC C NO

□ □ □ □ □ □ □ □

□ □ □ □ □ □ □ □

P7

P8

RS232 0V
0V 0V 0V 0V

These RS232 0V terminals are for use with the RS232 wiring associated with IOC fitted in Card slots 1 to 4.

□ □ □ □ □ □ □ □

P12



Quick release terminals

Commissioning instructions

Pre power-up checks

- ☐ The mains cable is the only external cable that is required to be connected at this initial stage. Other external circuit cables are left disconnected, but are connected and tested later, they include:
- all loop circuits (not applicable for Vigilon Plus Network node)
 - clean contacts
 - auxiliary circuits
 - master alarm circuits, has the end-of-line resistors (22K Ohm) fitted to the terminals on the terminal card to inhibit a master alarm circuit fault indication.
 - and RS232/RS485



A networked system is commissioned after all the individual standalone systems are fully commissioned.

- ☐ Ensure all cards are securely fitted into their appropriate slots on the backplane.
- ☐ Ensure all ribbon cables are securely fitted into their respective sockets.

Mains supply wiring

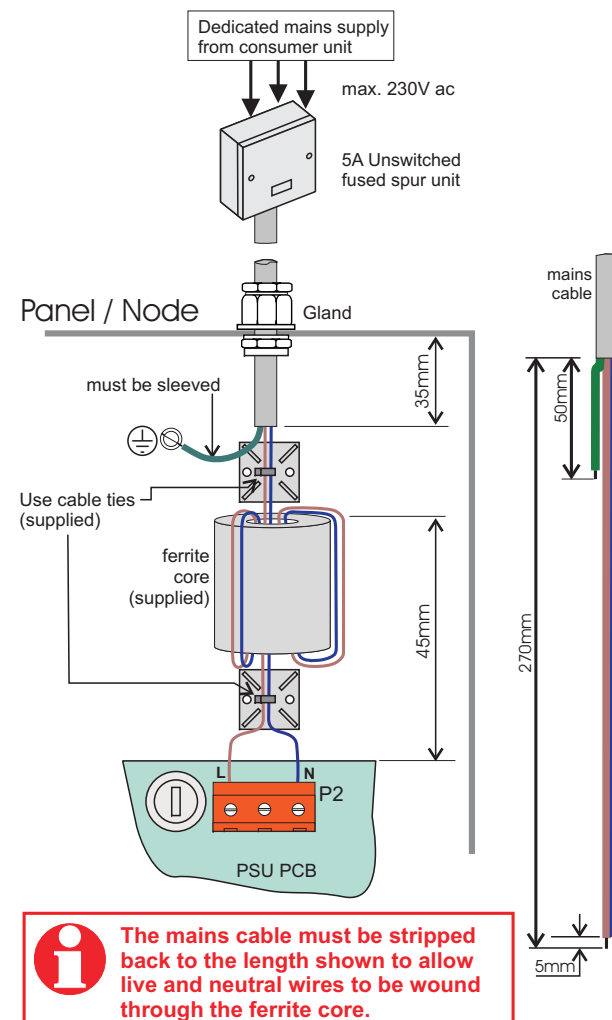


1. The fire alarm system products are NOT designed to be powered from IT Power systems.

2. All mains powered equipment must be earthed.

Ensure the mains supply cable enters the equipment via a dedicated cable entry point, which is located adjacent to the mains terminal block and is also segregated from other loop wiring.

- ☐ Mains supply to any fire alarm control and indicating equipment must be via an unswitched 5A fused spur unit. A disconnect device must be provided to disconnect both poles and must have a minimum gap of 3mm. The 'disconnect device' should be available as part of the building installation and must be easily accessible after installation is complete.
- ☐ The fused spur isolator unit cover should be marked:
FIRE ALARM - DO NOT SWITCH OFF
- ☐ The Mains power is switched on after battery is installed and connected.



The mains cable must be stripped back to the length shown to allow live and neutral wires to be wound through the ferrite core.



Hazardous voltage remains after operation of a protection fuse. Take appropriate action to guard against the risk of equipment having exposed live mains supply.

Battery installation



The panel makes use of sealed lead acid type batteries which can have a useful life of 5 years or more from the date of manufacture. It is strongly recommended that batteries are replaced after 4 years of use. All batteries must be disposed of as per recommendation made by battery manufacturer.

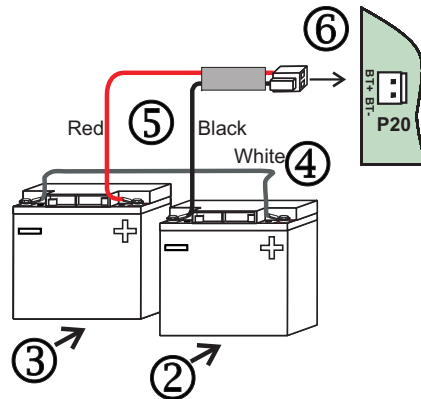
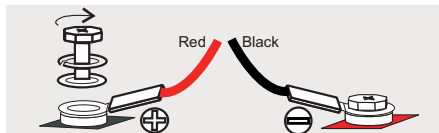


Always use the recommended replacement battery. As there is a risk of an explosion if incorrect batteries are used.

VIGPLUS-24 panel and VIGPLUS-NODE battery installation



Ensure the batteries are held in place while they are being wired. Ensure the battery terminals do not come into contact with the metal enclosure.



It is recommended that the mains supply to the panel/node is switched **Off** during battery installation.

- ① Fit the bolt, spade connector, washer, spring washer to each battery terminal, as shown.
- ② Insert the battery on the right into the back box.
- ③ Insert the battery on the left into the back box.
- ④ Fit the 'White' link lead to outer + and - spade connectors on the two batteries, as shown.
- ⑤ Fit the battery lead assembly (red & black) to the remaining + and - spade connectors on the two batteries, observe polarity.
- ⑥ Plug the plug on the battery lead to connector P20 located on the bottom left of power supply PCB.

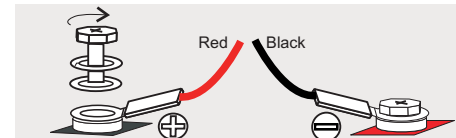


The panel will only power up after the mains supply is switched **On**.

VIGPLUS-72 panel battery installation

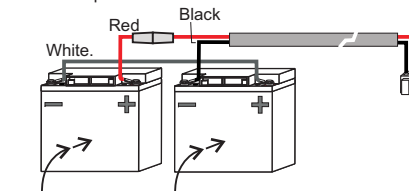


It is recommended that the mains supply is switched **Off** during battery installation.



Ensure the batteries are held in place while they are being wired. Ensure the battery terminals do not come into contact with the metal enclosure.

- ① Place the two batteries on the lower shelf and lay them horizontally with terminals facing outwards.
- ② Fit the bolt, spade connector, washer and spring washer to each battery terminal as shown above.
- ③ Fit the white link lead and then fit the red/black fused lead to the battery terminals, as shown above.
- ④ Raise the two batteries to an upright position and push them back into the enclosure.



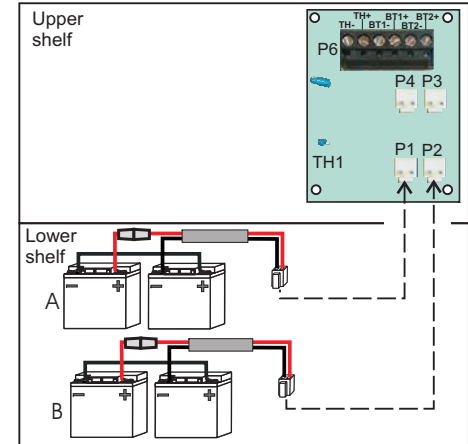
- ⑤ Route the battery red/black lead through hole in the shelf of the battery box and fit the connector end of the lead to the PCB, located on the top right side.
- ⑥ Place two more batteries 'B' onto the lower shelf and repeat procedures ① to ⑤.
- ⑦ Where required, add further four batteries onto the upper shelf, use procedures ① to ⑥. The only exception is that the red/black lead is directly connected to the respective upper connectors on the PCB.



The panel will only power up after the mains supply is switched **On**.

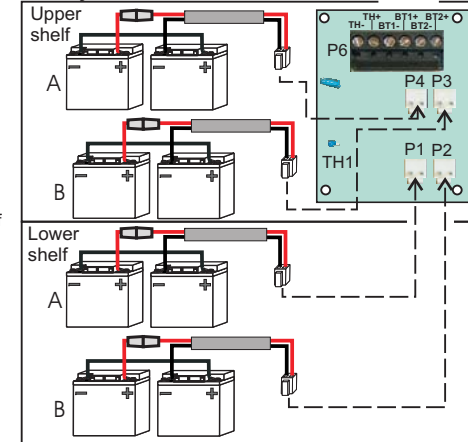
4 - 12V 21Ah batteries

Battery Box



8 - 12V 21Ah batteries

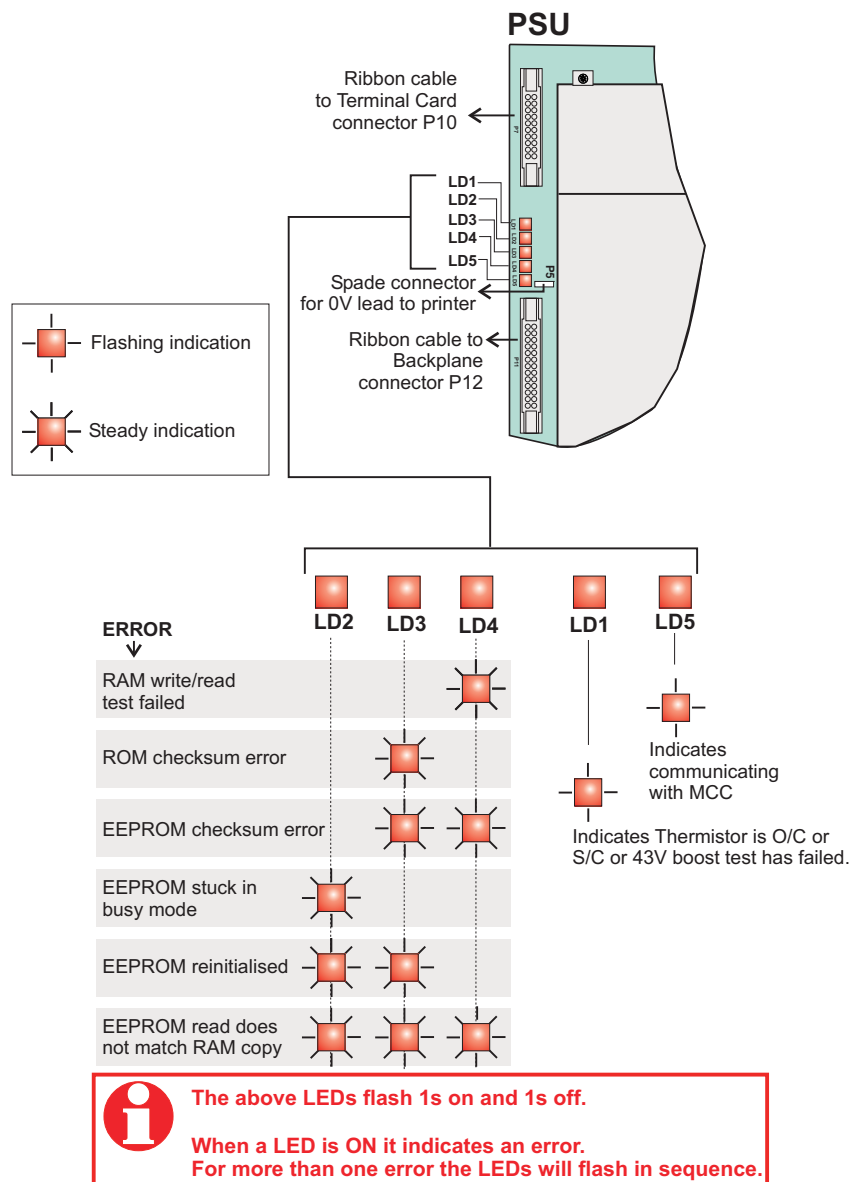
Battery Box



In-line fuse rated 10A QB ceramic 20mm x 5mm

Commissioning instructions

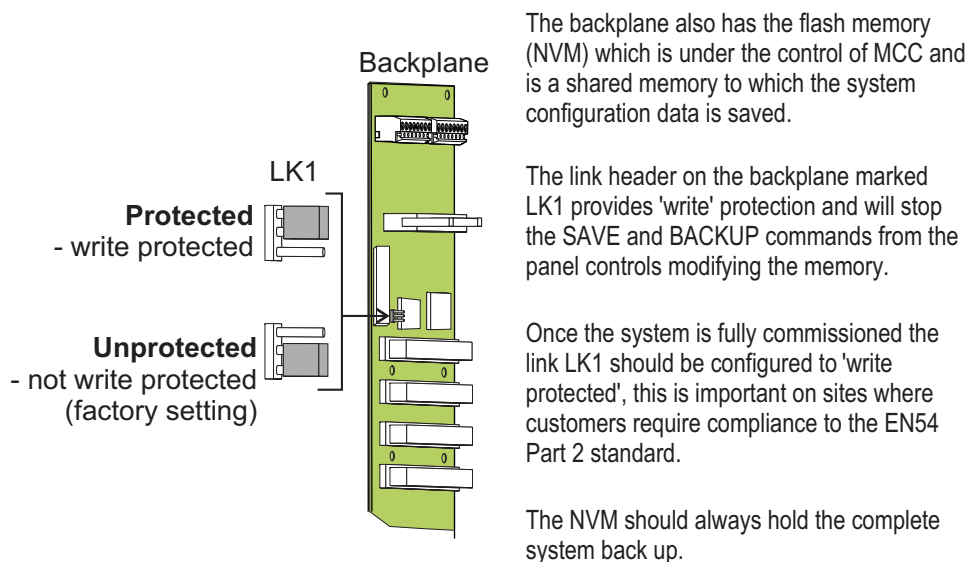
PSU LED indications



Write protect link on backplane

The backplane assembly is fitted inside the left side of the backbox.

The backplane has the card slots to facilitate interconnection of plug-in cards, such as the Main Controller Card (MCC), Loop Processor Cards, IO Cards and Network Cards.

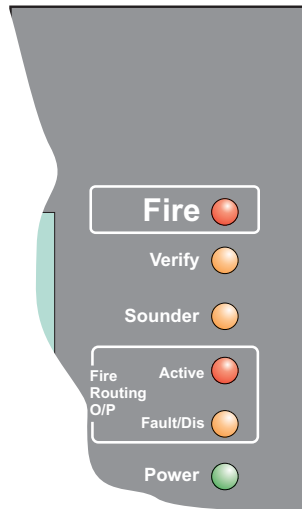


Fire Routing LEDs (Post mid 2015)

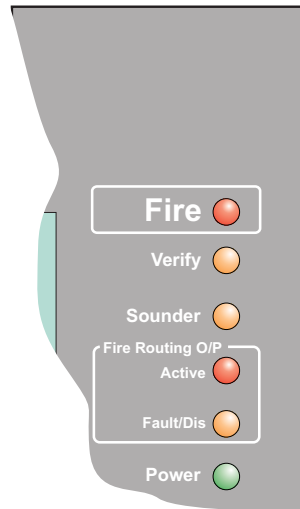
The Fire alarm routing LEDs are fitted on Vigilon panels post March 2015 in place of Command Build LEDs.

The Fire Alarm Routing Equipment FARE red LED works with FARE Interface configured to operate as a FARE device. The FARE LEDs illuminate to indicate status of FARE, with Fault/Dis LED showing fault or disablement status.

Post Q3 2016
Vigilon Plus panel



Post mid 2015
Vigilon panel



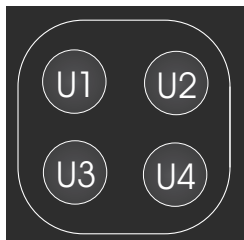
Commissioning instructions

How to configure the 'U' buttons

The U1, U2, U3 and U4 buttons are active at access level 2, that is the buttons are accessible by opening the panel outer door.

Post Q3 2016

Vigilon Plus panel
Vigilon Compact Plus panel
Vigilon Plus Network Node
Vigilon Compact Plus Node



Parts of the Vigilon keyboards showing the U buttons.

To operate a 'U' button first press the 'Shift' key and then the respective 'U' button. Each 'U' button is linked with a command build:

- ☐ U1 button will trigger command build number 251
- ☐ U2 button will trigger command build number 252
- ☐ U3 button will trigger command build number 253
- ☐ U4 button will trigger command build number 254

During commissioning the required command build action must be configured.

Example

This example shows how to configure the U1 button such that pressing it will start the master alarm sounders and pressing 'U1' button again or pressing the **Silence alarm** button will stop the sounders.

The following procedures assume a password entry is not required.

- will appear if a PIN is setup.

Press **Menu On/Off** button and select **[SetUp]** -> **[Usercode]#**, momentarily press **<etc>** to select **[SetUp]** -> **[Build]** and type in the command build number **251**, select **[Action]** -> **[Start MA]** -> **[Enter]** -> **[Reversbl]** -> **[Enter]**. The U1 button is now configured to start the master alarm sounders.

Factory settings

An EN Vigilon Plus 4/6 loop control panel has the following factory settings:

System configuration

- ☐ All devices are assigned to Sector 1
- ☐ All devices are assigned to Zone 1

Port settings

- ☐ P3 - USB (on MCC)

Options:

P0 - RS232 - (Terminal card - terminals P4) when an optional IOC is fitted in slot P2 of backplane
P1 - RS485 - (Terminal card - terminals P4) when an optional IOC is fitted in slot P2 of backplane

Addresses

- ☐ Domain address - **1** for EN panel
- ☐ Panel (Node) address - 1

NVM protect

- ☐ Software 'write' allowed (NVM protect - disabled)
- ☐ Hardware 'write' allowed (NVM protect -disabled)

How to fit the outer door

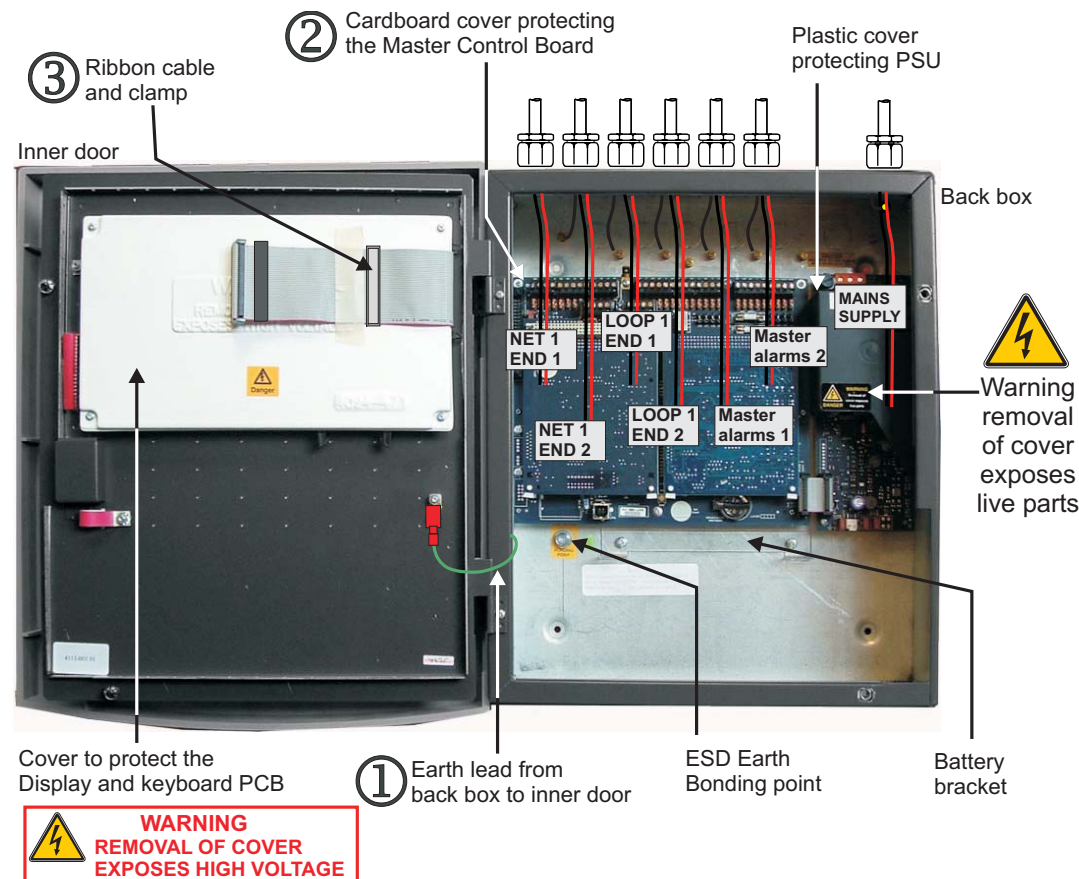
To fit the outer door you will first need to close the inner door and secure it to the backbox using the two fasteners located on the right edge of the inner door.

Fit the Outer door to the enclosure by hooking it onto the side hinge pins on the backbox located on the left edge. The outer door may be secured to the backbox using the key lock.

Vigilon Compact Plus Panel and Compact Plus Network node

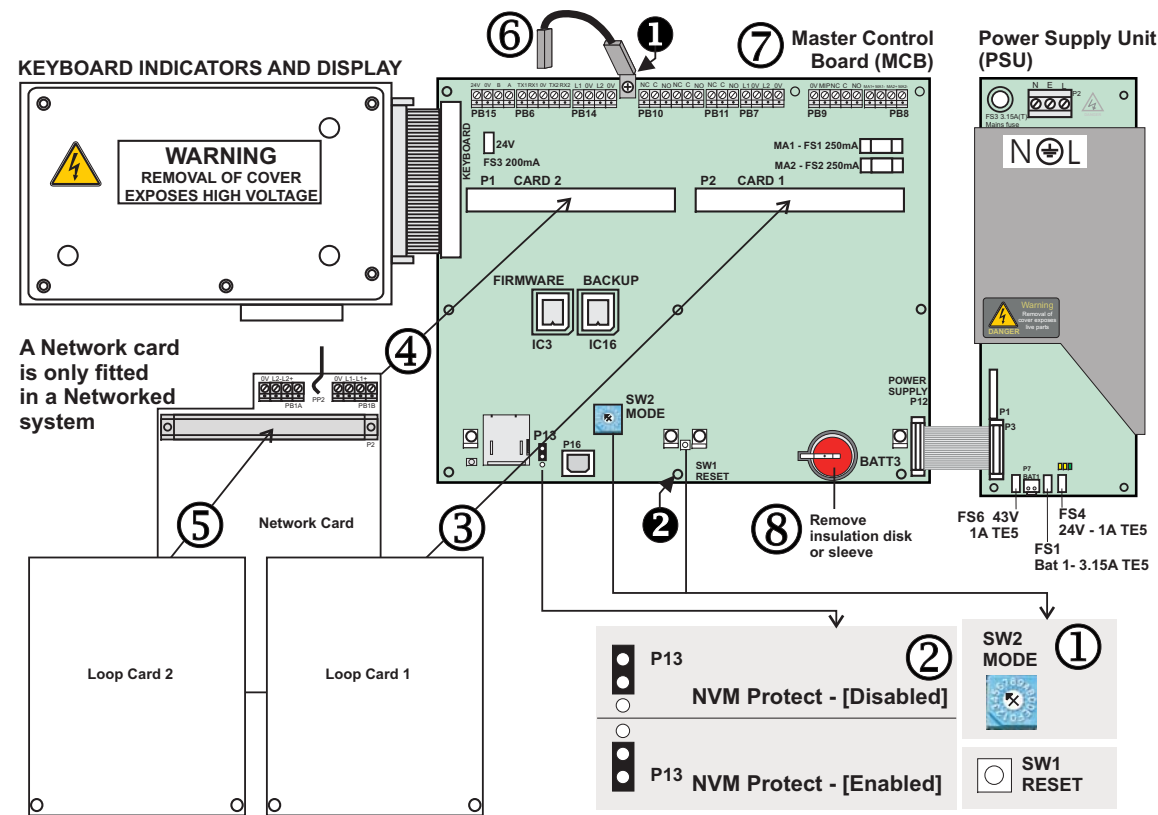
The following procedures assume the Vigilon Compact Plus panel or Compact Plus network node is installed, with cables terminated at the backbox and with the inner and outer doors fitted.

- ☐ Connect the flying earth lead ① from the back box to the inner door spade connector.
- ☐ If not already done remove the protective cover ② fitted over the Master control board inside in the backbox.
- ☐ Connect the ribbon cable ③ from the Display KEYBOARD (DKC) to the socket marked 'KEYBOARD' on the Master control board. Secure the ribbon cable to the side of the enclosure using the cable clamp provided.



Commissioning instructions

Cards and internal cables of the panel



Battery

The lithium battery ⑧ is disconnected on leaving the factory by means of an insulation disk or sleeve over the top connector. The insulation disk or sleeve must be removed before powering up the system.

Terminals

The terminal blocks ⑦ on the top edge of the MCB are used for wiring external circuits. The terminal block on the top edge of the PSU is used for wiring the mains supply to the panel. The connector P7 'Bat 1' located on the bottom edge of the PSU board is used to connect the battery supply.

Earth Link

The earth link lead ⑥ is supplied with the Network Card. This link lead must be fitted to the spade connector on the top edge of MCB with the other end to the spade connector on the Network Card.

Set the Rotary switch SW2

This switch is set to the required position for the panel build on leaving the factory. Before installing the Loop cards and Network card onto the *Master control board MCB* ensure the switch ① SW2 is set to position 6, for panel installation in a legacy system select relevant setting.

Firmware in socket IC3 of MCB being replaced	① Applicable settings of switch SW2 on replacement MCB
2211-148	0
2211-146	1
2211-136	2
2211-127	3
2211-180-4	6

NVM hardware link P13

The NVM can be enabled or disabled by setting the hardware link ② on the MCB. If the NVM protect is hardware disabled then it is also possible to software enable or disable the NVM using the [Protect] option under the [Setup] menu at the panel.

Unprotect: Normally during commissioning the NVM is disabled (unprotected) and writing to NVM is allowed.

Protect: Once the configuration is backed up to the NVM the hardware link must be in the enabled position to disallow writing to the NVM.

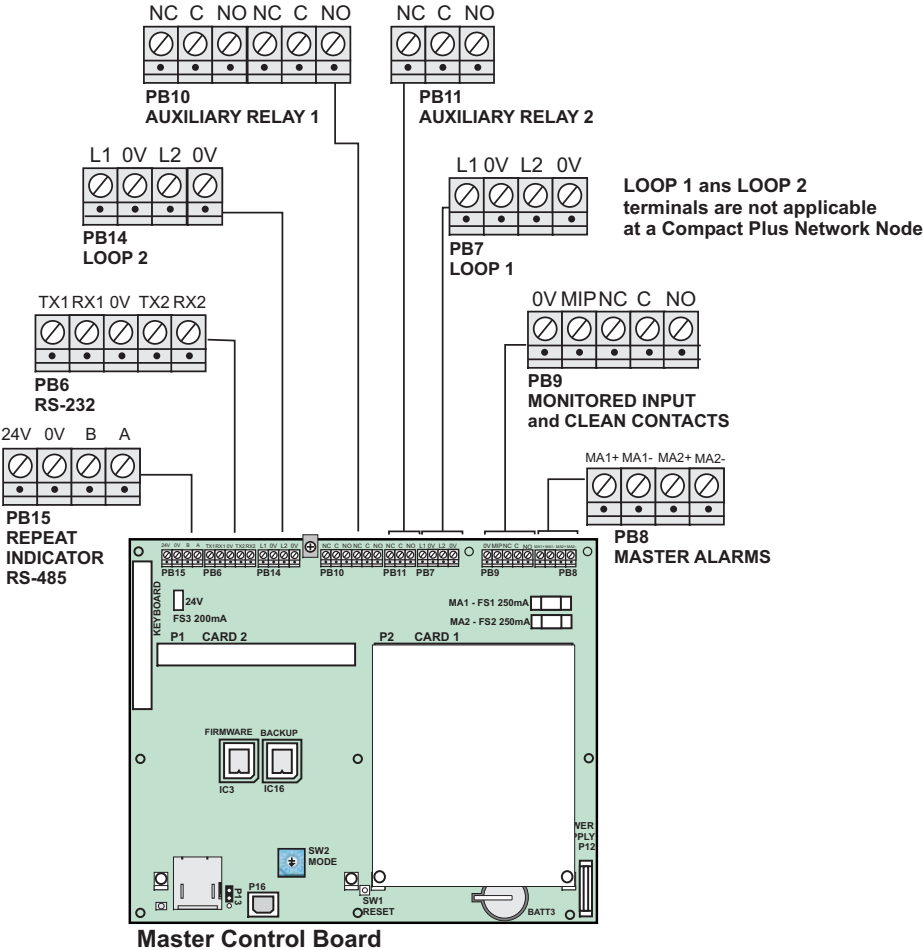
Installing the Cards

The MCB can accommodate two Loop Cards (Loop cards are applicable for Vigilon Plus Compact panel only). One Loop card ③ can be fitted into the slot labelled CARD1 and the other Loop card ⑤ into the slot labelled CARD2.



A networkable system has a Network Card ④ fitted into the slot CARD 2 on the MCB and additionally the Network Card can accommodate the second Loop card.

Terminals on the Master Control Board (MCB)



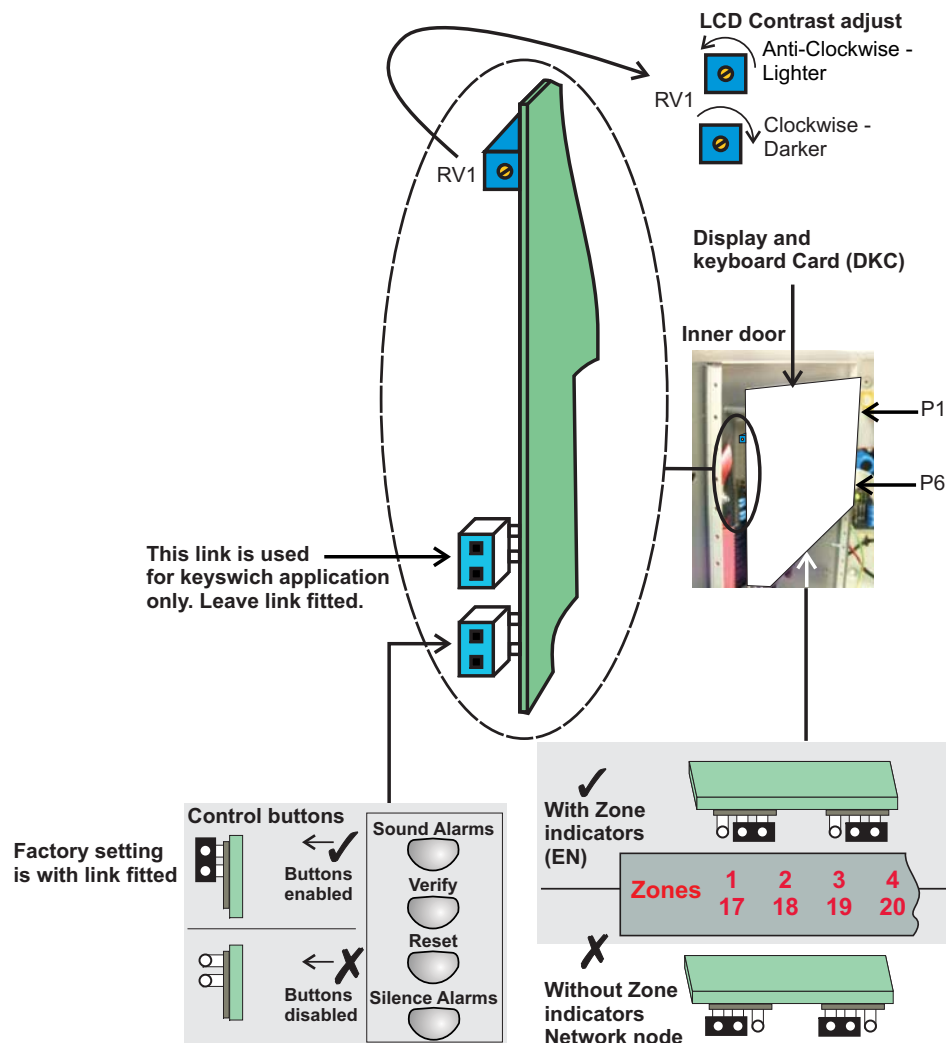
Descriptions of MCB terminals

Terminals	Description
24V, 0V, B and A	These terminals accept the connection of a repeat indicator panel (Port 0). Also the power terminals may be used on a Network node to supply BACnet Gateway.
NC, C and NO Aux 1- 2-sets Aux 2 - 1set	These are auxiliary relay contacts. The auxiliary relay 1 is normally de-energised and operates with any fire event. The auxiliary relay 2 is normally energised and operates (de-energises) with any fault event.
0V, TX, A, RX and B	These are terminals for RS232 or RS232 (Port 1 or 2 respectively).
L1, 0V, L2, 0V (Loop 1 and Loop 2)	These terminals are only applicable for Vigilon Compact Plus panel. These terminals accept the connection of system devices on a loop circuit that starts at L1 and ends at L2. Devices that are connected to the loop circuit include addressable fire sensor/sounders, manual call points, interface units and repeat/mimic panels.
MA1+, MA1-, MA2+ and MA2-	These terminals can accept two master alarm circuits that can operate 24V conventional alarm devices.
0V and MIP	These are monitored input terminals that can accept the connection of a switch. An active input will trigger Command build 250.
NC, C and NO (clean contact)	These are voltage free relay contacts (output) that operate with a fire event.

Commissioning instructions

Settings on the DKC

The links and pot on the DKC are factory configured as shown below.



Control buttons



The emergency control buttons **MUST** always be left in an enabled state, with the 'control buttons' link fitted.

There is no indication given at the panel/node when the links are set for inactive controls.

Display Contrast

- ☐ The display contrast is set using the Pot RV1 on the DKC, which is factory adjusted for optimum contrast. You can also adjust the contrast using the menu option under [Test/Eng].

Address and baud rate



The node and domain addresses can be changed from the factory settings to a required address using the panel [Test/Eng] menus.

Zone indicators

The Zone indicators are enabled for normal operation at the control panel and must NOT be disabled.

The Zone indicators are not applicable on a Network node, so check to ensure the zone indicator links are set in the disabled position on these products.

Pre Power up checks

- ☐ The mains cable is the only cable required to be connected, with other external cables left disconnected at this stage of commissioning:
 - loop circuits
 - clean contacts
 - auxiliary circuits
 - master alarms (the 10K Ohm end-of-line resistor fitted to inhibit a fault indication).
 - Monitored input (the 10K Ohm end-of-line resistor is fitted to inhibit fault indication).

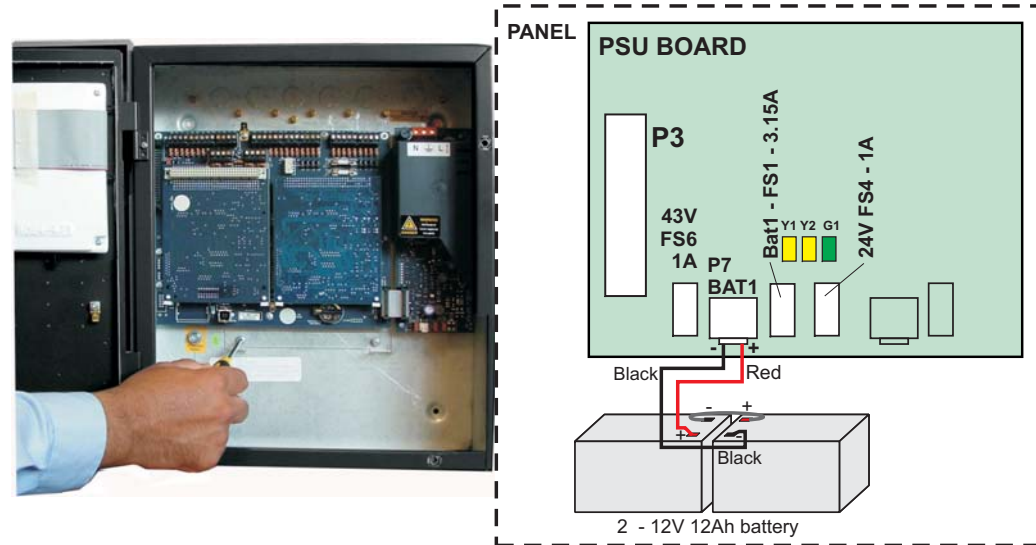


Ensure the mains cable to the panel is securely connected to the mains terminal block on the Power supply unit (PSU).

- ☐ Ensure all cards are securely fitted into their appropriate slots on the MCB.
- ☐ Ensure all ribbon cables are securely fitted into their respective sockets.

Battery Installation

- ☐ Remove the battery support bracket from the backbox.
- ☐ Fit the battery lead to the PSU.
- ☐ Fit the batteries in their correct orientation inside the backbox, see illustration.
- ☐ Refit the battery bracket to secure the batteries in place.



The panel makes use of sealed lead acid type batteries which can have a useful life of 5 years or more from the date of manufacture. It is strongly recommended that batteries are replaced after 4 years of use. All batteries must be disposed of as per recommendation made by the battery manufacturer and in accordance with local regulations.

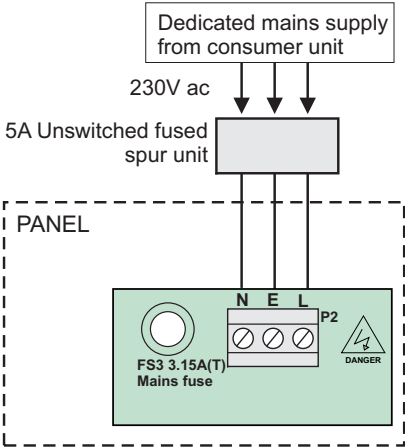


Always use the recommended replacement battery, as there is a risk of an explosion if incorrect batteries are used.

- ☐ Connect the battery leads as shown but leave the final circuit connection until after the Mains supply is switched On.

Commissioning instructions

Mains supply



 **Ensure the mains supply cable enters the panel enclosure through a dedicated cable entry point.**

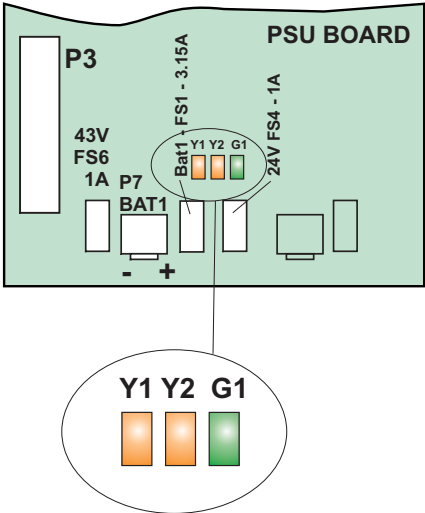
 **These fire alarm system products are NOT designed to be powered from an IT Power system.**

All mains powered equipment must be earthed. Mains supply to any fire alarm control and indicating equipment must be via a dedicated unswitched 5A fused spur unit, which should be clearly labelled **FIRE ALARM - DO NOT SWITCH OFF**.

A 'disconnect device' must be provided to disconnect both poles and must have a minimum gap of 3mm. The device should be available as part of the building installation and must be easily accessible after installation is complete.

 **Hazardous voltage remains after operation of a protection fuse. Take appropriate action to guard against the risk of equipment having exposed live mains supply.**

PSU Indicators



Description	Y1 (yellow LED)	Y2 (yellow LED)	G1 (green LED)
Normal condition (no faults in the system)			fast flash
Mains out of limit			slow flash
24VDC failure only		fast flash	fast flash
Battery 1 failure only		slow flash	fast flash
Battery 1 and 24VDC failure		slow flash	fast flash
43VDC failure only	fast flash		fast flash
No power to PSU			
Earth fault only	On	On	On

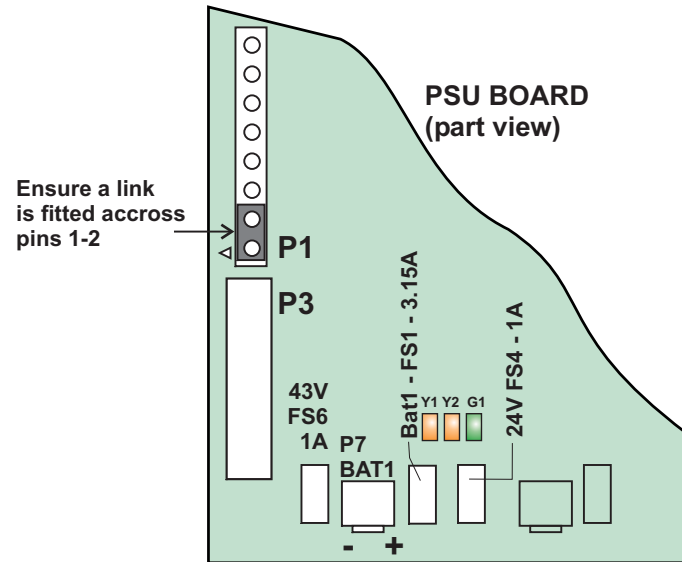
 **On initial power-up all three PSU LEDs will switch ON for approximately 1 second.**

PSU EEPROM link

Ensure a shorting link is fitted to the programming header P1 located on the PSU. The link must be fitted such that it shorts pins 1 and 2. This enables the EEPROM hardware write protect.



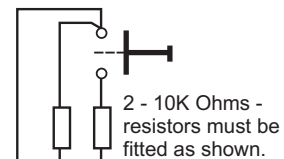
The link is supplied fitted on the PSU board manufactured after September 2007.



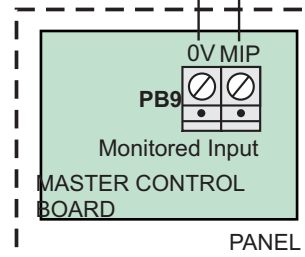
How to configure the monitored input

Normally open contacts

These contacts can be a push button switch, fire/fault relay contacts from another panel or contacts from a timer.



2 - 10K Ohms - resistors must be fitted as shown.



An active input will trigger the Command Build No 250

The Command Build action is configured during commissioning.

The monitored input at the fire panel is activated by an external switch installed a maximum of 100m cable distance away from the fire panel. The input is monitored for both short and open circuit faults. When the input is active it triggers the command build number 250 of the fire panel.

Example

The following example shows how to configure the monitored input at the panel to provide an output sound signal 1 alarm in sector 2 of loop 1 as a reversible action. This means on operating or "closing" the monitored input the panel will start alarms in sector 2 and on releasing or "opening" the monitored input the panel will stop all the alarms from sounding.

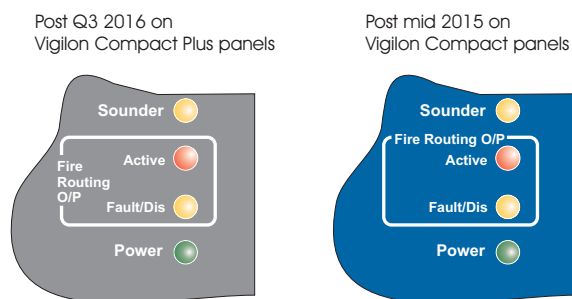
- will appear if a PIN is setup.

The following procedures assume a password entry is not required.

Press **Menu On/Off** button and select **[SetUp]** -> **[UserCode]#**, momentarily press **<etc>** to select **[SetUp]** -> **[Build]** and type in the command build number **250**, select **[Action]** -> **[UserCode]**, momentarily press **<etc>** to select **[Sector]**, type in **2** for sector 2 and select **[Loop]** and type in **1** for Loop 1 -> **[Action]** -> **[Signal 1]** -> **[Enter]** -> **[reversb]** -> **[Enter]**.

Commissioning instructions

Fire Routing LEDs (post mid 2015 and post Q3 2016)

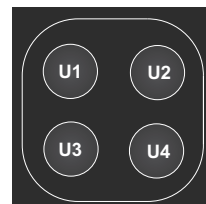


The Fire alarm routing LEDs are fitted on Vigilon panels post mid 2015 in place of the Command Build LEDs.

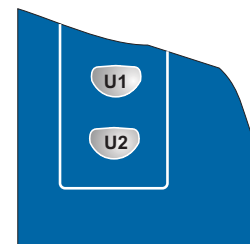
The Fire Alarm Routing Equipment FARE **red** LED work with FARE Interface configured to operate as a FARE device. The FARE LEDs illuminate to indicate the status of FARE.

How to configure the 'U' buttons

Post Q4 2016 on
Vigilon Compact Plus panels



Pre Q4 2016 on
Vigilon Compact panel



Part of the Keyboards showing the U buttons.



The 'U' buttons are accessible by opening the panel outer door.

On operating the U1, U2, U3 or U4 at the panel will cause triggering of command build number 251 (for button U1) to command build number 254 (for button U4). During commissioning the required command build action must be configured.

Example

This example shows how to configure the U1 button such that pressing it after holding down the Shift button it will start the master alarm sounders. On releasing the U1 button it will stop the sounders from sounding.

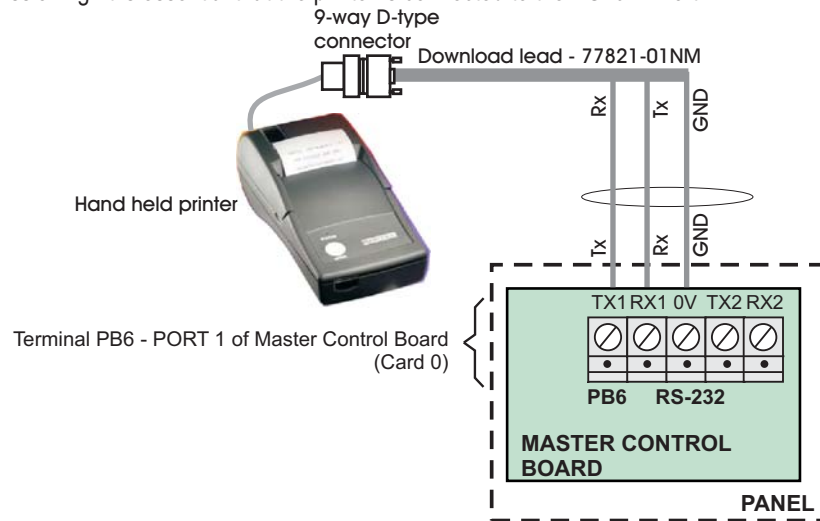
The following procedures assume a password entry is not required.

- will appear if a PIN is setup.

Press **Menu On/Off** button and select **[SetUp] -> [Usercode]#**, momentarily press **<etc>** to select **[SetUp] -> [Build]** and type in the command build number **251**, select **[Action] -> [Start MA] -> [Enter] -> [Reversbl] -> [Enter]**. The U1 button is now configured to start the master alarm sounders.

External printer

An external printer (part number PRINTER-HAND) may be connected to the control panel during commissioning. It is essential that the printer is connected to the RS232 - Port 1.



PORT 1 of the Master Control Board (Card 0) is configured during commissioning to 'Printer mode' with an appropriate 'baud rate' setting, can be 1200 baud.

- will appear if a PIN is setup.

How to configure Port 1 mode to the printer

To configure Port 1 to the Printer mode: Press **Menu On/Off** button and then select **[Test/Eng]-> [UserCode]#**, select **<etc>** and then select **[Config]**. Momentarily press **<etc>** to select **[Comms]**. Then select **[Card]** and enter **0** on choosing the master control card. Then enter **1** to select Port 1. Select **[Mode]** and using the **[Previous]** and **[Next]** buttons scroll to **Printer** and select by the **[Enter]** option.

To switch ON the printer

If the printer is not working and is switched Off then you can switch it On. To switch On the printer: Press **Menu On/Off** button and then select **[Control] -> [UserCode]#** and select **[Printer]** and then select **[On]** and **[Enter]**.

Factory settings

An EN Vigilon Compact Plus Control panel has the following factory settings:

System configuration

- ☐ All devices are assigned to Sector 1
- ☐ All devices are assigned to Zone 1

Port settings

- ☐ P0 - RS485, Baud: 1200, Mode: Repeat
- ☐ P1 - RS232, Baud: 38400, Mode: Standard
- ☐ P2 - RS232, Baud: 38400, Mode: Standard
- ☐ P3 - USB

Addresses

- ☐ Domain address - 1 for EN panel
- ☐ Panel (Node) address - 1

NVM protect

- ☐ Software 'write' allowed (NVM protect - disabled)
- Hardware 'write' allowed (NVM protect -disabled)

Address allocation

Resonance Search¹

Where S-Quad sounder devices manufactured post April 2014 are installed in a system then these devices automatically check that the components used to produce the sound output are working at their resonant frequency, this is the optimum point where the sound output is at its loudest for the lowest power consumption.

During a resonance search the sounders will slowly change the tone frequency to confirm the resonant point. This change in tone frequency will be **heard** as a slight tone shift that lasts for about **1 minute**.

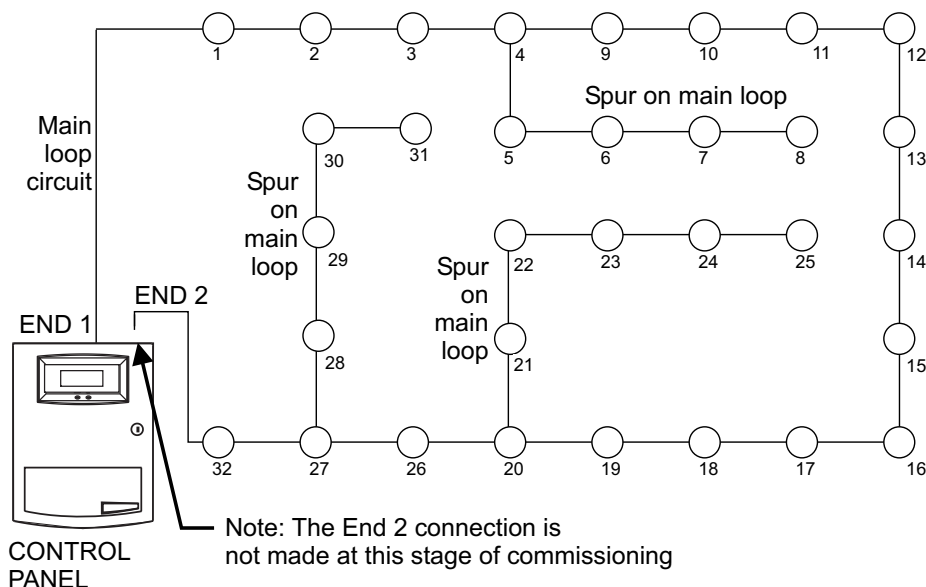
After each allocation of loop circuit from a VIGILON PLUS or VIGILON Compact PLUS panel and subsequent **first** operation of sounders such as when operating the **Sound Alarm** function then there is one time only post allocation *Resonance Search performed*.



Always power-up the panel with the mains supply first and then connect the battery. The power-down sequence should be in the reverse order.

How addresses are allocated to devices on the loop

Each device connected to a loop circuit is allocated a software address on a sequential lowest unused value basis. The allocation of addresses starts from one End of a loop circuit in a numerical order. If both ends of the loop circuit are connected then the allocation of addresses start from End1. On reaching a T-breaker the devices on the spur circuit are allocated addresses. On completion of allocation of addresses to the spur circuit the process continues along the main loop.



Address allocation with Loop circuit end 1 connected



Ensure the panel is first powered-down before connecting the external loop cables to the panel.

It is recommended that each external circuit is wired up to the terminals at the panel, one circuit at a time and tests conducted to ensure that each circuit is functioning correctly. We start by connecting End 1 of the loop only.

- ☐ Ensure **end-2 of the loop cable remains disconnected**.
- ☐ Switch ON the mains and battery supply to power-up the panel, you will initially see the powering up messages. If there are no device 'address allocation' faults the panel will start the loop.

Allocating Loop n

Allocation: OK at Card X : Allocated Y

- ¹ Resonance search has the advantage that ensures sounders operate at maximum efficiency, especially in extremes of temperature and to take account of potential ageing of the sounder parts with time. It is important for resonance search to be conducted during annual maintenance. Note also sounder & VAD devices are monitored for failure. Also EN54 Part 3 speech sounder has an attention tone volume that is lower than the volume output from a standard sounder. All speech sounder devices will not perform the resonance search unless this mode is exited and set to maximum volume.

Starting Loop n

Loop started OK at Card X: Started Y

- ☐ Even if all the devices are wired correctly and there are no allocation faults, there may still be some device faults. The panel *fault buzzer* will sound and the Fault LED will be lit.
- ☐ To stop the local buzzer from sounding press the *Cancel Buzzer* button.



In practice there may be system wiring and hardware faults during the address allocation stage, these faults are also displayed as messages, for further information on what they mean and possible action that can be taken, see Appendix B - Message action.

Allocation faults

During allocation of addresses to the devices connected to a loop circuit there may be faults detected and these are displayed at the panel accompanied with LED indications.

Allocation : HW Fault Card x number y Loop z
 Allocation : Tx Fault Card x number y Loop z
 Allocation : Double Allocated Card x number y Loop z
 Allocation : Map Error at Card x number y Loop z

- ☐ An *allocation fault* that has been rectified will not be recognised until after re-allocation of the loop.
- ☐ A loop with *allocation faults* will not be able to distinguish between a point type sensor and a call point.
- ☐ When a short circuit fault is found on allocation of addresses, the loop re-allocates addresses to the device before the short circuit. The device loop breaker remains open and the device LED is lit.

During start up

- ☐ A loop is started after allocation. Each device is set up to start operating normally based on its type, when the analogue channels are read.

Typical starter fault

Device Battery OC, number X Loop Y

The above fault will only be displayed if for example an interface unit is powered up from mains supply.

How to re-allocate addresses to a loop circuit

It is possible to selectively commence the address reallocation to devices on a loop circuit:.

- ☐ Press the **Menu On/Off** button and then [Test/Eng] -> [UserCode] -> type your password if one is set up and then select [Loop] -> [Allocate], now enter the CARD number of the loop circuit and select [Enter].



After successful allocation of addresses to all the devices on a loop, the new loop map is checked against the last map on the NVM, if found to be different then a *warning* indication is given.

Address allocation with Loop circuit End 2 connected

Before changes are made to the wiring inside the panel first power-down the respective loop circuit.

- ☐ Disconnect End 1 of loop 1 and connect the End-2 of the loop and power up to commence reallocating of addresses to devices on the loop, this time from End 2. Ensure addresses are allocated to all the devices on the loop.
- ☐ On completion of this test, power down the loop and connect both ends of the loop and ensure addresses are allocated to all the devices on the loop from End 1.
- ☐ Repeat these tests on the other loops if fitted.

Plexus Devices

After soft allocation of addresses to loop devices the Plexus devices will need to be allocated addresses using the Plexus tool.

The Plexus tool will need to be connected to the VIGILON panel for LRT binding data to be created. The binding data is then applied to each Plexus device so that it is associated with the LRT and the Plexus devices is given a SAFE device address.

For full information on how to pre commission and pre allocate a Plexus system see publication titled 'Preallocation & Pre-Commissioning - User Guide (4188-1011_Part 2).

SAFE Addressing

A SAFE address is an address given to a device during commissioning, the value of which is stored in the non volatile memory within the electronics module of the device, the SAFE address is therefore carried with the device.

A SAFE address can be given to any device on a loop circuit:

- ☐ Individually
- ☐ in a consecutive range or
- ☐ to all the devices on a loop circuit.

To SAFE address a device

- will appear if a PIN is setup.

Press **Menu On/Off** button and then **[Set Up] -> [Usercode]#** and then momentarily press **<etc>** to select **[Set Up]** and then again momentarily press **<etc>** to select **[Device]**, now type in the soft address of the device and then select **[Loop]**, now type in the loop number on which the device resides and then select **[Safe]** and finally type in a safe address and select **[Enter]**.



It is possible to SAFE address devices using the commissioning tool. If SAFE addressing a range of devices, it is quicker to do this using the controls at the panel.

To convert from SAFE to soft address

If for any reason the SAFE address given to a device is not required and the device needs to be converted for soft addressing, then:

- will appear if a PIN is setup.

Press **Menu On/Off** button and select **[Set Up] -> [Usercode]#** and then momentarily press **<etc>** to select **[Set Up]** and then again momentarily press **<etc>** to select **[Device]**, now type in the SAFE address of the device and then select **[Loop]**, now type in the loop number on which the device resides and then select **[Safe]** and finally types in **0** for the safe address and select **[Enter]**. This will convert the device back to soft addressing mode.

Checking a loop map

A *loop map* is checked against the *as fitted wiring drawings*. This will confirm the exact location of each system device and its address.

It is possible to display or print a loop map.

- ☐ Press the **Menu On/Off** button and then **[Info] -> [Display]/[Print]**, note print will only appear if a printer is connected to the panel. Momentarily press **<etc>** to select **[Loop Map]** (note the menu is different if a network card is fitted in the panel as there will be a [Net Map] option), type in a loop number and select **[Enter]**. The loop map is either displayed or printed, a typical example is shown below.



Map information for Loop 2						15:45
Os	Prev	Next	Common		Position	
1	End1	2			Main Loop	
2	1	3			Main Loop	
3	2	4			Main Loop	
4	3	9	L2	5	Spur	
5	4	6			Spur	
6	5	7			Spur	
7	6	8			Spur	

Address of the device

Address of previous device

Address of next device

Where the device is located, either on the main loop or spur

To find devices on a loop circuit

Checking physical location of devices on a loop map is made easier by having two people in communication with each other. While one person walks around the site and mark off the devices on the *as fitted wiring drawings*, the other person operates the panel controls.

The **'find'** device is a function that gives visual or audible indication at the respective device on a loop circuit. The following procedure assumes no password access is required.

- ☐ Press the **Menu On/Off** button and then **[Test/Eng] -> [Usercode] -> [Loop] -> [Find Dev]**, now enter CARD number n of loop circuit n, select **[Start]** and type in a device number (usually device 1 if starting from the beginning) and then select **[Enter]**. The display shows the device and loop number:

Number 1 on Loop 1

- ☐ Press **[Next]** to find the next device on the loop and at any stage if the previous device needs to be found then select **[Previous]**.

A found device will provide visual or audible indication.

S-Squad Sensors and MCP

- ☐ A *fire sensor* or system *manual call point* will operate its LED for 0.5 second *On* and 0.5 second *Off* repeated.

Sounders and S-Cubed

- ☐ Each system alarm sounder on *S cubed* or *S-Squad* device will provide an audible and / or visual indication for 0.5 second *On* and 0.5 second *Off* repeated.

Mains powered DAU and micro DAU - in Legacy system only

- ☐ A Mains powered DAU will switch *On* an LED located on the main board, which changes from blink to 1s on 1s off when it is found.

Interface units

Ensure the outputs are not connected to external equipment at this stage of commissioning.

- ☐ Each *interface output* will in turn be activated for 0.5 second *On* and 0.5 second *Off*, this process will repeat.

Panels on loop circuits

- ☐ A *mimic and repeat panel* on a loop circuit will display its device number.

NVM and SD Cards

The NVM card and SD Card are located inside the panel on the *Backplane* of VIGILON panel and on the *Master Control board* of VIGILON Compact panels. The NVM should hold the latest system configuration data.



Do not attempt to write protect using the link P13 while a back up to the flash chip or initialisation of the memory is in progress.

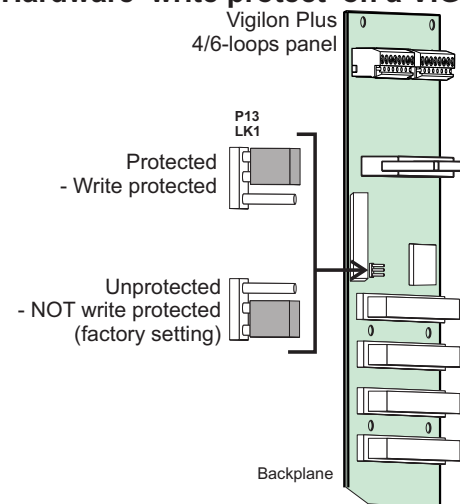
The link header **P13** on the Backplane provides hardware 'write' protection and will stop **[Save]** and **[Back-up]** commands from the panel controls modifying the memory. The link is factory fitted to allow card data to be saved to the memory.



Where the system is required to be EN54 Part 2 compliant, then the link P13 must be left in protect enabled position, that is to disallow write to NVM and SD card on completion of system commissioning.

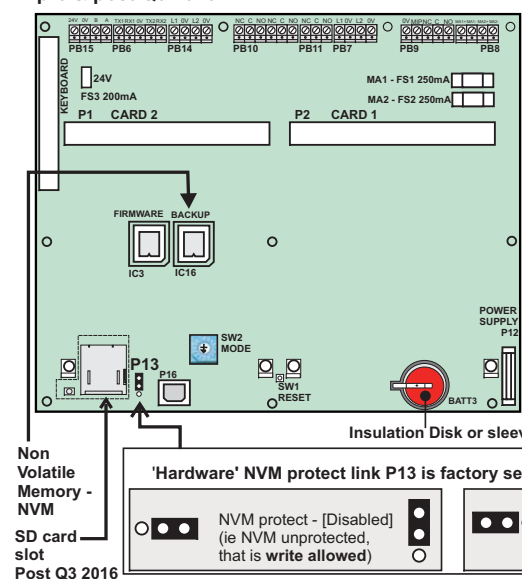
NVM / SD card - Hardware Write protect

Hardware 'write protect' on a VIGILON Plus 4/6 loop panel



Hardware 'write protect' - VIGILON Compact Plus panel

Master Control Board (MCB) - networkable
pre & post Q3 2016



There is also 'software' NVM protect enable and disable feature, see menu maps.

'Hardware' NVM protect link P13 is factory set to NVM protect disabled

☐ NVM protect - [Disabled]
(ie NVM unprotected,
that is **write allowed**)

☐ NVM protect - [Enabled]
(ie NVM protected,
that is **write not allowed**)

NVM / SD Cards software - 'write protect' and 'unprotect'**To 'software' write protect NVM**

Once the NVM / SD Cards are protected, it is not possible to 'write' or 'backup' card data to NVM card or SD Card. To enable write protect:

- ☐ Press the **Menu On/Off** button and then **[SetUp]** -> **[Usercode]#**, momentarily press **<etc>** to select **[Protect]**, select to **[NCM card]** / **[SD Card]** and then select **[Enable]** -> **[Enter]**. # - will appear if a PIN is setup.

A confirmation will be given on the display:

NVM is write protected at card 14

To 'software' write unprotect NVM card / SD Card

Once the NVM is unprotected, it is possible to 'write' or 'backup' card data to the NVM. To write to NVM:

- ☐ Press the **Menu On/Off** button and then **[SetUp]** -> **[Usercode]#**, momentarily press **<etc>** to select **[Protect]**, select to **[NCM card]** / **[SD Card]** and then select **[Disable]** -> **[Enter]**. # - will appear if a PIN is setup, if so enter a PIN.

A confirmation will be given on the display and the warning light may be illuminated:

NVM is not write protected at card 14

To back up loop data to NVM or SD Cards

A fully allocated loop should be backed up to the Non Volatile Memory (NVM). The following procedure assumes no password access is required and write protect is disabled.

- ☐ Press the **Menu On/Off** button and then **[SetUp]** -> **[Usercode]#**, momentarily press **<etc>** to select **[Backup]** and select the loop number n to be backed up to the NVM. Select **[NCM card]** / **[SD Card]** -> **[Enter]**. # - will appear if a PIN is setup.

A confirmation will be given on the display:

Loop card n Backed up

Checksum written to card 14



It is also important to back up data held in Card 0 (MCC), to do this carry out above procedures except after the **[Backup]** command enter '0', which signify card 0.

Commissioning instructions

To recover loop data from NVM / SD Cards

- ☐ On power-up the information stored on the Non Volatile Memory is automatically recovered to the *Local Controller (MCB)* and *Loop processor cards*.

If there is a recovery failure you will get the following message:

Recovery failed at card x

- ☐ It is possible to recover data previously backed up to NVM card or SD Card. Press the **Menu On/Off** button and then **[SetUp] -> [UserCode]#**, momentarily press **<etc>** to select **[Recover]** and select the card number *n* to be recovered from the Memory. Select **[NVM card] / [SD card] -> [Enter]**. # - will appear if a PIN is setup.

A confirmation will be given on the display:

Loop card 1 Recovered



A warning will be given if the loop map is different to that previously backed up to the memory (NVM). For information on how to 'hardware' write protect of NVM Cards and SD Card, see page 46.

How to electrically erase the NVM / SD Cards



Under normal circumstances erasing the NVM is not necessary as there is only one back up of the complete system configuration data held at the panel. The NVM Initialisation should only be performed by a trained and qualified engineer, as executing this command will erase all the data backed up at the panel.



To completely erase the NVM you will first need to switch On the freeblock and disable the hardware and software write protect. This will allow access to the initialisation command.

To switch On freeblock

The following procedures assume no password access is required.

Press the **Menu On/Off** button and then **[Test/Eng] -> [UserCode]#** momentarily press **<etc>** and select **[Test]** and then select **[Freeblock]**, press **[Next]** for controller and select **[Enter]**, which will display numbers on the top left of the display to show the freeblock is On. # - will appear if a PIN is setup.

To initialise the Non Volatile Memory

The following procedures assume no password access is required.

Press the **Menu On/Off** button and then **[Test/Eng] -> [UserCode]#** and then select **[Card] -> [Init] -> [NVM] / [SD Card]** and then **[Enter]**. The following confirmation message will be displayed:

NVM Card Being Initialised NVM card 14 : 512K bytes

To switch Off freeblock

Press the **Menu On/Off** button and then **[Test/Eng] -> [UserCode]#**, momentarily press **<etc>** and select **[Test]** and then select **[Freeblock]**, press **[Previous]** for Off and select **[Enter]**. On completion do a display test to clear the freeblock number display.

Conduct tests on 2Km Loop circuits using the LDT

Tests using the Loop diagnostic tool

It is important to conduct tests using the Loop Diagnostic Tool (LDT) to ensure the panel loops are working correctly. Tests are applicable where the panel is fitted with VIG-LPC-EN or COMPACT-LPC-EN cards. The LDT may be used when 2km loop cards are fitted in a VIGILON Plus panel or VIGILON Compact Plus panel and additionally the loops must have compatible devices.

For information on the Loop Diagnostic Tool (LDT), see the associated User guide.

During certain tests using the Loop Diagnostic Tool and when directed by the tool it may be necessary to repair a loop circuit after a test.

How to [Repair] a loop circuit

When a loop circuit is tested for open circuit fault or if a genuine wiring open circuit fault, then rectification action must be taken. To rectify the open circuit fault, remove the fault and process the repair command at the control panel.

Press the **Menu On/Off** button and select **[Test /Eng]**, **[Usercode]** and if a password access is required, type in the PIN and press the **Enter** button. Select **[Loop]** and then momentarily select **<etc>** and select **[Repair]** and type in the loop circuit you want to repair, for example loop **1or 2** and then select **[Enter]**.

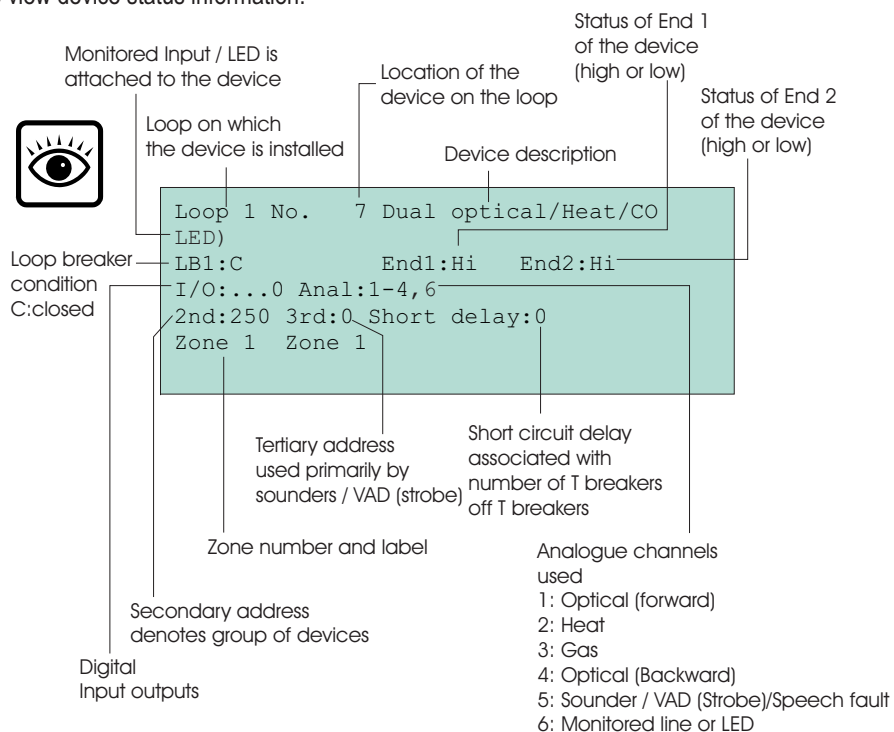
Loop resistance as measured by the panel

The loop cards VIG-LPC-EN or COMPACT-LPC-EN measures and displays the loop resistance on the panel screen during power up of the loop circuits. The measured resistance can be viewed at any time after allocation using the **[Info]** menu under the Loop Card status at the panel.

Checking device STATUS

Loop Device Status

To list the STATUS of a device on an allocated loop at the Control Panel: Press **Menu On/Off** button and then select **[Info]**, momentarily press **<etc>** to select **[Status]** -> **[Device]**, enter the device number, select **[Loop]**, enter the loop number n on which the device resides, select **[Enter]** to view device status information.



Analogue Channel 1 is forward scatter optical smoke sensor

Analogue Channel 2 is heat sensor

Analogue Channel 3 is for CO gas sensor

Analogue Channel 4 is for backward scatter optical smoke sensor

Analogue Channel 5 reports sounder/flasher/speech faults to the control panel

Analogue Channel 6 reports monitored input/monitored output faults to the control panel

O - Output, I - Input and . signify - not used

Device	digital I / O	channels
S4 Dual OH/Red VAD (HP)	O . . O	1, 2, 4, 5
S4 Heat/Voice Sen Snldr/Red VAD (HP)	OOOO	2, 5
S4 Dual OH/Voice Sen Snldr/Red VAD (HP)	. OOO	1, 2, 4, 5
S4 Dual OH/CO/Voice Sen Snldr/Red VAD (HP)	OOOO	1, 2, 3, 4, 5
S4 Dual OH/White VAD (Std)	O . . O	1, 2, 4, 5
S4 Dual OH/White VAD (HP)	O . . O	1, 2, 4, 5
S4 Dual OH/Voice Snldr/White VAD (Std)	OOOO	1, 2, 4, 5
S4 Dual OH/Voice Sen Snldr/White VAD (HP)	OOOO	1, 2, 4, 5
S4 Heat/Voice Snldr/White VAD (HP)	OOOO	2, 5
S4 Dual OH CO/Voice Sen Snldr / White VAD (HP)	OOOO	1, 2, 3, 4, 5
S4 Dual OH/Voice Sounder	. OOO	1, 2, 4, 5
S4 OH Sensor Sounder	. OOO	1, 2, 5
S4 Dual OH Sensor Sounder	. OOO	1, 2, 4, 5
S4 Heat Sensor Sounder	. OOO	2, 5
S4 Dual OH CO Sensor	. . . O	1, 2, 3, 4, 5
S4 Heat Sensor	. . . O	2
S4 Heat Sensor Sounder	. OOO	2, 5
S4 Dual Optical Heat Sensor	. . . O	1, 2, 4
S4 Dual Optical Heat Sensor CO	. . . O	1, 2, 3, 4, 5
S4 Dual Optical Heat Sensor Sounder	. OOO	1, 2, 4, 5
+ Monitored input		6
+ Monitored output		6
Other devices		
Beam transmitter	. . . O	2
Beam receiver	. . . O	1- raw data 2- average data

Device	digital I / O	channels
MCP	. . . O	6
Repeat panel	. . . O	-
Zonal mimic	. . . O	-
Mimic panel	. . . O	-
T-breaker	. . . O	
Loop interface	all possible I - input O - output	1-channel 1 2-channel 2 3-channel 3 4-channel 4 5
Interface unit (mains powered)	all possible I - input O - output	1-channel 1 2-channel 2 3-channel 3 4-channel 4 5-battery 6-mains
S Cubed	OOOO 1 - IR control 2 - Tone 3 - Power 4 - VAD (Strobe)	5

Device	digital I / O	channels
The legacy products below are for reference		
S-Quad Dual Optical Heat Sensor Strobe	O . . O	1, 2, 4, 5
S-Quad Dual Optical Heat Sensor Sounder	. OOO	1, 2, 4, 5
S-Quad Dual Optical Heat Sensor Speech Strobe	OOOO	1, 2, 4, 5
S-Quad Dual Optical Heat Sensor CO Speech Strobe	OOOO	1, 2, 3, 4, 5
Slave LED	. . . O +L	
Sounder	. OO . 3 - Low freq. 2 - On/Off	
Repeat sounder	. OO .	
Optical	. . . O	1
Optical heat	. . . O	1,2
Heat	. . . O	4
Single channel interface unit	I O . O	1,2,5
Loop powered zone module	I . . O	1,5
Optical/Heat Sounder	. OOO	1,2



The correct device type may not be displayed if the loop circuit has allocation faults.

- ☐ Check that the device is of the correct type and is suitable for the area in which it is installed.
- ☐ Check the digital status of all devices.
- ☐ On successful allocation check that the total number of devices found equals the number installed. Also note there will be a loop voltage on the unconnected end of the cable.

Plexus Device Status

At the Vigilon Plus Control panel select **[Info]** -> momentarily select **<etc>** in order to find and select **[Status]** -> **[Device]** and enter the device number or range whose status is required, then select **[Loop]** and enter the loop number on which the Plexus devices reside and then select **[Enter]**. The panel displays the status of device(s):

Radio Call Point

```
Loop 2 No.    43 Radio Call Point    15:45
I/O:...O Anal:5, 6
2nd:0 3rd:0 LRT:41
Zone 2 Zone 2
S:000005098244 ID:3040
P1:42 RSSI:48 P2:41 RSSI:49 Batt:40
```

This is the RSSI which is a measure of the link quality with the linked **Parent** device having address 42

This is the network ID
Months of charge remaining dependent on use.
A device battery fault indication is given closer to end of useful battery life.

This is the RSSI which is a measure of the link quality with the linked **Parent** device having address 41

This is the serial number given of the Radio call point, which is the same number as the one printed on a label inside the Call point on its PCB.

Radio Sounder Strobe

```
Loop 1 No.    47 Radio Sounder Strobe 15:47
I/O:...000 Anal:5,6
2nd:0 3rd:0 LRT:41
Zone 4 Zone 4
S:010010485766 ID:3040
P1:43 RSSI:47 P2:42 RSSI:43 Batt:35
```

Radio Optical detector

```
Loop 1 No.    51 Radio optical    15:48
I/O:...O Anal:1,5
2nd:0 3rd:0 LRT:41
Zone 4 Zone 4
S:039100006577 ID:3040
P1:47 RSSI:41 P2:51 RSSI:44 Batt:59
```

Radio Heat detector

```
Loop 1 No.    45 Radio Heat    15:46
I/O:...O Anal:2, 5
2nd:0 3rd:0 LRT:41
Zone 2 Zone 2
S:136110645425 ID:3040
P1:44 RSSI:45 P2:47 RSSI:47 Batt:59
```

Radio Dual Optical Heat detector

```
Loop 1 No.    51 Radio Dual optical/Heat
I/O:...O Anal:1,2,3,4
2nd:0 3rd:0 LRT:41
Zone 2 Zone 2
S:0471000064440 ID:3040
P1:50 RSSI:44 P2:43 RSSI:42 Batt:59
```

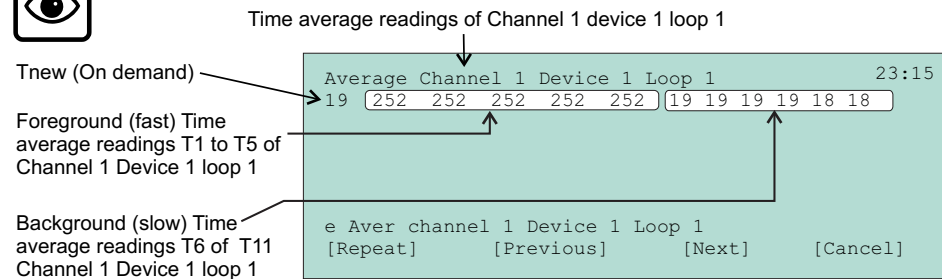
Device checks



Ensure all dust covers have been removed from the sensor heads and the system is allowed to operate for at least 24 hours to obtain accurate time average and condition code readings.

Checking the time averages

Press the **Menu On/Off** button and then **[Info]**, momentarily press **<etc>** to select **[Time Avg]** and type in the required channel number from the range **1-6** whose time average is required, select **[Device]** and type in the device number, select **[Loop]** and type in the loop number of the device **->[Enter]**. The display shows time averages for the device.



	Foreground (fast) time average readings					
Tnew	T1	T2	T3	T4	T5	
On demand	80mS	320mS	1.28S	5.12S	20.48S	
	Normally reads 252, unless there is a threshold.					
	Background (slow) time average readings					
Tnew	T6	T7	T8	T9	T10	T11
On demand	1.36min	5.46min	21.84min	1.45hr	5.82hr	23.3hr

Product	Analogue Channels	Time average at manufacture Tnew
S4 Heat Sensor	2 - heat	66
S4 Heat Sensor Sounder	2 - heat	66
S4 Dual Optical Heat Sensor	1 - optical (forward) 2 - heat 4 - optical (backward)	25 66 40
S4 Dual OH/Red VAD	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Heat/Voice Sen Sndr/Red VAD	2 - Heat	66
S4 Dual OH/Voice Sen Sndr/Red VAD	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Dual OH/CO/Voice Sen Sndr/Red VAD	1 - Optical (forward) 2 - Heat 3 - CO 4 - Optical (backward)	25 66 25 40
S4 Dual OH/White VAD	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Dual OH/White VAD (HP)	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Dual OH/Voice Sndr/White VAD (Std)	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Dual OH/Voice Sen Sndr/White VAD (HP)	1 - Optical (forward) 2 - Heat 4 - Optical (backward)	25 66 40
S4 Heat/Voice Sndr/White VAD (HP)	2 - Heat	66
S4 Dual OH/CO/Voice Sen Sndr/White VAD HP	1 - Optical (forward) 2 - Heat 3 - CO 4 - Optical (backward)	25 66 25 40

Commissioning instructions

S4 Dual OH/Voice Sounder	1 - Optical (forward)	25
	2 - Heat	66
	4 - Optical (backward)	40
S4 OH Sensor Sounder	1 - Optical (forward)	25
	2 - Heat	66
S4 Dual OH Sensor Sounder	1 - Optical (forward)	25
	2 - Heat	66
	4 - Optical (backward)	40
S4 Heat Sensor Sounder	2 - Heat	66
S4 Dual OH CO Sensor	1 - Optical (forward)	25
	2 - Heat	66
	3 - CO	25
	4 - Optical (backward)	40
Beam Transmit		0
Beam Receive	2	150 - 170

Typical new Time average values for legacy 34K sensors

Device type 34K	Analogue Channel number	Time average Tnew
Optical	1	200 - 235
Heat	2	180 - 210
Heat	2	180 - 210

Checking the Exception codes

'Exception' condition codes were previously known as 'Subfault'. These codes provide information about a sensor device. A code indicates small changes in the environmental condition, sensor mechanism and how the sensor performs in the system. To ensure sensor exception codes are meaningful, all existing codes must be cleared and the system left undisturbed for at least 24 hours.

Code definition

There are ten different conditions possible for each sensor type, although not all of them are defined. Each condition code has a range of 0-3.

Code	Meaning
0	This condition is OK. This code is not displayed unless another non zero code exists.
1 or 2	Suggests preventive maintenance is required, where necessary, to avoid possible false alarms.
3	The sensor is faulty. A fault condition indication is given at the control panel. A message reading 'Sensor is faulty' Pre MCC V4.45 this message read 'Sensor out of Specification'.

A CODE 3 is automatically disabled by the system to prevent false alarms.

Time and date when Exceptions or Condition codes were read



```

Time: 15:54.07  Fri 20 December 2002  23:15
Exception
Number 2 on Loop 1
condition 3 3 3 3 3 3 3 3 3 3
  
```

Location of the device residing on this loop whose exceptions are shown.

These condition codes indicate the device is probably disconnected

0 - position
10th - position
For definition of these codes see the table

To view the exception/condition codes of a device

Press to select **Menu On/Off** button and select **[Info] -> <etc> -> [Event] -> <etc> -> [Exception] -> [Enter]**. To view all the currently active exceptions on the system use the next and previous menu options to scroll through.

How to clear 'Exceptions' (# - will appear if a PIN is setup)

This can be done by removal and replacement of the *sensor chamber*, by *re-allocation of the loop having the sensor or*, by using *[Clear]* function at the panel:

Press **Menu On/Off** button to select **[Test/Eng] -> [Usercode]# -> <etc> -> [Config] -> [Clear]-> [Exception] -> [Enter]**. Note cleared codes will return if the conditions are still true.

Exceptions /Condition codes	Meaning	Action
1 0 0 0 0 0 0 0 0 or 0 1 0 0 0 0 0 0 0 or 1 1 0 0 0 0 0 0 0	This is the <i>sub-fire</i> band and if set as shown it should be taken as showing that the sensor is at its optimum sensitivity for its location.	No action need be taken.
2 0 0 0 0 0 0 0 0 or 0 2 0 0 0 0 0 0 0 or 2 2 0 0 0 0 0 0 0	A <i>sub-fire</i> has been generated which would suggest that the sensor was either too sensitive for its environment or that the sensor type may be incorrect for the location.	Action should be to check location and alter sensor state or type as required. This should only be done with the knowledge of the customer. Remember to re-backup all changes to states onto the NVM.
0 0 1 0 0 0 0 0 0 or 0 0 2 0 0 0 0 0 0	These codes indicate that the sensor has sensed small signals. Code Level 1 shows one event, Level 2 shows greater than one event.	The location should be checked and a change in siting made, if required.
0 0 0 0 1 0 0 0 0 or 0 0 0 0 2 0 0 0 0 or 0 0 0 0 0 1 0 0 0 or 0 0 0 0 0 2 0 0 0	This shows that the sensor time averages are close to acceptable limits.	On commissioning the sensor should be replaced. A sensor with code 2 is worse than code 1. On maintenance sensors with code 2 should be replaced.
0 0 0 0 0 0 1 0 0 or 0 0 0 0 0 0 2 0 0 or 0 0 0 0 0 0 0 1 0 or 0 0 0 0 0 0 0 2 0	This shows that high frequency noise events have been detected by the system. A code 1 shows one event and a code 2 shows more than one event.	The device should be replaced. On maintenance, sensor with code 2 should be replaced. If it reoccurs then check the environment.
0 0 0 0 0 0 0 0 1 or 0 0 0 0 0 0 0 0 2	This shows device hardware faults. 1-fault code 1 and more than one fault code 2.	The sensor should be replaced.
0 0 0 0 0 0 0 0 1 or 0 0 0 0 0 0 0 0 2	This shows that transmission faults are being noted. Code 2 is worse than code 1.	The device should be replaced. If it reoccurs then check the environment.

Condition Codes for S-Quads (Exception / Sub Fault codes)

Gen type	Pos. No.	Description	Condition codes			
			Normal band 0	Sub fault band		Fault band 3
				1	2	
E N V I R O N M E N T	1	Optical subfire	None	Small signal sensed [Check location, state & type]	Subfire [Check location, state & type]	
	2	Heat subfire	None	Small signal sensed [Check location, state & type]	Subfire [Check location, state & type]	
	3	Gas subfire		Small signal sensed [Check location, state & type]	Subfire [Check location, state & type]	
	4					
S E N S O R	5	Optical/Gas channel drift or out of range	OK		Close to acceptable limit	Out of limits [Clean/replace]
	6	Heat channel drift or out of range	OK			Out of limits [Clean/replace]
	7	Optical/Gas channel noisy (High freq)	OK	Single HF noise event detected	Multiple HF noise seen (Check location and report)	
	8	Heat channel noisy (High frequency)	OK	Single HF noise event detected	Multiple HF noise seen (check location and report)	
D E V I C E	9	device firmware	OK	Isolated fault [Note/report]	Repetitive fault [Note / report / replace]	Total failure [Replace]
	10	device transmission	OK	Low error rate	Medium error rate [Report]	High error rate [Replace]

Exception (or Condition) Codes for S-Quad Beam

Gen type	Pos No.	Description	Exception codes			
			normal band 0	sub fault band		fault band 3
				1	2	
E N V I R O N M E N T	1st	Subfire	None			
	2nd	Subfire	None			
	3rd					
	4th	Total beam obscuration	OK			Obscured for > 1 minute or Tx failure
S E N S O R	5th	Out of range - high	OK			Gain not set-up or faulty set replace
	6th	Drift out of range - low	OK	T5 time average < 14 seen	Low resolution [Clean / check / gain, align / replace]	Below lower limit [Check gain / replace]
	7th	Noisy (high frequency)	OK	Single HF noise event detected	Multiple HF noise seen [Check location]	-
	8th					
D E V I C E	9th	Outstation firmware	OK	Isolated fault [Note / report]	Repetitive fault [Note / report / replace]	Total failure [Replace]
	10th	Outstation transmission	OK	Low error rate	Medium error rate [Report]	High error rate [Report / Replace]

S-Quad Sensor with VAD - STATES (current range of devices)

The STATE in which an S-Quad sensor operates can be changed from the default factory set STATE to another STATE during commissioning using the Commissioning tool. The environment in which an S-Quad device is installed will determine what STATE is used. # - Factory Default STATE ~ EN54 : Part 7 :2000 * EN54 : Part 5 :2002



The S-Quad with VAD devices are compatible with Vigilon Plus control panel software (MCC/MCB ≥ V4.52 and LPC ≥ V4.48).

See page 13 for information on LPCB approved STATES.

S-Quad Dual Optical Heat & Optical Heat sensor STATES

STATE	Definition / Class	Application in / Suitable for:	Sensitivity a-high- to-k-none
STATE 3	High sensitivity optical, Class A1 heat	Clean area or environment, where early detection is required.	a
STATE 4	Normal sensitivity with no optical spike protection, Class A1 heat	General area as STATE 0 with less immunity to steam and dust for ease of testing using artificial smoke in normal operating mode, when test / commissioning mode is not active.	b
STATE 0#	Normal sensitivity optical, Class A1 heat	General area.	c
STATE 8	Delayed normal sensitivity optical, Class A1 heat	General application in area with transient steam, dust or smoke present.	d
STATE 7	Normal sensitivity optical, Class A2S heat	General application in area with low dust, smoke or steam present, for fixed temperature operation that is totally unaffected by transient temperature variation	e
STATE 5	Normal sensitivity optical, Class B heat	Area having high ambient temperature with low dust, smoke or steam present.	f
STATE 2	Low sensitivity optical smoke ◇, Class A1 heat	Areas with moderate dust, smoke or heat during working hours when used with timing function.	g
STATE 11	Low sensitivity optical smoke ◇, Class B heat	Area having high ambient temperature, plus where moderate dust, smoke or steam present.	h
STATE 6	Low sensitivity optical smoke ◇, Class BS heat	Area having high ambient temperature that changes, plus moderate dust, smoke or steam present.	i
STATE 12	Class A1 heat only	Area where smoke, dust or steam occurs at times.	j
STATE 15	No detection		k

- factory default STATE. NOTE: All the Sensor STATES in the above table are LPCB approved to EN54-5 & EN54-7 with the exception of STATES 2, 6 and 11 which are only approved to EN54-5.

◇ Low sensitivity optical smoke sensing is not approved but may still be useful in certain applications. The STATE 15 is a no detection state and is not part of LPCB approval.

S-Quad Dual Optical, Heat & CO sensor STATES

STATE	Definition / Class	Application in / Suitable for:	Sensitivity a-high- to-i-none
STATE 1	High sensitivity optical, Class A1 heat, high sensitivity gas	Clean area or environment where early detection is required with false alarm reduction.	a
STATE 4	Normal sensitivity optical with no spike protection, Class A1 heat, normal sensitivity gas (CO)	General area as STATE 0 with less immunity to steam and dust for ease of testing using artificial smoke in normal operating mode, when test / commissioning mode is not active.	b
STATE 0#	Normal sensitivity optical, Class A1 heat, Normal sensitivity gas (CO)	General area where false alarm reduction is required for comprehensive fire detection.	c
STATE 7	Normal sensitivity optical, Class A2S heat, normal sensitivity gas (CO)	General area with moderate dust, smoke (tobacco) or steam, operates at a fixed temperature and is totally unaffected by transient temperature variation.	d
STATE 2	Normal sensitivity optical, Class A1 heat, low sensitivity gas (CO)	Applications with moderate dust, smoke (tobacco) or steam.	e
STATE 9	Normal sensitivity gas (CO), Class A1 heat	Area where high levels of smoke, dust or steam is often present.	f
STATE 11	Normal sensitivity optical, Class B heat,	Area with high ambient temperature, plus moderate levels of dust, smoke or steam present.	g
STATE 6	<i>Low sensitivity optical smoke</i> ♦, Class B heat, normal sensitivity gas (CO)	Area with high ambient temperature, plus normal levels of dust, smoke or steam present.	g
STATE 12	Class A1 heat only	Area where smoke, dust, steam or gas occurs at times.	h
STATE 15	No detection		i

- factory default STATE. NOTE: All the Sensor STATES in the above table are LPCB approved to EN54-5 & EN54-7 with the exception of STATE 6 which is only approved to EN54-5.

♦ Low sensitivity optical smoke sensing is not approved but may still be useful in certain applications. The STATE 15 is a no detection state and is not part of LPCB approval.

Commissioning instructions

S-Quad Heat sensor STATES

STATE	Definition / Class	Application in / Suitable for:	Sensitivity a-high- to-f-none
STATE 0#	Class A1 heat	Area having high levels of smoke, dust or steam.	a
STATE 13	Class A2 heat	Area where there is moderate temperature changes plus dust, smoke or steam present.	b
STATE 7	Class A2S heat	Area where fixed temperature operation is required, but is totally unaffected by transient temperature variations.	c
STATE 5	High temperature Class B heat	Area having high ambient temperature plus dust, smoke or steam present.	d
STATE 6	High temperature Class BS heat - with no rate of rise component	Area where there is rapid temperature changes plus dust, smoke or steam present.	e
STATE 15	No detection		f

- factory default STATE. NOTE: All the Sensor STATES in the above table are LPCB approved to EN54-5. The STATE 15 is a no detection state and is not part of LPCB approval.

S-Quad Optical sensor STATES

STATE	Definition / Class	Application in / Suitable for:	Sensitivity a-high- to-f-none
STATE 3	High sensitivity optical	Clean area or environment where early detection is required.	a
STATE 4	Normal sensitivity with no spike protection	General area as STATE 0 with less immunity to steam and dust to facilitate witness test.	b
STATE 0#	Normal sensitivity optical	General area.	c
STATE 8	Delayed normal sensitivity optical	General area with transient steam, dust or smoke present.	d
STATE 2	Low sensitivity optical smoke ♦	Area with moderate dust, smoke or steam present.	e
STATE 15	No detection		f

- factory default STATE. NOTE: All the sensor STATES in the above table are LPCB approved to EN54-7 with the exception of STATE 2.

♦ Low sensitivity optical smoke sensing is not approved but may still be useful in certain applications. The STATE 15 is a no detection state and is not part of LPCB approval.

Beam sensor STATES

States verses distance

Sensor STATES	Ideal range minimum to maximum
4 or 5	2m to 30m
2 or 3	5m to 100m
0 or 1	12m to 100m

State	Definition	Application
State 0#	Default detection	A fire is detected when there is a 50% (3dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1second and is maintained for 40 seconds, then a fault is registered. This allows the Control panel to differentiate between a fire and a fault signal caused by accidental obscuration.
State 1	Normal sensitivity	A fire is detected when there is a 50% (3dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1 second and is maintained for 40 seconds then a fire is registered.
State 2	Medium Sensitivity	A fire is detected when there is a 25% (1.3dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1 second and is maintained for 40 seconds then a fault is registered.
State 3	Medium Sensitivity	A fire is detected when there is a 25% (1.3dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1 second and is maintained for 40 seconds then a fire is registered.
State 4	High Sensitivity	A fire is detected when there is a 10% (0.5dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1 second and is maintained for 40 seconds then a fault is registered.
State 5	High Sensitivity	A fire is detected when there is a 10% (0.5dB) fall in signal level, however if the fall is by 90% (10dB) in less than 1 second and is maintained for 40 seconds then a fire is also registered.
State 15	No detection.	This is a total disablement of the sensor.

- factory default STATE

Interface Unit STATES

Single channel interface input STATES

These STATES apply to loop powered **single channel interface**.

STATE	Definition	Application
State 0	Default - all Inputs enabled	Normal use
State 1	Input 1 disabled	Selective disablement and enablement of interface input circuit. Where the input circuit is used for fire detection application, the call points on the circuit will NOT remain operational on disablement of the circuit.
State 15	All inputs disabled	Interface input disablement

4 - Channel Interface Unit STATES

The 4 - channel interface units have input STATES for configuration, which are applicable for both mains and loop powered interface units.

STATE	Definition	Applications
State 0	Default - all Inputs enabled	Normal use
State 1	Input 1 disabled	 Where an input circuit is used for fire detection application, the call points on the loop circuit will remain operational on disablement of the loop circuit.
State 2	Input 2 disabled	
State 3	Inputs 1 and 2 disabled	
State 4	Input 3 disabled	
State 5	Inputs 1 and 3 disabled	
State 6	Inputs 2 and 3 disabled	
State 7	Inputs 1, 2 and 3 disabled	
State 8	Input 4 disabled	
State 9	Inputs 1 and 4 disabled	
State 10	Inputs 2 and 4 disabled	
State 11	Input 1, 2 and 4 disabled	interface input disablement
State 12	Inputs 3 and 4 disabled	
State 13	Inputs 1, 3 and 4 disabled	
State 14	Inputs 2, 3 and 4 disabled	
State 15	All inputs disabled	

Installed equipment tests

Preparation

- ☐ Check to ensure *access* will be provided to areas where installed equipment is to be tested, such as to locked or secure areas.
- ☐ Ensure all sensor *dust covers* are removed from sensor devices.
- ☐ Tests may be made easier by having: sensor extractor tool, smoke poles, smoke canister, heat gun, beam obscuration filter and MCP test key, plus keys to open system equipment. Where applicable read the instructions supplied with the test kit.

Communication to site occupants

- ☐ Before undertaking any of these tests and to prevent unnecessary building evacuation, ensure:
 - all affected personnel on the site are informed, via a responsible person that the fire alarm system is being commissioned.
 - where there is a link to an alarm receiving centre, the appropriate action should be taken to ensure they are informed that tests are being carried out on the system.

Commissioning Tool

It is much easier for the panel and system to be configured from the Commissioning tool, see the Commissioning tool manual.

Plant equipment



Prior to any functional tests being carried out on the system, all the plant equipment interfaced to the fire alarm system should be isolated.

- ☐ Tests should be carried out following customer consent. It may also be necessary to obtain third-party consent dependent on what plant equipment is connected to the panels auxiliary contacts.
- ☐ All the tests on plant equipment should be as agreed during the project design stage.
- ☐

Zone 'Test' mode

The Zone Test mode is applicable for EN VIGILON Plus 4/6-loop panel and VIGILON Compact Plus panel only and is used when testing devices in a zone. It allows the engineer to test zones without having to return to the panel to silence alarms and reset the system.

While the panel is in the Zone test mode, following a fire event the alarms sound for 5 seconds and then automatically the alarms are silenced followed by a panel reset. The panel inhibits the operation of delay block functions, network functions, auxiliary and clean contact relay's operation in Zone test mode.

- ☐ All zones of the system that are not in zone test mode will operate normally.
- ☐ To put the control panel into or out of the 'zone test mode': Press **Menu On/Off** button and select **[Test/Eng]** -> **[UserCode]** and enter the Engineering password, this need only be done if an Engineering password is set up. Press **<etc>** and select **[Test]** and then select **[Zone]** and **[On]** / **[Off]**. Now enter the zone number and select **[Enter]**. The 'Test' LED is lit: **Test** 



On completion of all tests, switch Off the [Test] mode and notice the 'Test' LED also switches Off.

Fire Sensors



1. When testing heat sensors **DO NOT** use a heat gun for the test in a hazardous environment.
2. Manufacturer recommended test equipment and methods must be used to fire test flame detectors connected via interface unit.
4. When smoke testing fire sensors using artificial smoke, avoid excessive spray to prevent accumulation of sticky residue on sensor, see instructions on the smoke canister. Ensure the 'Test mode' is used when performing smoke test on an S-Quad sensor
5. The beam sensors should be tested using obscuration filters to simulate smoke at the appropriate sensitivity STATE.

The BS5839:Part 1 recommends that all sensors are tested for correct operation.

- ☐ Unless otherwise instructed all sensors should be tested. Each Sensor should also be checked for any physical obstacles that would inhibit the operation of the sensor in the event of a fire.
- ☐ Each Sound and Speech part of the S-Quad and S-Cubed should output the correct signal at the appropriate volume level in accordance with the Standard requirements and to meet the site specific needs as agreed with the customer.
- ☐ The VAD/Strobe part of S-Quad and S-Cubed should operate at the required flash rate in the event of a fire and is not obstructed.
- ☐ Each fire sensor should be tested for correct operation in the event of fire.
- ☐ Each speech/sounder should be tested for correct operation in the event of fire.

Manual Call Points

- ☐ Each *call point* should be tested for correct initiation of a fire event.

Interface Units



1. In some instances it may not be possible to functionally test equipment (plant equipment) connected to input / output circuits off an interface unit.
2. Ensure the contact rating of interface output circuits are adequate for the plant equipment load requirement.

Plant equipment

- ☐ Tests should be carried out on the operation of plant equipment, like lift, escalator, door release, fire dampers etc following customer consent. It may also be necessary to obtain third party consent.
- ☐ All tests on plant equipment should have been agreed during the project design stage.

Tests

- ☐ Where tests are performed on a interface unit check each circuit failure is indicated at the control panel, such as:
 - Battery disconnection*
 - Mains supply disconnection*
 - Input line open circuit and then short circuit
 - Input line detector and MCP fire#
 - Output line open circuit and then short circuit ~
 - Output line operation ~
- * - not applicable for the **loop powered interface units**.
 # - a fire from a **call point or detector** connected to an interface input cannot be differentiated.
 ~ - the output of a **single channel interface** operates with **sector** and the output is a set of **voltage free contacts**, the contacts are not fault monitored.

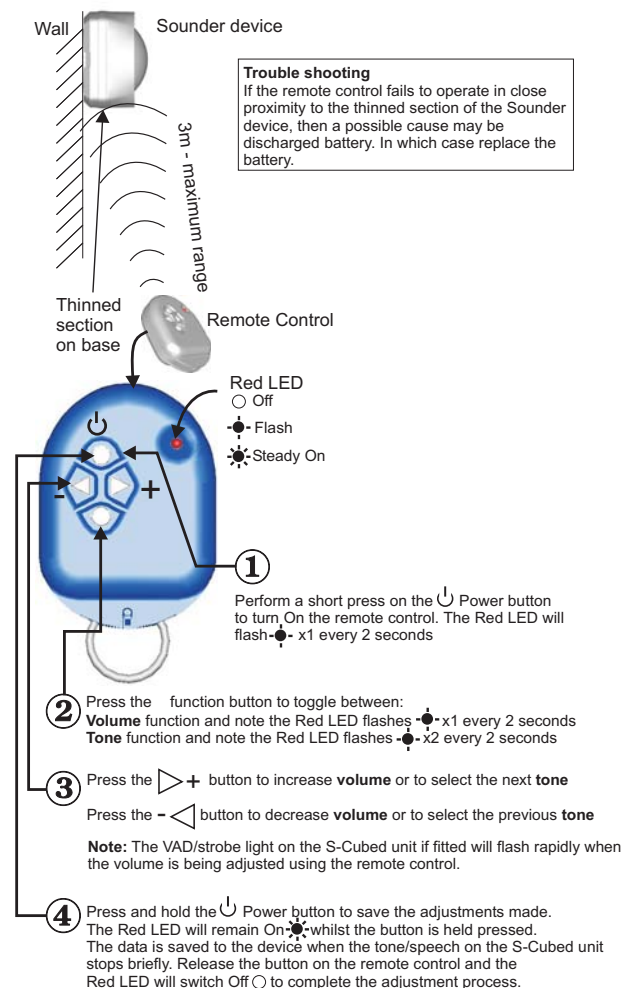
Keyswitches

- ☐ Where the interface unit input is from a keyswitch, then the keyswitch should be configured for correct operation and tested as per project specification.
- ☐ Check on operating a keyswitch the confirmation LED, where applicable, is lit to show operation.

S Cubed



The remote control is only operable when the panel is in the Zone Test mode for an EN system or Commission mode for a BS system.



- ☐ Depending on the type of S-Cubed device being commissioned check to ensure correct output sound tone, speech message and VAD/strobe rate are given. The S-Cubed devices should have been setup using the commissioning tool.
- ☐ The volume of the sounder S-Cubed device can be adjusted by using a remote control.
- ☐ Each S-Cubed should be tested for correct operation in the event of fire.
- ☐ The sound levels in the areas should be tested in accordance with the British Standard requirements and to meet the site specific needs, as agreed with the customer.

Repeat panel (Loop connected)

Each loop *repeat panel* should be tested for the following:

- ☐ System event messages and indications are given
- ☐ Local fault events, such as Battery and Mains supply disconnection
- ☐ Alarm controls & Cancel Buzzer operation is possible

Mimic Panel

Each Mimic indicator panel should be configured and tested for the following:

- ☐ To confirm fires are indicated
- ☐ Local fault events, such as Battery and Mains supply disconnection
- ☐ Lamp Test
 - Can be performed on an **A3 Mimic panel** by pressing the button on the centre face of the unit

Deviations from standards

- ☐ The results of system tests carried out must be in accordance with the *relevant standards and project specification*.
- ☐ Any deviations must be documented and reported for approval.

Vigilon Compact Plus Network

A VIGILON Compact Plus network can consist of a number of control panels and network nodes wired together in a secure loop. A secure loop is achieved by installation of a network card in each VIGILON Compact Plus panel and VIGILON Compact Plus Network Node to facilitate the network interconnection of panels and nodes.

Each standalone system is first commissioned before being networked together with other panels and nodes.

The network commissioning process involves systematic introduction of each standalone system to the network with tests to ensure the network is working.

The whole networked system is then configured and tested to ensure alarms and global controls operate to site requirements.



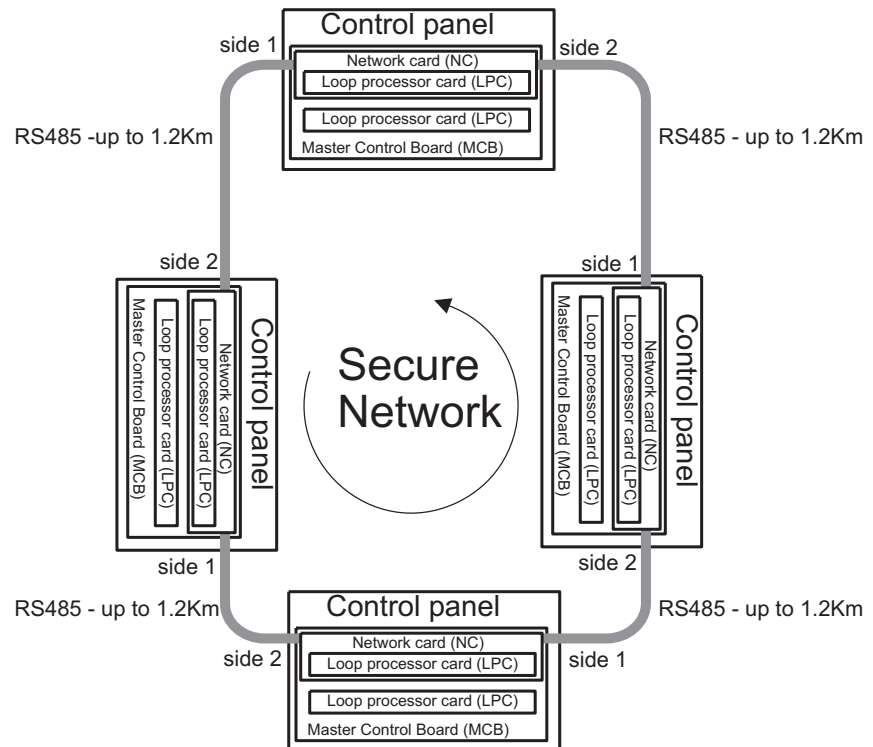
When setting the address switches on the 'network card', ensure the network controller which can be the network node or panel is given the lowest address.

A network controller is a node that can be connected to a Central monitoring system.

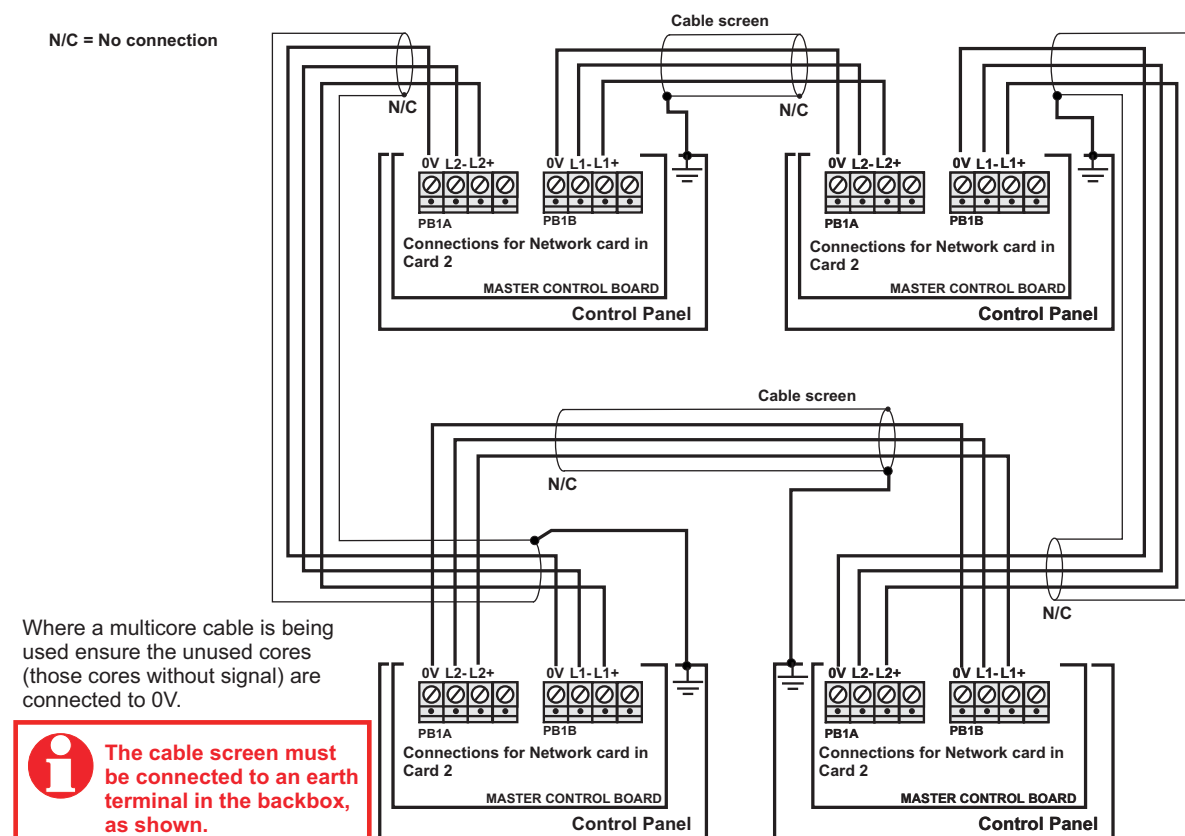
All panels in a networked system must operate at the same baud rate, normally this is 38.4K where copper network card is being used.

Single Network

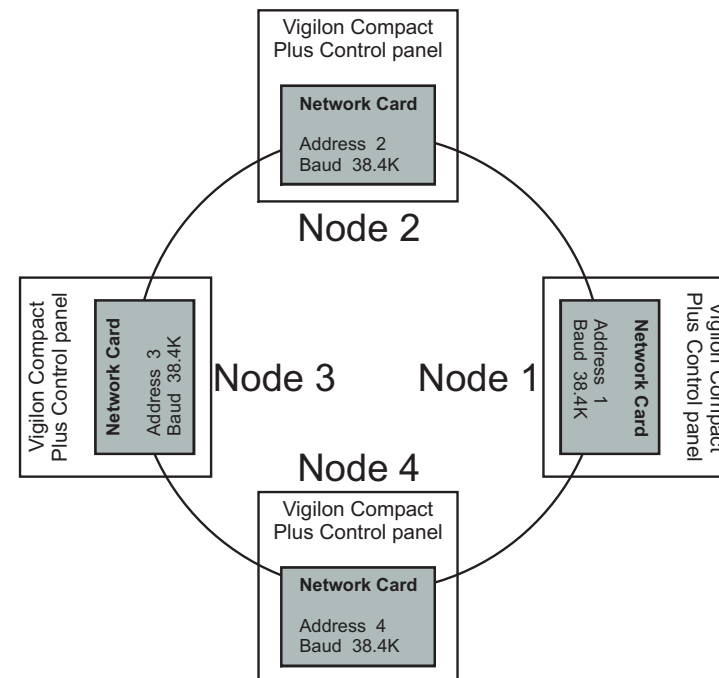
- ☐ Each standalone system must first be fully commissioned individually.
- ☐ Check that the correct cable is used to wire the network, the installer's manual lists all the approved cables.



Wiring the single network



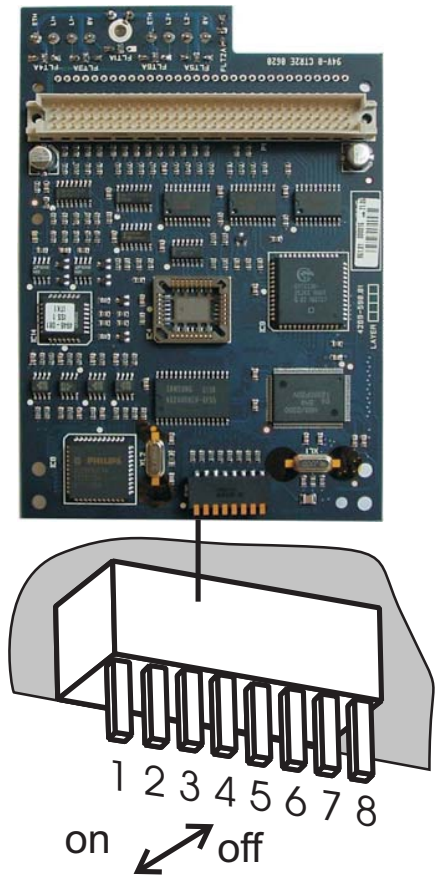
Single network without domain bridge



Network Card for VIGILON Compact Plus - baud and node address switch settings

 The copper network card for use in a Vigilon Compact Plus panel is factory set to 38.4K baud with node address 4. Both node address and Baud rate can be changed to another setting by configuring the switches on the Network card.

Component side



Node address							Baud rate		
	8	7	6	5	4	3	2	1	
64	off	off	off	off	off	off	off	off	2400
1	on	off	off	off	off	off	on	off	9600
2	off	on	off	off	off	off	off	on	19.2K
3	on	on	off	off	off	off	on	on	38.4K
4	off	off	on	off	off	off			
63	on	on	on	on	on	on			
	Address						Baud		
	- factory settings								

Commissioning instructions

Powering-up the Network

- ☐ Add one panel at a time starting from the *network controller* (Node 1) at **side 1**.



It can take a few minutes for the system map to update.

- ☐ For each panel powered-up the *network controller* will automatically try to establish communication with the connected equipment.
- ☐ The *network controller* will display the status of the network as being started, secure or non-secure and from now on will monitor and control the network communications.
- ☐ Use *disable comms* to isolate a panel from the rest of the network, ie it operates at the network card in an echo mode, select **Menu On/Off** -> **[Control]** -> **[Disable]** -> **[Comms]** and enter the Network Card always card 10 on a Compact panel.

How to check a Network map

- ☐ Using the **Menu On/Off** -> **[Info]** -> **[Map]** -> **[NetMap]** option to enter the network card address 10.

The display will provide a list in cabling order the addresses of panels in the networked system.

These are examples of different networks. The lowest address '1' has been given to the network controller.

```
Map information for Network at card 10
Nodes on side 1
1  9  7  6  5
Nodes on side 2
1  3
```

This example shows the network is **not** secure

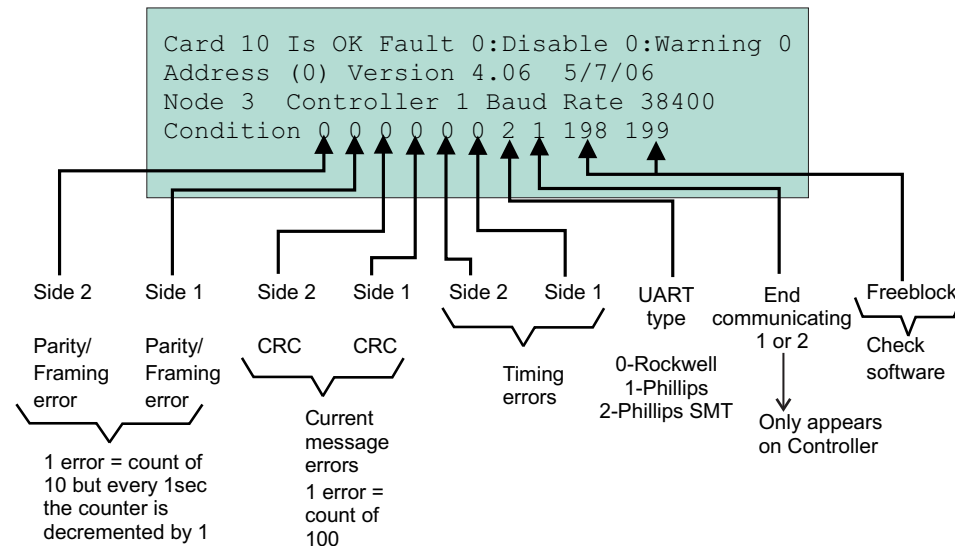
```
Map information for Network at card 10
Nodes on side 1
1  9  7  6  5  3  1
Nodes on side 2
```

This example shows the network is secure

How to check Network Card status

Using the **Menu On/Off** -> **[Info]** -> **[Status]** -> **[Card]** and enter the network card address in this case it is 10'. The display will confirm:

- ☐ the address of the network card
- ☐ the address of network controller
- ☐ and a set of condition codes



Errors	meaning
Parity	Parity is incorrect, data corrupted.
Framing	8- bit transmitted incorrectly and the data line does not return to logic 1 at the end of the transmission.
CRC	Cyclic redundancy code. When a message consisting of a number of 8 bit transmissions is sent, a calculation is carried out to check that data corruption has not occurred.
Time out error	Where an 8 bit transmission block is not fully received, the network card will time out ready to receive the next message. A time out error will also be generated for each parity/ framing error.

Errors	meaning
Freeblock	Free memory blocks available to temporarily store messages waiting to be processed. The maximum number = 199, although it may drop as low as 175 - 180 on a highly populated network. A lower value may indicate a high error value and will require rectification.

Fault Finding

- ☐ The communication path in a secure network will alternate between end 1 (side 1) and end 2 (side 2) every minute.
- ☐ The network will be non secure until the last panel is connected and powered-up.

High errors

- ☐ Check cable lengths and type used.
- ☐ Check wiring, connections and earthing arrangements. Check the network cable is correctly earthed to the equipment.
- ☐ Replace network card at point of failure, where required.
- ☐ Check the addresses and baud rates of the network cards of all the networked panels.
- ☐ It may be necessary to investigate and:
 - look at the noise voltage between conductors, screen and earth.



Always reset the Network Card first when rechecking for errors, using [Test/Eng] -> [Card] ->[Reset] -> Network Card 4.

On completion of the above the networked system may now be configured for network operation, for master sectors and global controls to be actioned as per site requirements.

Single Vigilon Plus Network

A VIGILON Plus network consists of a number of control panels of standalone systems and network nodes wired together in a secure loop using copper or fibre network cards.

A secure loop is achieved by installation of a copper or fibre network card in each VIGILON Plus panel and VIGILON Plus Network Node to facilitate the interconnection of panels and nodes. Each standalone system is first commissioned before being networked together with other panels and nodes.

The commissioning involves systematic introduction of each standalone system to the network and tests to ensure the network is working.

The whole networked system is then configured and tested to ensure alarms and global controls operate to site requirements.



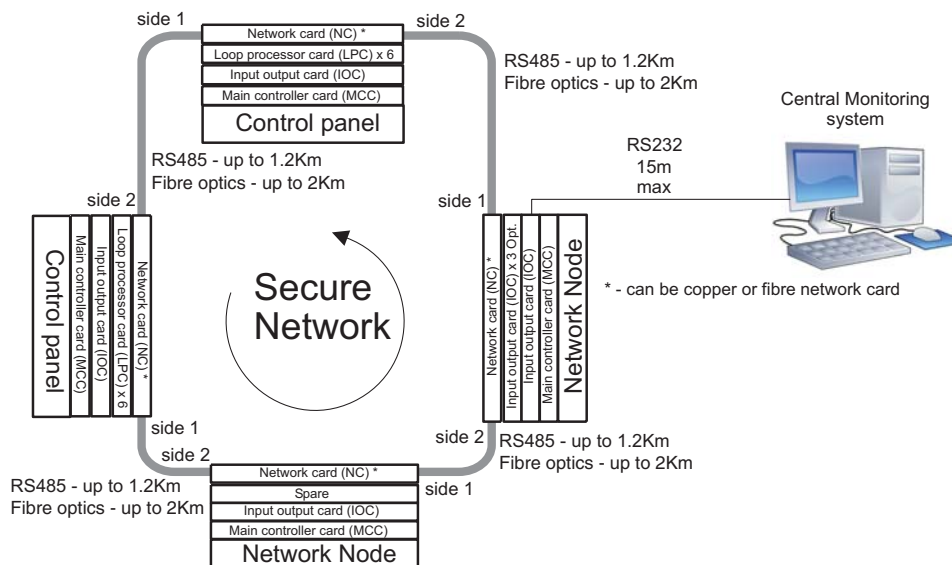
When setting the address switches on the 'network card', ensure the network controller is given the lowest address.

A network controller is a node/panel that can be connected to a Central monitoring system.

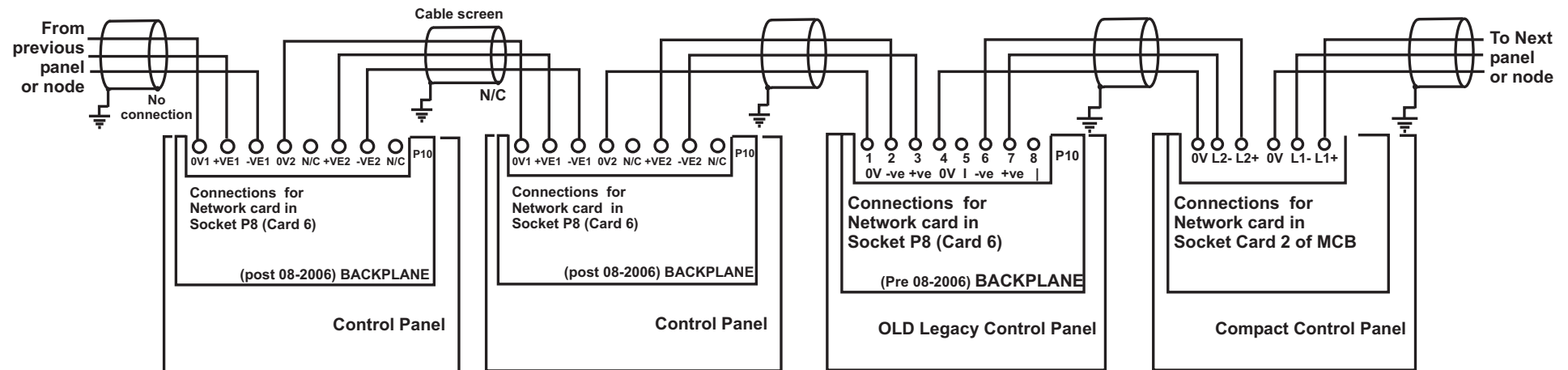
All panels in a networked system must operate at the same baud rate, normally this value is 38.4K where copper network cards are being used and 115.2K where Fibre network cards are being used.

Single Network

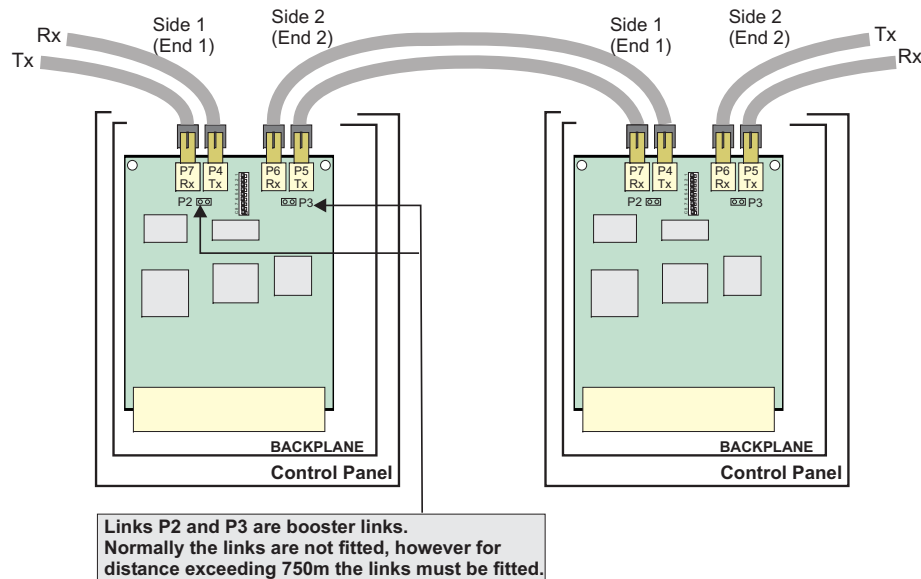
- ☐ Each standalone system must first be fully commissioned individually.
- ☐ Check that the correct cable is used to wire the network.



Wiring a Copper network

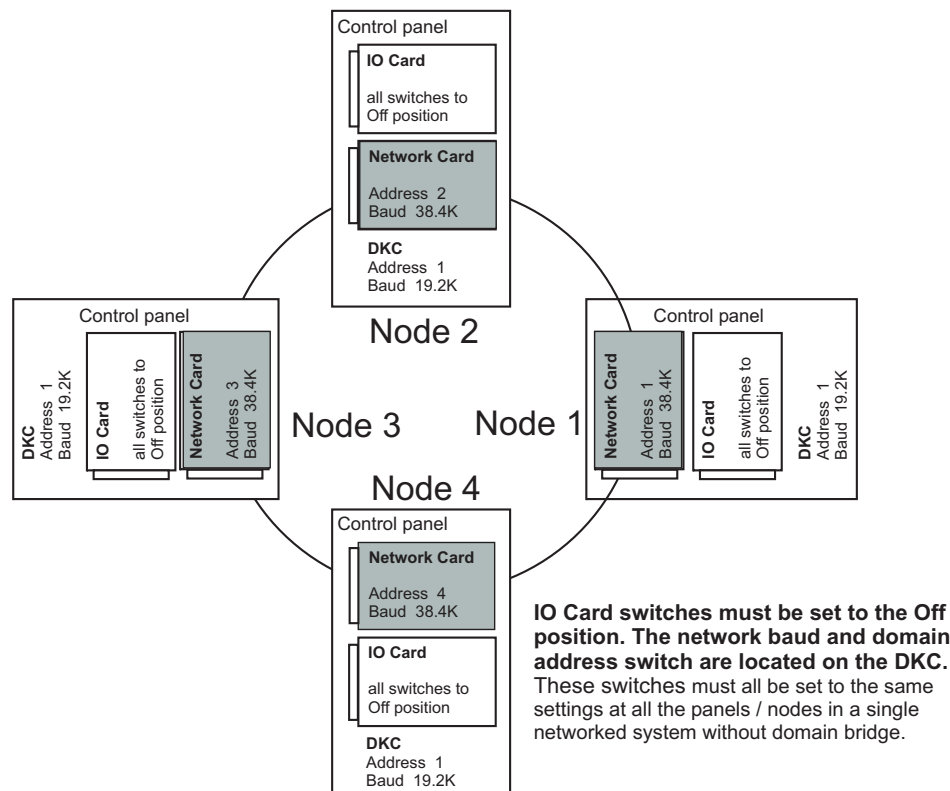


Wiring a Fibre network



Commissioning instructions

Single network without domain bridge



Powering-up the Network

- ☐ Add one panel at a time starting from the *network controller*, adding one panel/node at a time to **side 1**.



It can take a few minutes for the system map to update.

- ☐ For each panel powered-up the *network controller* will automatically try to establish communication with the connected equipment.
- ☐ The *network controller* will display the status of the network as being started, secure or non-secure and from now on will monitor and control the network communications.
- ☐ Use *disable comms* to isolate a panel from the rest of the network, ie it operates at the network card in an echo mode, select **Menu On/Off -> [Control] -> [Disable] -> [Comms]** and enter the Network Card address, usually card 6 but can also be card 15.

How to check a Network map

- ☐ Using the **Menu On/Off -> [Info] -> [Map] -> [NetMap]** menu, enter the network card address, usually card 6 (may be card 15) and select **[Enter]**.

The display will provide a list in cabling order the addresses of panels/nodes in the networked system.

Here are examples of different networks. The lowest address '1' is normally the network controller.

Nodes on side 1 of a SECURE NETWORK

1 10 4 24 3 6 2 22 7 23 1

Nodes on side 1 of a NON-SECURE NETWORK

1 10 4 24 3 6

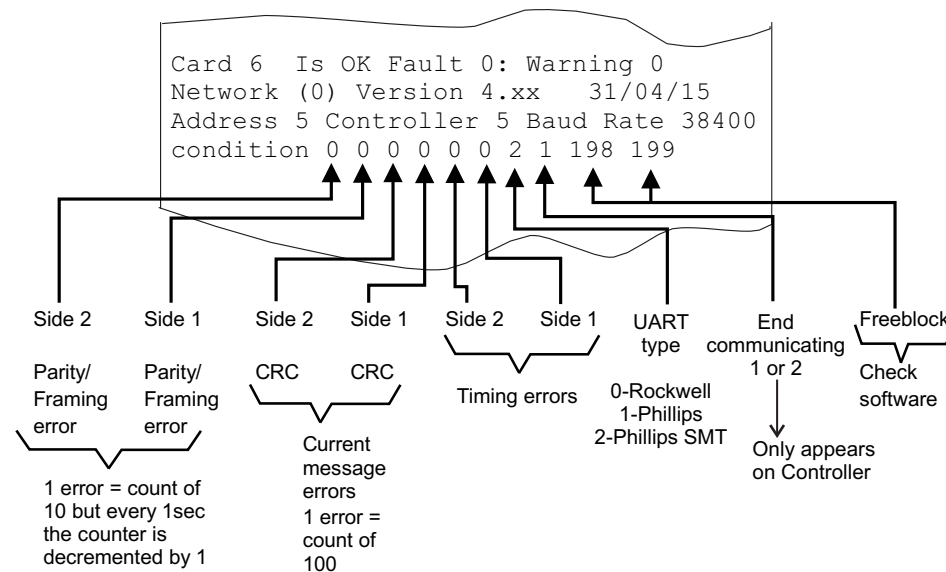
Nodes on side 2 of NON-SECURE NETWORK

1 23 7 22 2

How to check Network Card status

Using the **Menu On/Off** -> **[Info]** -> **[Status]** -> **[Card]** and enter the network card number, usually this is card number 6 and select **[Enter]**. The display will confirm the:

- ☐ address of the network card
- ☐ address of network controller
- ☐ and a set of condition codes



Errors	meaning
Parity	Parity is incorrect, data corrupted.
Framing	8- bit transmitted incorrectly and the data line does not return to logic 1 at the end of the transmission.
CRC	Cyclic Redundancy Code. When a message consisting of a number of 8 bit transmissions is sent, a calculation is carried out to check that data corruption has not occurred.

Errors	meaning
Time out error	Where an 8 bit transmission block is not fully received, the network card will time out ready to receive the next message. Time out error will also be generated for each parity/ framing error.
Freeblock	Free memory blocks available to temporarily store messages waiting to be processed. The maximum number = 199, although it may drop as low as 175 - 180 on a highly populated network. A lower value may indicate a high error value and will require rectification.

Fault Finding

- ☐ The communication path in a secure network will alternate between end 1 (side 1) and end 2 (side 2) every minute.
- ☐ The network will be non secure until the last panel is connected and powered-up.

High errors

- ☐ Check cable lengths and type used.
- ☐ Check wiring, connections and earthing arrangements. Check the network cable is correctly earthed to the equipment.
- ☐ Replace network card at the point of failure.
- ☐ Check the addresses and baud rates of the input / output and network cards of the networked panels.
- ☐ It may be necessary to escalate an investigation to look at the noise voltage between conductors, screen and earth.



Always reset the Network Card first when rechecking for errors, using [Test/Eng] -> [Card] -> [Reset] -> Network Card number, normally Card 6 and select [E].

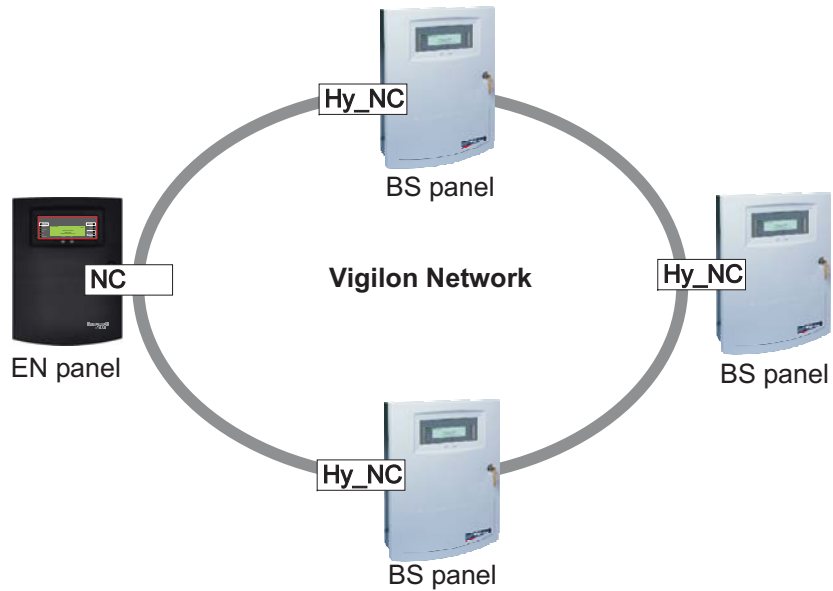
The networked system may now be configured for network commands to be actioned, such as master sectors and global controls as per site requirements.

Hybrid Network

The Vigilon network can make use of Hybrid EN/BS Network card (VIG-NC-HYBRID) installed in BS networked panels where controlled introduction of EN panels is required. The use of this card permits EN panels to co-exist with BS panels while keeping the network functionality.

Hy_NC BS/EN Hybrid Network Card (VIG-NC-HYBRID)

NC Network Card (VIG-NC)

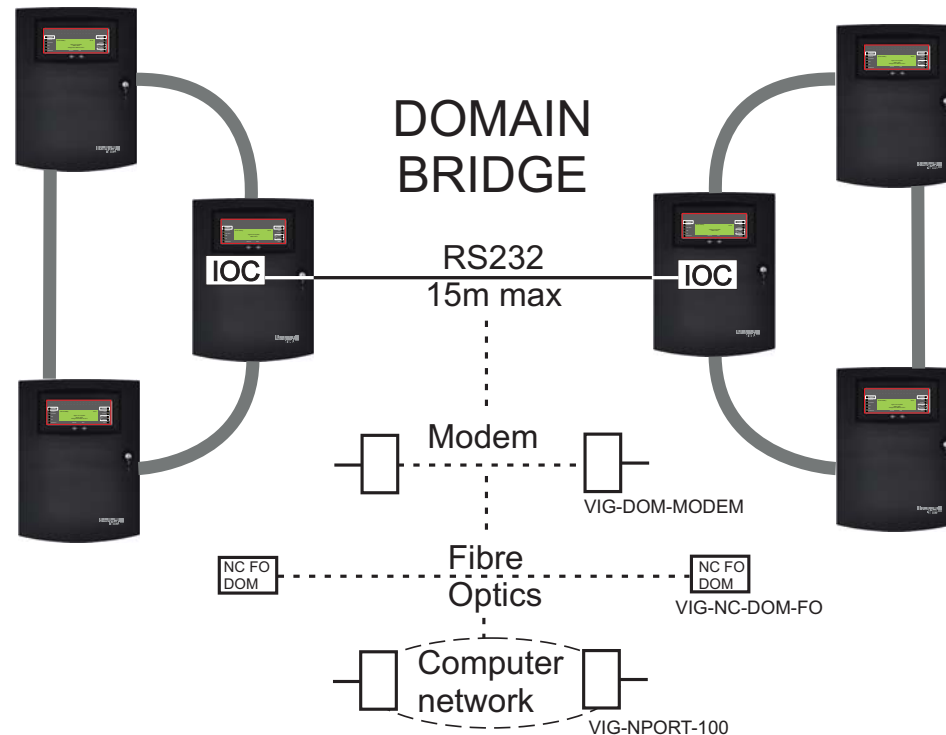


Multiple Vigilon Plus Networks

Domain Bridge using Input Output card

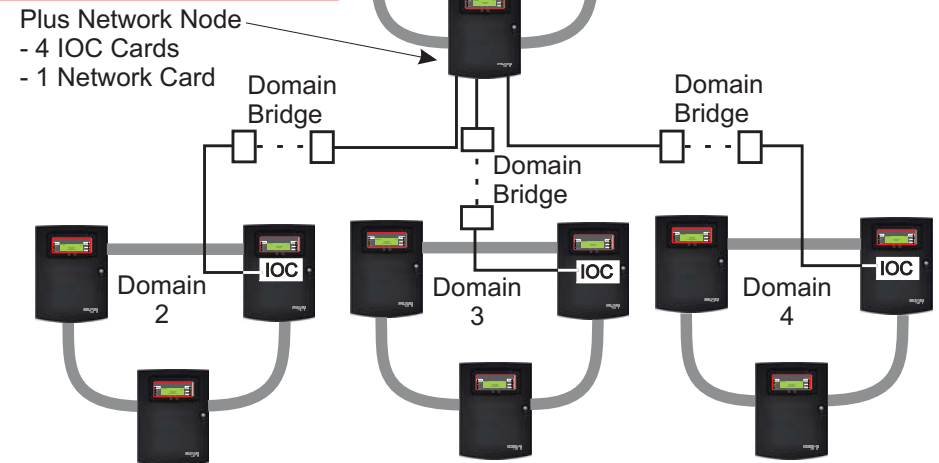
Two or more VIGILON Plus networks can be connected together at domain bridge IO cards by having a direct RS232 connection, Modems, Fibre Optics units or NPORT units. This allows the display of events at any *control panel/Network node* in the connected networks.

Two networks using Domain bridge IO card

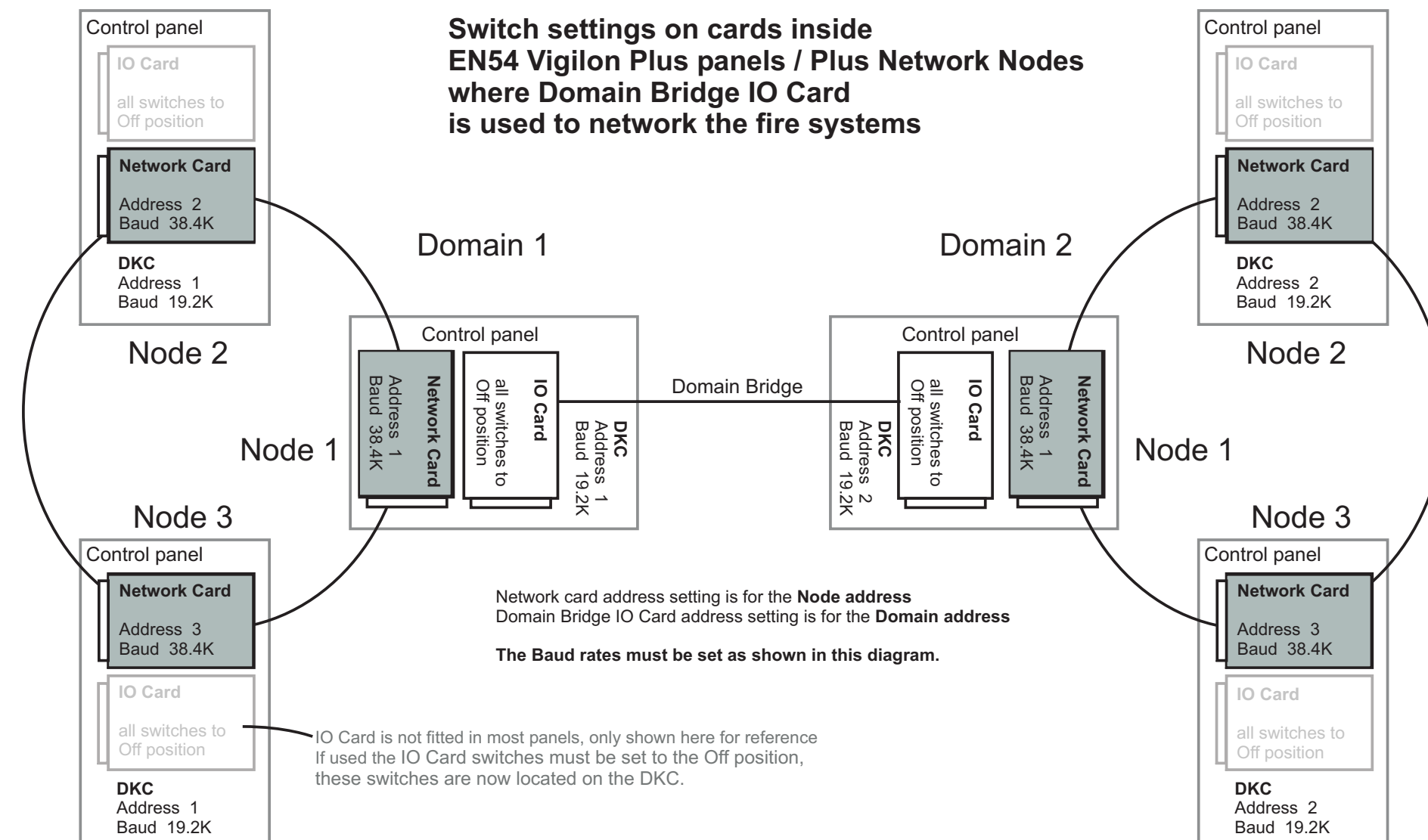


Star network using Domain bridge IO cards

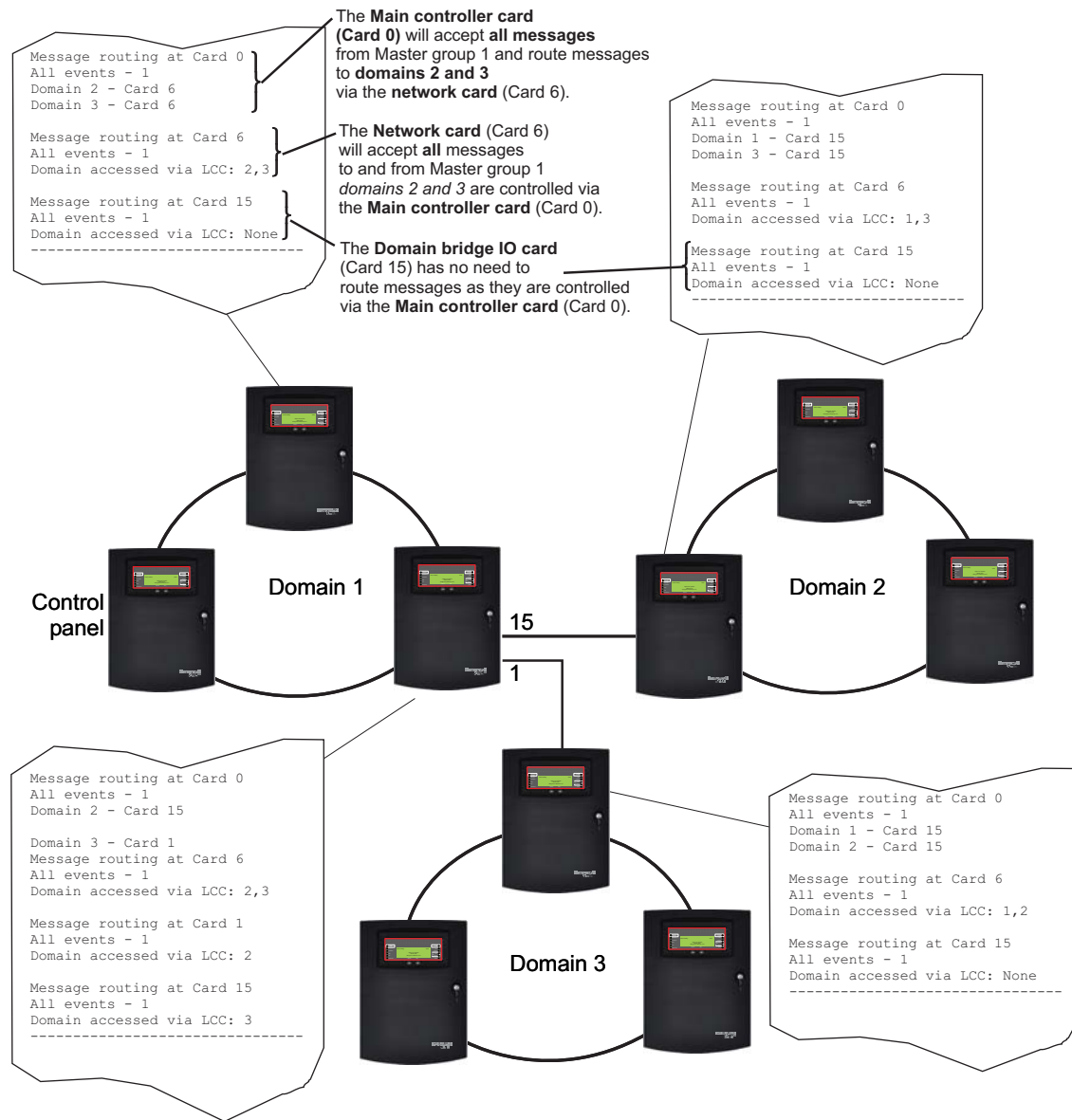
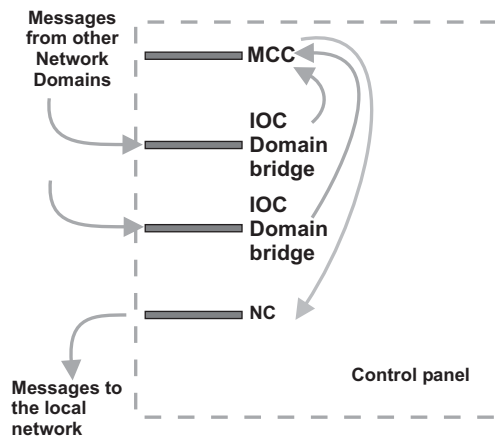
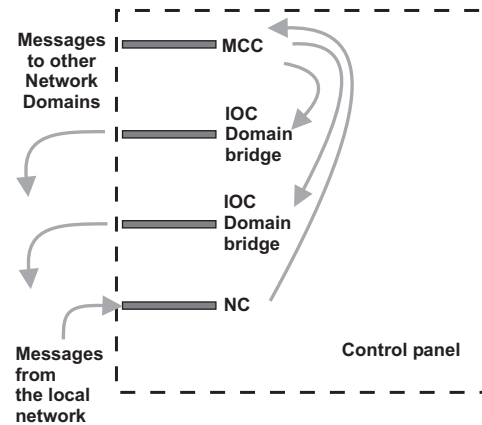
If remote sites are residential it is not permissible to rely on domain bridge link to call fire brigade



IO domain bridge network switch settings



Message routing



Commissioning instructions

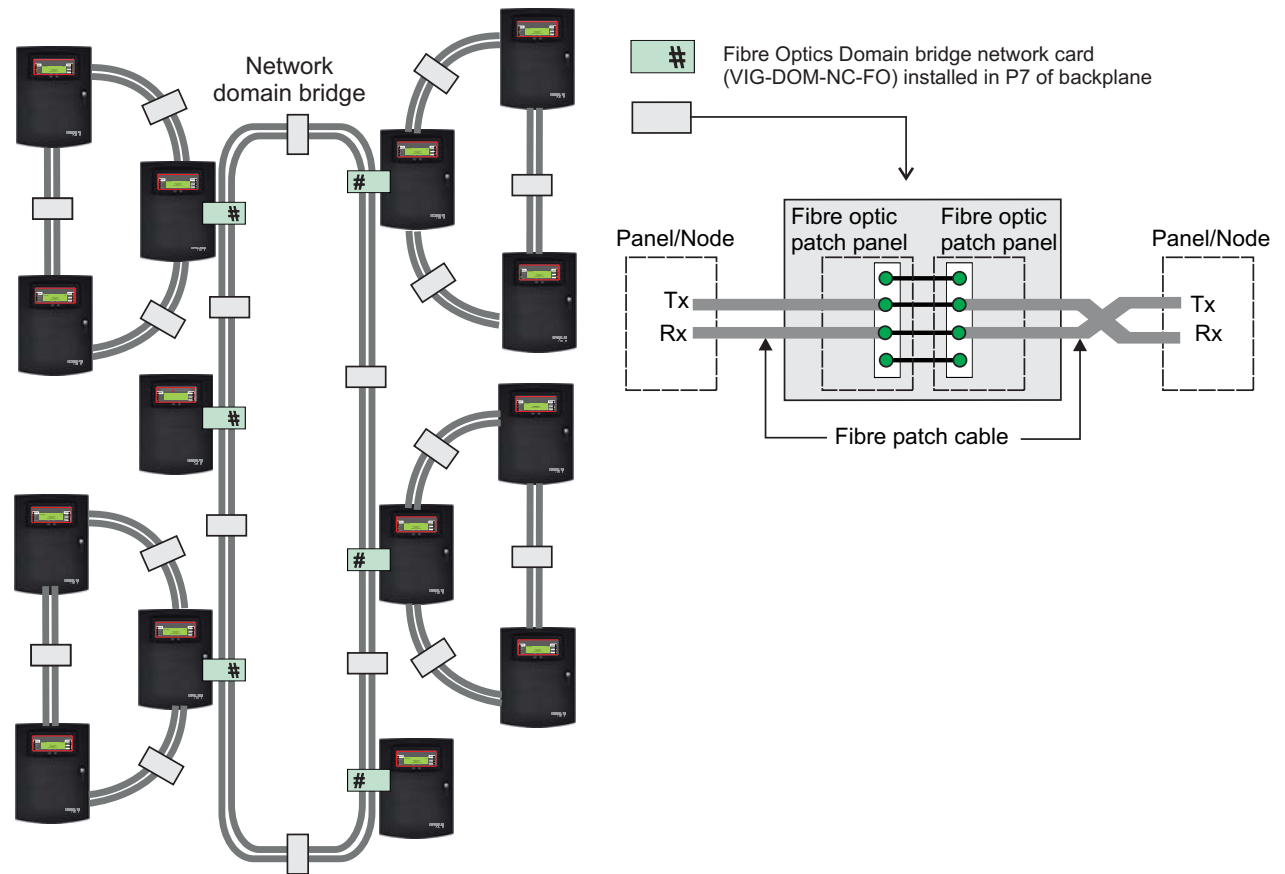
Domain bridge message passing tests

To check the domain bridge connections and to ensure messages can be passed between networks the following must be done. Trigger an event in a network, such as a fault.

- ☐ Check the event is passed on to the connecting networks and is displayed at the networked control panels
- ☐ Repeat the test at each network to establish message passing between networks
- ☐ Finally back up the cards of the panel including the *Network card* and *IO card* to the *Memory*. This ensures the panels of each network knows the location of network controller, ie the domain bridge connection point.

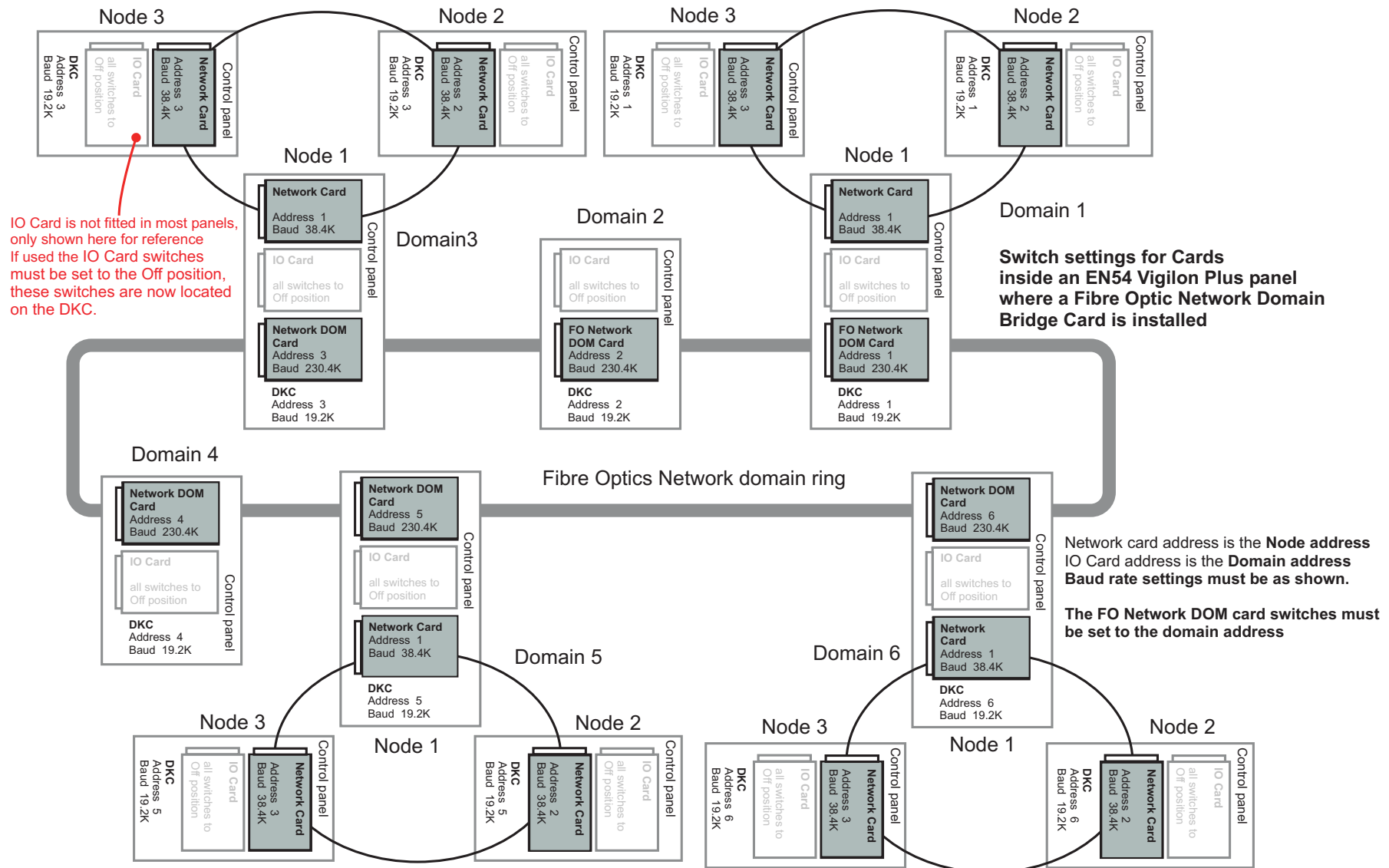
Domain bridge using Fibre Optic network card

Up to 64 small VIGILON Plus networks can be connected together in a secure loop by using domain bridge fibre optics network card (VIG-NC-DOM-FO), with the card installed in socket P7* of the backplane. This allows the display of events at any *control panel* in the connected networks. (* can be card 15)



The above diagram shows small fibre networks connected together by a fibre network domain. It is equally possible to mix small copper networks and small fibre networks together in the manner shown above to a network domain.

FO Domain Network switch settings

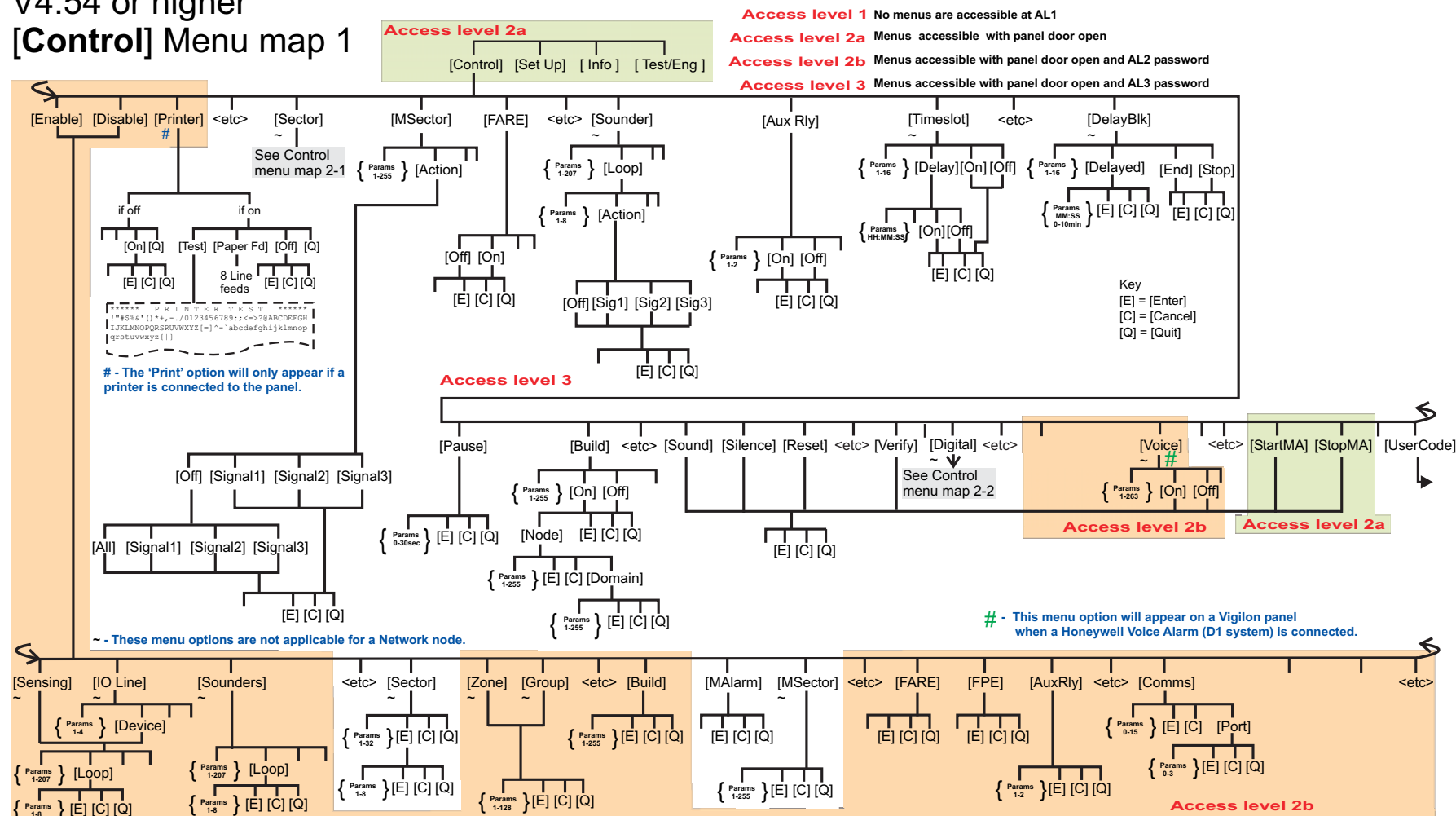


Appendix A - Menu maps for EN54 (V4) Vigilon Plus panels

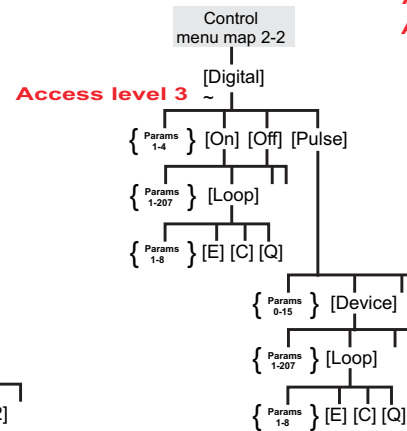
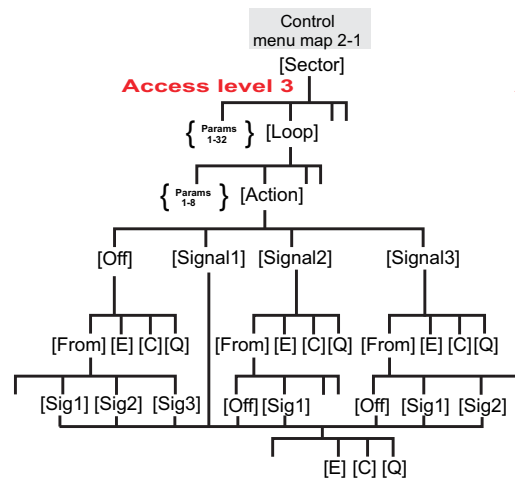
The menu options [Control], [SetUp], [Info] and [Test/Eng] are accessible on pressing the MENU ON/OFF button.

EN Vigilon Plus Control panel with MCC/MCB at V4.54 or higher

[Control] Menu map 1



EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher [Control] Menu map 2-1 to 2-2



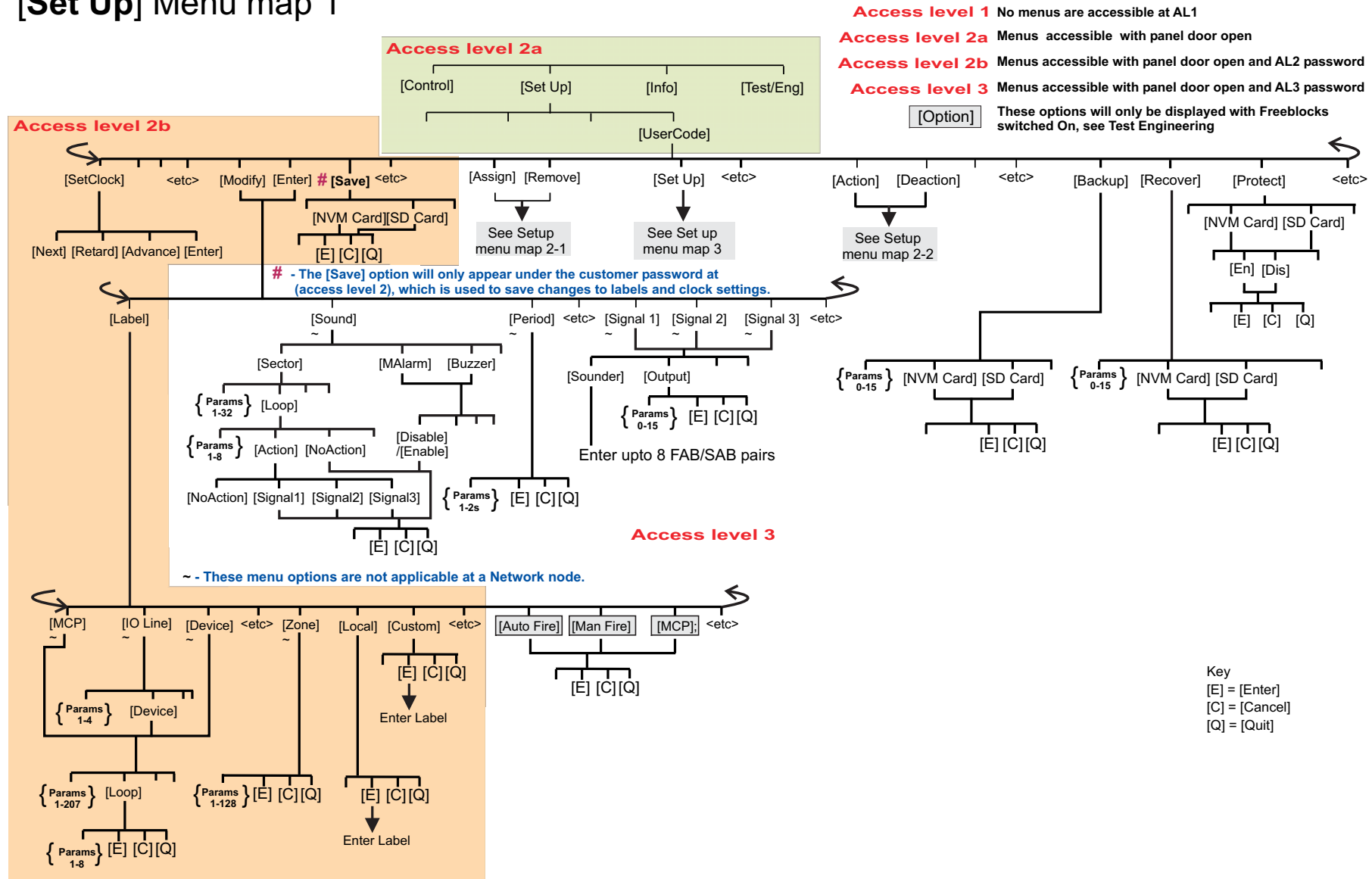
- Access level 1** No menus are accessible at AL1
- Access level 2a** Menus accessible with panel door open
- Access level 2b** Menus accessible with panel door open and AL2 password
- Access level 3** Menus accessible with panel door open and AL3 password
- [Option]** This option will only be displayed with Freeblocks switched On, see Test Engineering

Key
[E] = [Enter]
[C] = [Cancel]
[Q] = [Quit]

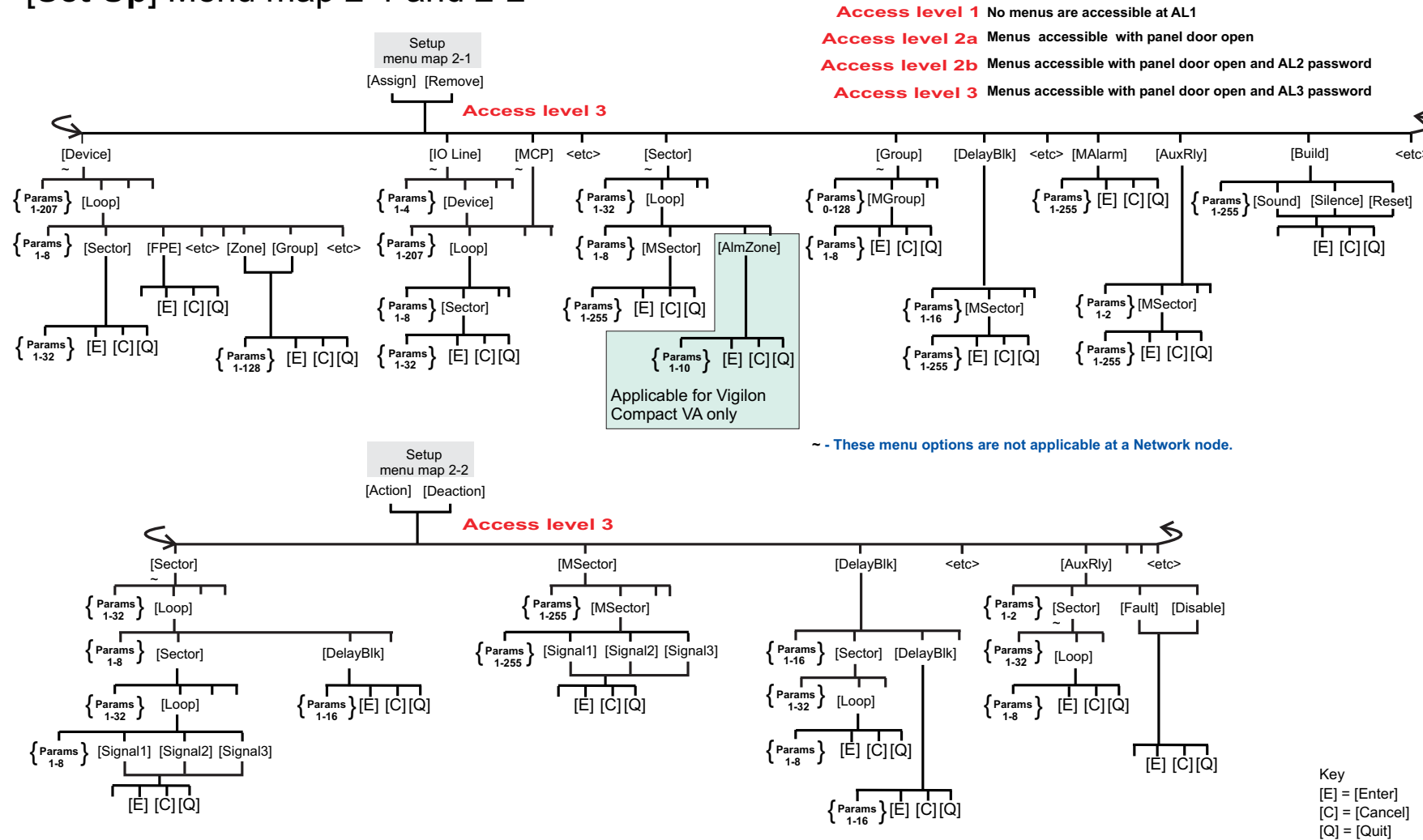
EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher

[Set Up] Menu map 1

Access level 1

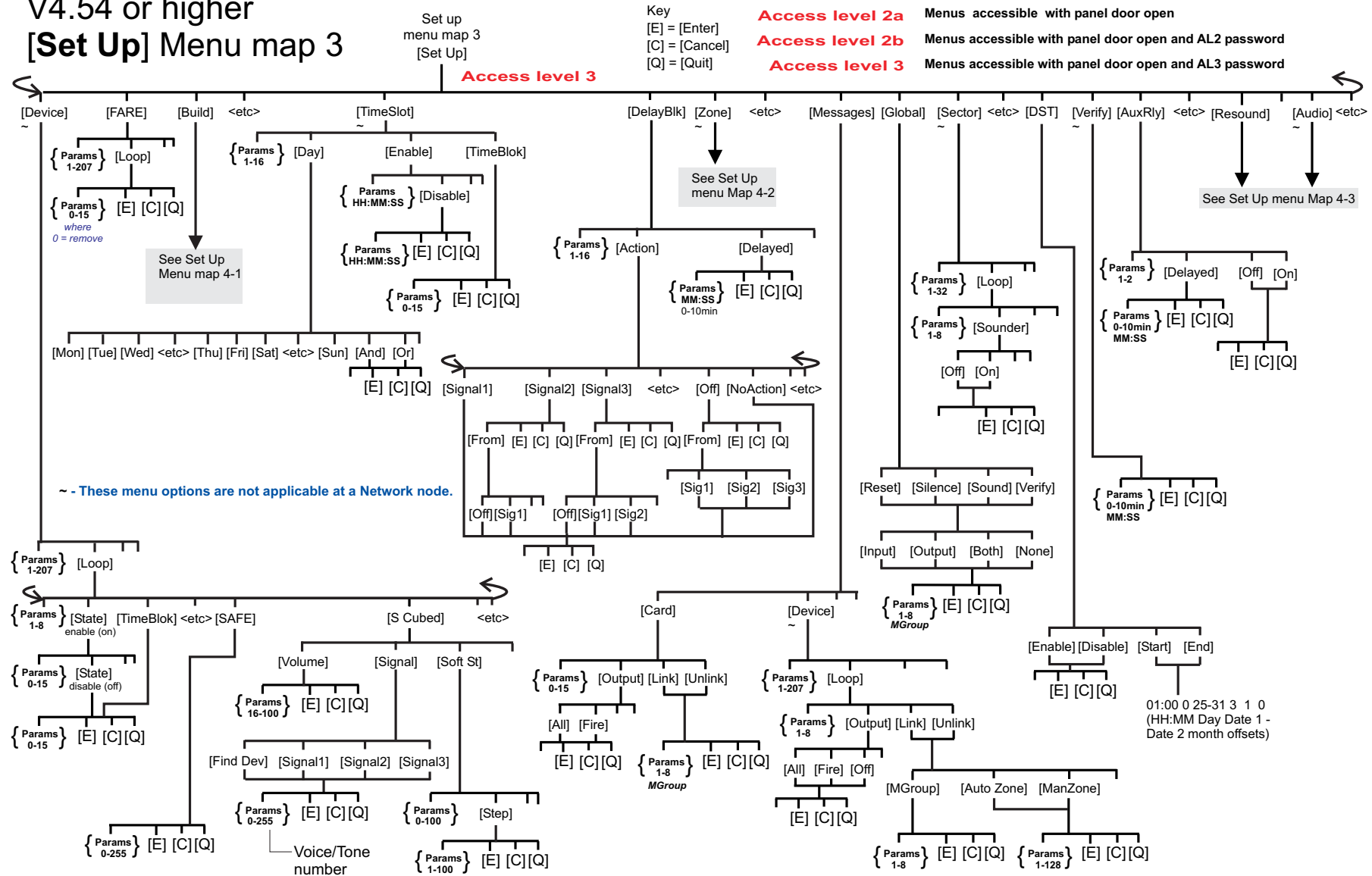


EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher [Set Up] Menu map 2-1 and 2-2



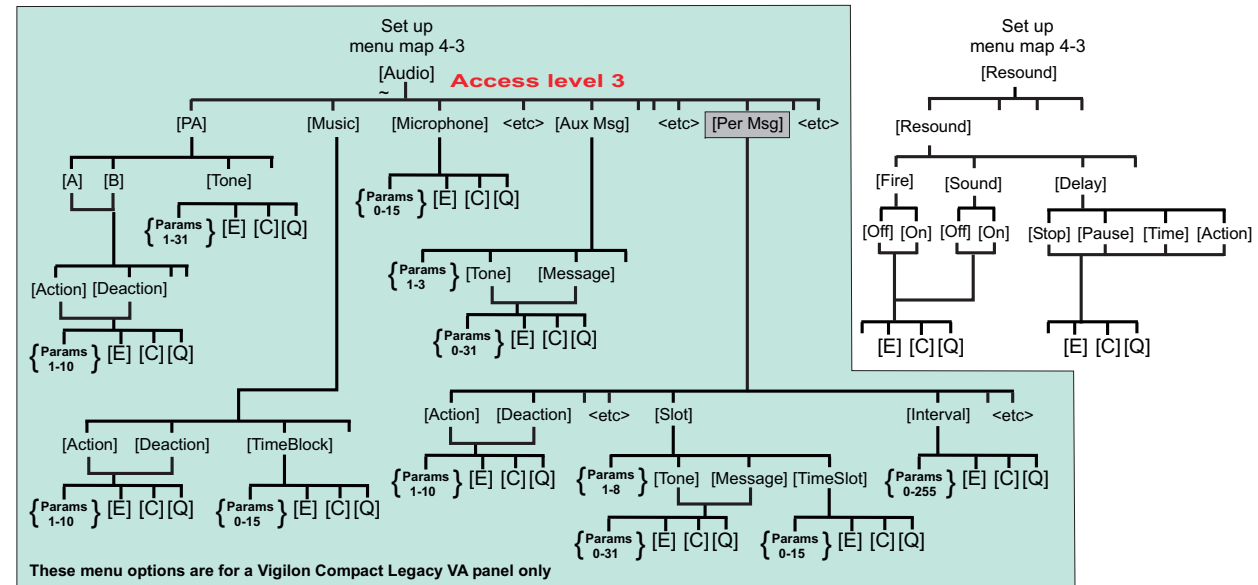
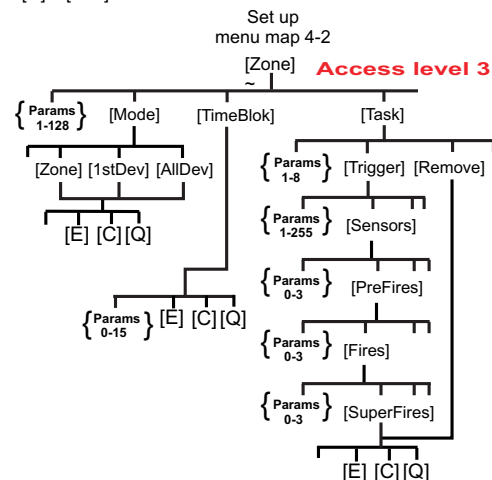
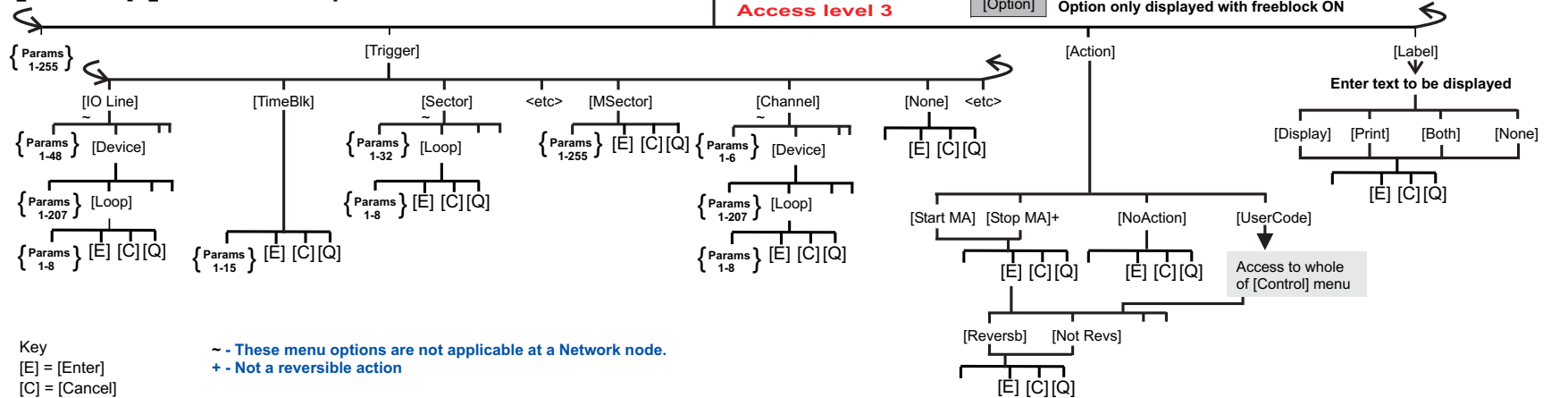
EN Vigilon Plus Control panel with MCC/MCB at V4.54 or higher

[Set Up] Menu map 3

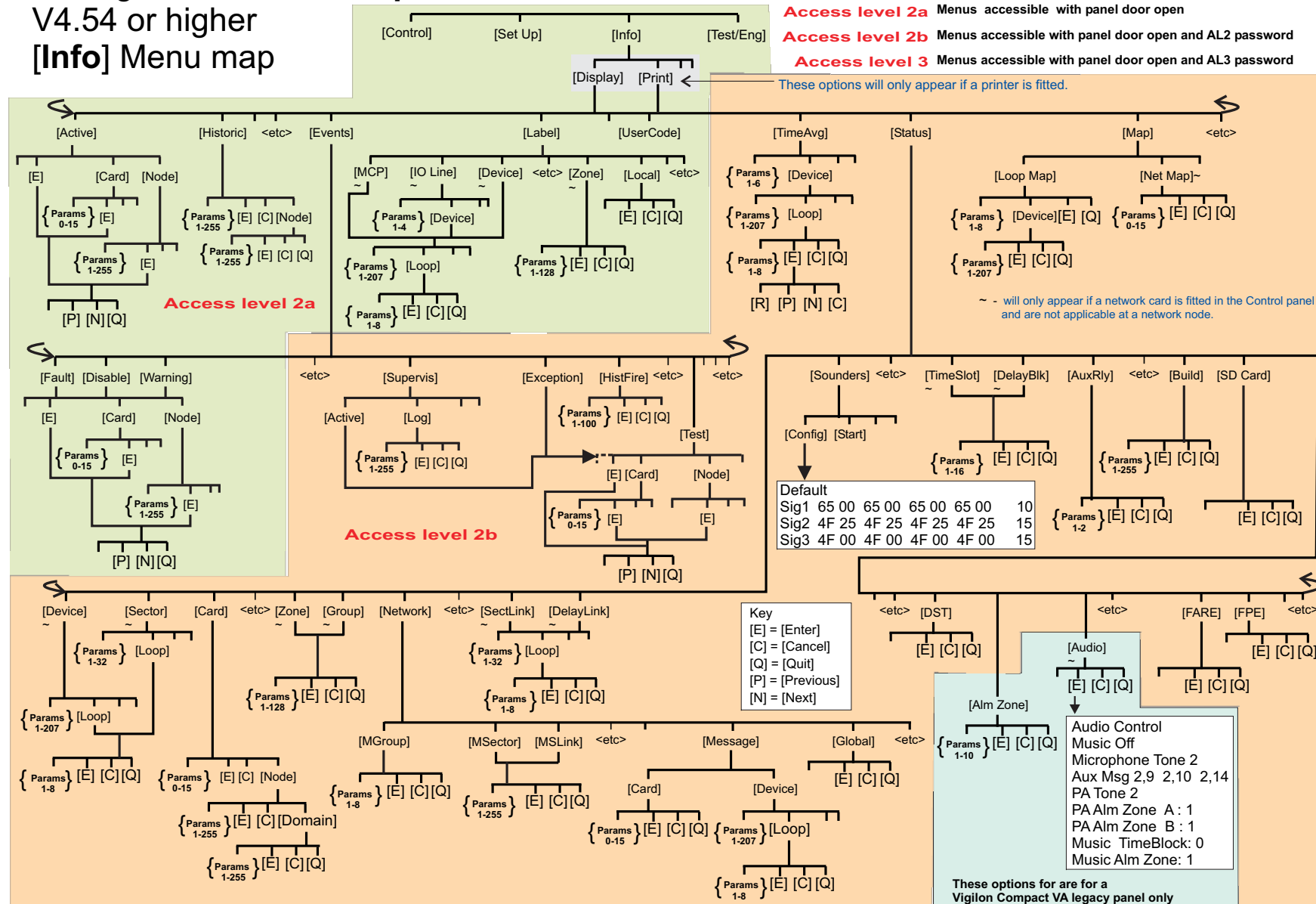


EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher

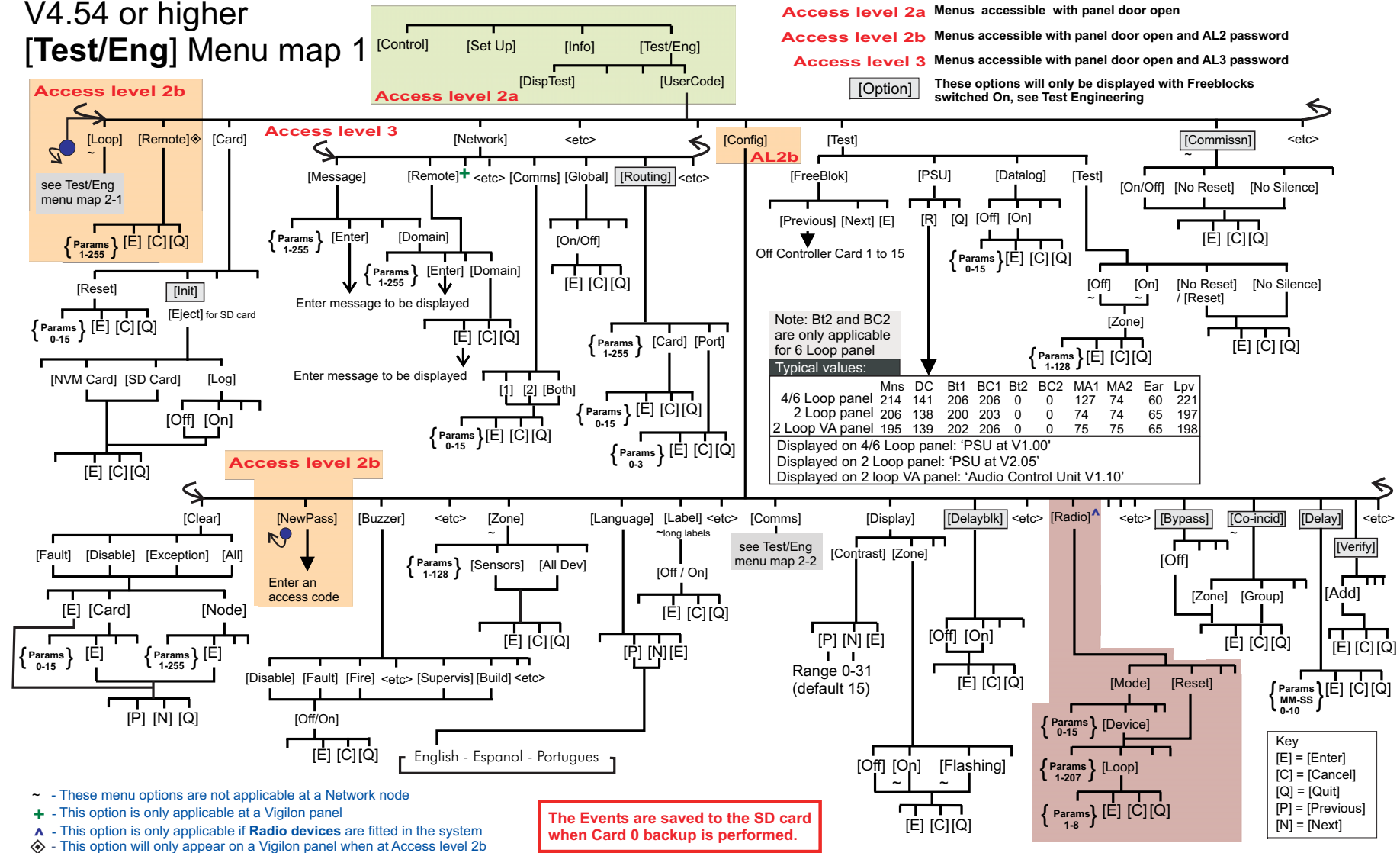
[Set Up] Menu map 4-1, 4-2 and 4-3



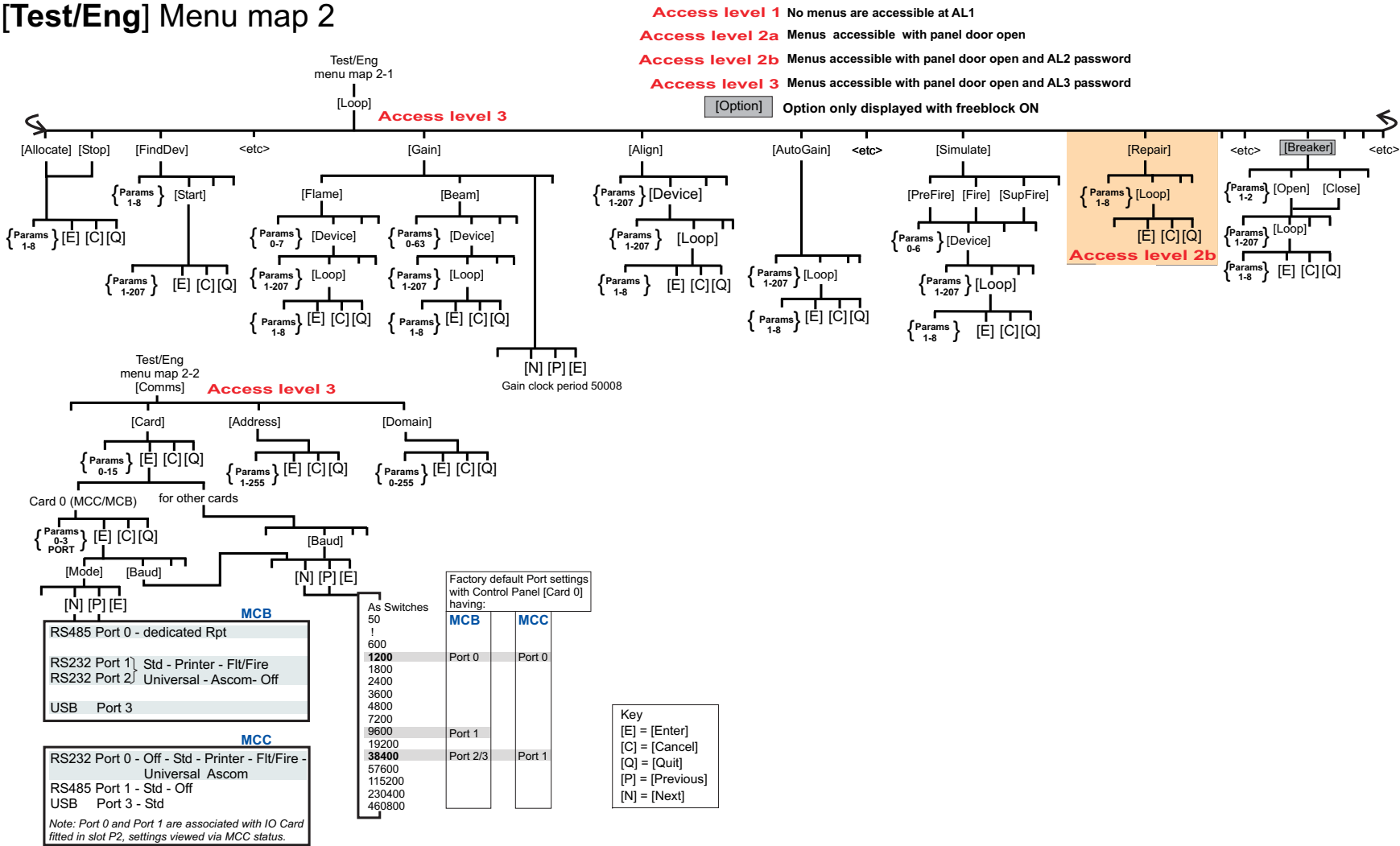
EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher [Info] Menu map



[Test/Eng] Menu map 1



EN Vigilon Plus **Control panel** with MCC/MCB at V4.54 or higher
[Test/Eng] Menu map 2



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Appendix B - Message Action List

This appendix lists all the messages that are likely to be displayed at the *Vigilon Plus control panel / Network Node*. There are some fault events that are not self clearing and will require manual intervention.

Clearable fault events

The following faults are identified as clearable fault events.

- ☐ FLASH corrupt
- ☐ Invalid device configuration
- ☐ Too many errors
Can be cleared but will not reset the counters.
- ☐ Unrecoverable Tx fault
- ☐ Checksum error
FLASH Configuration data and FLASH in sensors
- ☐ NVM recovery failures or map mis-match

How to clear a clearable fault event

To **clear** the **fault** at the control panel press the **Menu on/Off** button -> **[Test/Eng]** -> **[UserCode]** -> enter your PIN (where necessary) -> **<etc>** -> **[Config]** -> **[Clear]** -> **[Fault]** -> **[Enter]**, then scroll through to the fault you wish to clear and select **[clear]**.
If the fault is current then it will reappear and return to the display.

Latching fault events

The following faults are identified as latching events.

- ☐ Slave devices lost
- ☐ Loop allocation fault
- ☐ Invalid loop configuration

How to clear a latching fault event

To **clear** a **latching fault** it is necessary to **re-allocate** the **loop circuit**.

Repairable fault events

Repairable fault events are those that require manual intervention to clear the fault, for example after a Ground break require Repair action.

How to clear a repairable fault event

To **repair** the **fault** at the control panel press the **Menu on/Off** button -> **[Test/Eng]** -> **[UserCode]** -> enter your PIN (where necessary) -> **[Loop]** -> **<etc>** -> **<etc>** -> **[Repair]** -> **[Loop]** -> enter the loop number on which the fault resides -> **[Enter]**.

Message Action list

The messages displayed at the *control panel, network node or loop repeat panels* are given here in an *alphanumerical* order to provide guidance:

- ☐ on a 'fault message'
- ☐ the likely meaning of the message
- ☐ along with suggestions on what initial actions may be taken to rectify the problem.



Only the messages that are applicable will appear on the display of the control panel, network node or loop repeat panels. Additionally the display will identify the panel where the event was first detected. Not all the messages are applicable as this list covers all the panels including the legacy range.

- ☐ The term device was previously referred to as outstation.

Message	associated with..	meaning..	..possible action
A spurious FIRQ at card x	MCC / MCB (main control card/board)	There is glitch on the MCC hardware that appears on the backplane / MCB.	Clear and ignore single occurrence.
A spurious IRQ at card x	MCC / MCB (main control card/board)	There is glitch on the MCC / MCB hardware.	Clear and ignore single occurrence.
A spurious NMI at card x	MCC / MCB (main control card/board)	There is glitch on the MCC / MCB hardware .	Clear and ignore single occurrence .
A spurious SWI2 at card x	MCC / MCB (main control card/board)	There is glitch on the MCC hardware.	Clear and ignore single occurrence.
A spurious SWI3 at card x	MCC / MCB (main control card/board)	There is glitch on the MCC / MCB hardware.	Clear and ignore single occurrence.
Access fault at card x	Any Card	Card cannot talk to the MCC / MCB local controller.	Replace card if the fault keeps reoccurring. Could also be caused by excessive network messages. Investigate message passing.
ACIA Failed at card x	Network Card or MCC / MCB	Communication chip failure or the FABs and SABs are corrupt.	Reset the card and replace if it fault keeps reoccurring.
Alarms silenced	MCC / MCB (main control card/board)	Silence alarms button has been pressed.	-

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Message	associated with..	meaning..	..possible action
Alarms sounded	MCC / MCB (main control card/board)	Sound alarms button has been pressed.	-
Alarms Verified	MCC / MCB (main control card/board)	Alarms verify button has been pressed.	-
Alarm Zone Disabled / Enabled at card x	LPC (loop processor card)	-	
Allocation : Double Allocated card x number y on loop z	LPC (loop processor card)	Two devices are given the same address. Loop breaker (relay contacts) are fused 'micro welded' together (closed) or there is a bad connection.	Check the device wiring. The device may be faulty, replace if required.
Allocation : Hw fault card x number y on loop z	LPC (loop processor card)	Loop breaker relay has not closed or serial line End-2 is faulty or relay is stuck i.e. closed. The device may not be correctly fitted.	Check the device wiring. The device may be faulty, replace if required.
Allocation : Map error card x number y on loop z	LPC (loop processor card)	Loop wiring is incorrect, eg a sub loop may have been wired on a spur.	Check and rewire if necessary. This fault can also be device failure.
Allocation : memory Overflow at card x number y on loop z	LPC (loop processor card)	Loop processor has run out of memory during allocation / when starting a loop.	Software error or more likely a faulty RAM chip.
Allocation : not ready at card x	LPC (loop processor card)	An attempt to allocate a loop when it is already allocating.	Wait and if it takes longer than 10 minutes then reset the loop card.
Allocation : OK at Card n : Allocated x	LPC (loop processor card)	The loop circuit has been successfully allocated.	-
Allocation started from end 2 at card x	LPC (loop processor card)	No devices found on End-1 or End 1 of the loop is open circuit.	Check the wiring to the first device or last device on the loop.
Allocation : Too many card x number y on loop z	LPC (loop processor card)	Allocated more than 200 devices or the device is restricted for use with the particular software version.	Replace incompatible devices used on other systems ie:32000 or SMS device found in system.
Allocation : Tx fault card x number y on loop z	LPC (loop processor card)	Invalid reply from the device.	Check the device. Also check cable routing and length (measure resistance and capacitance of cable).

Message	associated with..	meaning..	..possible action
ASCII Device is faulty number x on loop y	LPC (loop processor card)	Failure of the Control panel to communicate with Repeat / Mimic panel. The power supplies to these panels may not be connected.	If applicable, connect the power supply or RESET the Master Repeat Card in the Repeat and Mimic panel. If necessary replace the card/software.
Back up	NVM (non volatile memory) on backplane or MCB	Data is being copied from cards and stored onto the NVM.	-
Back up required	NVM (non volatile memory) on backplane or MCB	Data must be saved to memory	Back up data to the memory.
Battery discharged	Power supply	This indicates the failure of battery load test. The ADC value should be equal to or less than normal - 9.	Check load test and the battery condition.
Battery disconnected/ restored	Power supply	A disconnection is indicated when the ADC value from the battery is equal to or less than 130.	Normally it should be = 234 (27.4V) To give a restored indication the ADC value should = $V_{batt} \times 1/6 \times 256/5$.
Baud rate x at card y	MCC / MCB / Network Card	Baud rate set has been read.	If required, change the baud rate .
being Initialised at card x	NVM (non volatile memory) on backplane or MCB	Memory is being set up by the MCC / MCB (main control card/board), ready for use.	-
Buffer full at card x	Any Card	Software error .	-
Buffers out of step at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Call point glass is broken number x loop y	Device	Call point has been operated when the device is disabled.	Replace the glass, if necessary.
Call point on interface unit operated / restored	Device	Mains powered interface input has been triggered by conventional MCP, value of input is equal to or less than 6V.	-

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Message	associated with..	meaning..	..possible action
Call point open cct number x on loop y	Device	Monitored base does not see end-of-line unit. There is a bad connection in MCP.	Check the internal wiring, there may be a loose connection.
Call point operated/restored number x loop y	Device	Break glass has been broken or operated with a test key.	Replace the break glass to restore the call point or remove the test key, whichever is applicable.
Call point short cct number x loop y	Device	Call point contacts are short circuit.	Check the call point contacts and internal wiring.
Card found/lost at card x	Any Card	The card has been found/lost. If lost then it is locked out.	There may be a hardware fault on card or backplane. Investigate and rectify.
Card n recovered /Card n backed up	NVM (non volatile memory) on backplane or MCB	Data recovered to a card from the NVM. A card data has been backed up to the NVM.	-
Chamber Removed /Replaced number x on loop y	Device	Device chamber has been removed.	Replacing the chamber will cause a replaced indication.
Charger Fault	Panel Power Supply Unit (PSU)	Charger circuit has failed.	Replace the PSU.
Charger Fault number x on loop y	EN54 System device power supply (Repeat, Mimic or mains powered interface)	Charger circuit has failed at a loop device.	Replace the board (having the power supply) on the loop device.
Charger Restored	Panel power supply	Charger circuit is working again.	-
Charger Restored number x on loop y	EN54 System device power supply (Repeat, Mimic or mains powered interface).	Charger circuit at the loop device is working again.	-
Chk=4.874 NVMC 41874 Flag=0 NVM is write protected at Card x	NVM (non volatile memory) on backplane or MCB	The memory is read only.	Disable the write protect using the [set up] menu.

Message	associated with..	meaning..	..possible action
Clock changed / Clock not set up	MCC / MCB (main control card/board)	Time and date has been altered or no time has been entered.	If necessary, set the clock using the set up menu.
CO sensor faulty	Device	There is a fault with the CO sensor in an S-Quad.	Replace device
CO sensor near end of life	Device	The CO cell is nearing the end manufacturer recommended life expectancy in an S-Quad.	Replace the device
CO sensor restored	Device	The CO sensor fault has been restored in an S-Quad	-
Command build activated / deactivated	MCC / MCB (main control card/board)	Command build has been switched ON (activated) or switched OFF (deactivated).	If necessary switch ON/OFF the command build using the [control] menu.
Command build enabled / disabled	MCC / MCB (main control card/board)	This is an automatic or manually controlled action.	If necessary, manually enable/disable the command build using the [control] menu.
Comms enabled / disabled at card x	MCC / MCB (main control card/board)	Card communications have been enabled or disabled.	If necessary, manually enable/disable the Comms using the [control] menu
	Network Card	Control panel has been isolated from the network.	
Comms Supply Fault	Network node	The communications power supply has failed.	Replace the power supply unit.
Comms supply Restored	Network node	The communications power supply has been restored.	-
Communications started / stopped at card x	MCC / MCB (main control card/board)	Communication to a particular panel or to external equipment linked to the panel (eg Laptop having the commissioning tool) has started / stopped.	Check the wiring.
Control passed at card x	Network	-	-
Corrupted NVM memory at card x	Any Card	Checksum failure on configuration at midnight.	Recover data and record event in the log book. If it reoccurs then replace the card.
DC Power fail / restored	Power supply	The ADC of output 27V regulator is normally=146(28.5V).	-

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Message	associated with..	meaning..	..possible action
DC too high / DC too low	Power supply	The DC supply is: Too high if greater than 164 (32V) Too low if less than 102 (20V)	Check the mains supply and battery wiring.
Delay block setup	MCC / MCB (main control card/board)	A Delay Block has been configured.	-
Delay blocks Cleared	MCC / MCB (main control card/board)	-	-
Detection Zone Disabled/Enabled at card x	LPC (loop processor card)	The detection zone has been enabled or disabled automatically or manually.	If necessary, manually enable/disable the detection zone using the [control] menu.
Detector on interface unit number x on loop y channel z	Device	Conventional detector connected to an input of a mains powered interface input has triggered. Value of the input is equal to or less than 10V.	-
Device Address changed from x number y on loop z	Device	Device SAFE address has changed.	If necessary rectify the device SAFE address.
Device Battery Fault	Radio (Plexus) Device	Radio (Plexus) device has a battery failure	Check the batteries at the Radio (Plexus) device using the Battery Condition Indicator tool and replace the batteries if necessary
Device batteries need replacing	Radio (Plexus) Device	The batteries inside a Radio (Plexus) device needs to be replaced	Replace the batteries in the Radio (Plexus) device
Device battery discharged	Radio (Plexus) Device	The batteries inside a Radio (Plexus) device are nearing end of life.	Replace the batteries in the Radio (Plexus) device
Device battery fault /restored	Device	The battery has failed the load test at the device, for a mains powered interface the value is equal to or less than normal - 2V.	Check and replace the battery if necessary.

Message	associated with..	meaning..	..possible action
Device Battery OC number x on loop y	Device	Failure of fuse or battery wiring is OC on products like mains powered interface unit. Normal value = 27.4V Value is equal to or less than 16V (OC) / greater than 17V (Restored).	Check and replace if necessary to restore.
Device Enabled / Disabled at Card x	LPC (loop processor card)	Device has been enabled or disabled manually or automatically.	If necessary manually enable or disable the device.
Device Fault number x on loop y	ACU Audio Control Unit	Fault input has been asserted.	Check remote equipment wired to the fault input.
Device Gain set to x number y on loop z	Beam devices (can also be a Flame device - no longer supported)	Confirmation of a change in gain setting.	-
Device inoperative	Radio (Plexus) Device	The Radio (Plexus) device batteries are too low for device to operate correctly	Use the Battery Condition Indicator tool to determine if the batteries in the Radio (Plexus) Device require replacing, if necessary replace the batteries.
Device Mains Failed / Restored number x loop y	Device	Failure of mains supply to mimic/repeat /mains powered interface unit. The value of rectified DC is equal to or less than 8V i.e. failed.	Check mains supply connections and the mains fuse. When the rectified DC is greater than 24V it is restored.
Device Mains too high / low number x loop y	Devices like: Mimic / Repeat / Mains powered interface unit	Failure of device mains supply. The value of rectified DC is too high / low i.e in the region of 30V / 17V.	Check the power supply and replace it if necessary
Device Moved	Radio (Plexus) Device	The Radio (Plexus) device has been tampered with	The Radio (Plexus) device has been moved from the original installed location or there has been changes in the area that affect the device operation.
Device power Fault / Restored number x on loop y	Device	The failure of internal power supply rail.	Replace the device.

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Message	associated with..	meaning..	..possible action
Device power too high/ restored too low /restored number x on loop y	Device	Internal power rail of mains powered interface unit has a value of rectified DC equal to or greater than 32V (high) / less than 30V (restored) DC equal to or less than 24V (low) / greater than 26V (restored)	Replace the device
Device Replaced is faulty number x on Loop y	LPC (loop processor card)	Tx fault re-finding device.	Check connections to device / base and also device type etc.
Device Replaced number x loop y	LPC (loop processor card)	The replaced device is now communicating with the panel.	-
Device Returned	Radio (Plexus) Device	Radio device has been returned to it's base	The moved Radio (Plexus) Device has been returned to its original installed position.
Device soft address; SAFE: number x on loop y	Device	Device primary address does not match SAFE address.	Check loop configuration.
Disables Cleared	MCC / MCB (main control card/board)	All disablements have been removed from the panel.	-
Duplicate SAFE Address error: number x on loop y	Device	Two devices on a loop have the same SAFE address.	Change one of the SAFE address.
Earth Fault/ Fault Cleared	Power supply	The ADC value to earth of the 8V regulator: is equal to or greater than 5 (for a +ve fault) or less than 3 (for a -ve fault).	A clear indication is normally given when ADC is equal to 64.
Earth Fault at Card n Loop n: 0v resistance is xxR, L is y Loop n: Earth is aaa, bbb (99)	LPC (2Km Loop Processor Card)	There is an earth fault present before allocation. Note: An Earth fault post allocation is indicated as Earth fault.	Use the Loop Diagnostic Tool to diagnose the earth fault condition and rectify it.
Enabled Aux relay x Disabled Aux relay x	MCC / MCB (main control card/board)	The auxiliary relay has been enabled or disabled automatically or manually.	If necessary, manually enable/disable the auxiliary relays using the [control] menu.

Message	associated with..	meaning..	..possible action
End switching normal at card x	Network	Network card is back to normal operation, (see also end switching disabled)	-
Exception number x on loop y	<i>(Exceptions also referred to as Sub-Fault)</i> Loop Processor Card	Change noted in the environment of the device	Check the condition codes.
Excessive transmission error rate Too many errors: lost device or loop split	LPC (loop processor card)	There were more than 10 reply errors in one day. (More than 16 reply errors will cause the device isolation.)	Check screening and electrical noise in area of operation + check loose connections to loop.
External Fire at panel n	MCC / MCB (main control card/board)	There is a Fire on another panel in a network.	Investigate the fire.
FARE Enabled / Disabled number z on loop y Zone x	Device	The FARE device has been enabled or disabled manually or automatically.	Manually enable or disable the FARE device if required.
FARE Activated / Deactivated number z on loop y zone x	Device	The FARE device has been activated / deactivated manually or automatically.	If appropriate manually deactivate the FARE device. The deactivation happens automatically on resetting the system after a fire event.
FARE Faulty/Restored number x loop y zone w	Device	A 'faulty' message is given when there is a fault associated with FARE equipment.	A 'restored' message is given when the FARE equipment fault is investigated and rectified.
FARE loop x, device y	Device	This information shows the FARE Interface is on loop x and is device y.	-
Fast scan started number x on loop y channel z	Data logging	Fire sensor has detected a change on its input.	The message is normally given when the data logger is connected.
Fault cleared Warning cleared	MCC / MCB (main control card/board)	Fault or warning has now gone.	-

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Message	associated with..	meaning..	..possible action
Fire number n loop y channel z	LPC (loop processor card)	Pattern match is equal to a Fire or MCP operated or conventional detection on IO line of interface operated.	-
Fire reset	MCC / MCB (main control card/board)	The Fire reset button has been pressed.	-
Flash memory checksum error	Device	Speech Memory Checksum Error	Replace the device
FPE Enabled / Disabled at card x	Device	All FPE Interface devices on loop x have been enabled or disabled manually or automatically.	Manually enable or disable all FPE Interface devices if required.
FPE w: x, y, z	Device	This information shows the FPE Interfaces on loop w and are devices x, y and z.	-
Full keyboard Removed / Fitted number x loop y	MCC/ MCB / DAC/ MRC	The keyboard has been 'removed' (is not fitted).	Check the cable connections to the keyboard to bring about the 'fitted' message.
Group Enabled / Disabled	MCC / MCB (main control card/board)	A group has been automatically or manually disabled or enabled.	If necessary, manually enable/disable the group using the [control] menu.
High Error rate at Card x	Network	High number of communication errors.	Check network wiring and connections.
Incompatible Card Version at card x	Any Card	There is a card software conflict.	Fit card with compatible software.
Interface Failed at card x	MCC / MCB (main control card/board)	The interface timer chip has failed.	Clear the fault and ignore single occurrence. Replace the card on multiple failures.
Interface input - fault / restored number x on loop y channel z	Device	IO line fault.	Check the wiring and restore the input for normal operation.
Interface Input Fire number x on loop y channel z	Device	Interface IO line triggered a fire event.	Check the wiring and restore the input for normal operation.

Message	associated with..	meaning..	..possible action
Interface input not Reset number x on loop y channel z	Device	The input line continues to trigger.	Check devices on the IO line.
Interface Input OC / restored	Device	The end-of-line is not seen.	Check the wiring. The device may have been removed, if so refit the device to restore.
Interface input operated / released number x on loop y channel z	Device	Fixed extinguishant interface unit input has been triggered (applicable for old systems only).	Follow the FE system reset procedure to release the input.
Interface input SC / restored number x on loop y channel z	Device	The interface wiring is short circuit. The value of input is equal to or less than 2V.	Check the interface wiring and rectify to restore the input.
Interface output SC or OC or Restored number x on loop y channel y	Device	The value of output is equal to or less than 50% for SC or is TBA for OC.	Check the wiring and restore the output circuit.
Intermittent Fault at card x	Any Card	A fault has been picked up by the MCC / MCB (main control card/board).	Replace the appropriate card if the fault keeps reoccurring.
Invalid configuration	LPC (loop processor card)	Loop recovered with freeblocks ON.	This is not allowed. Switch OFF the freeblocks and reallocate.
	Device	FLASH incorrectly programmed	Replace the device
Invalid configuration at card x	LPC (loop processor card)	Loop recovered with freeblocks ON or Loop configuration faulty, possibly after upload from Commissioning Tool.	This is not allowed. Switch OFF the freeblocks and reallocate.
Invalid message / reply received at card x	LPC (loop processor card)	The message/reply received cannot be understood. The communication can be from a Repeat, Mimic, Networked system.	Ignore single occurrence and record in the log book.
Invalid message/reply at card x	Network Card	The message/reply has not been understood from other panels.	Ignore for single occurrence and record in the log book.

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Message	associated with..	meaning..	..possible action
Invalid task delay at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Invalid task stage	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
IO line Enabled / Disabled at card x	LPC (loop processor card)	The IO line is either enabled or disabled automatically or manually.	If necessary, manually enable/disable the IO line using the [control] menu.
Loop map has changed card x	LPC (loop processor card)	The map is different to that previously allocated .	Check and confirm difference with backup map .
Loop n: High resistance	LPC (2Km Loop Processor Card)	High resistance on both lines.	Use the Loop Diagnostic Tool to diagnose the cause of high resistance on the loop and rectify it.
Loop n: High resistance on negative line	LPC (2Km Loop Processor Card)	High resistance on 0v lines.	Use the Loop Diagnostic Tool to diagnose the cause of high resistance on the negative line and rectify it.
Loop n: Overloaded	LPC (2Km Loop Processor Card)	The loop is overloaded, ie is taking too much current.	Use the Loop Diagnostic Tool to diagnose the fault condition and rectify it.
Loop n: Reply marginal	LPC (2Km Loop Processor Card)	The named device is at the marginal limits of correct communications (timing or level)	Use the Loop Diagnostic Tool to diagnose the cause of communication issue and rectify it.
Loop power restored	Power supply	Normal loop supply restored: Normal ADC = 202 (44V). Value = Vsupp x 1/11X256/5.	-
Loop Started not ready at card x number y loop z	LPC (loop processor card)	As per message.	Wait and if it takes longer than 10 minutes then reset the loop card.
Loop stopped at Card x	LPC (loop processor card)	The loop has been aborted or is still being allocated.	Re-allocate the loop.

Message	associated with..	meaning..	..possible action
Loop voltage too high/low	Power supply	The ADC value too high. or The ADC value is too low.	Check PSU values in the [test/eng] menu, normal reading is 221. Check fuses and replace PSU if necessary.
Lost Command number x Loop y	LPC (loop processor card)	Lost FAB/SAB.	Ignore single occurrence and record in the log book.
Lost Device number x on loop y	LPC (loop processor card)	There is no reply on ends 1&2 of the device.	Check the device connections.
Lost slave Device number x on loop y	LPC (loop processor card)	There is no reply on ends 1&2 of the device.	Check the slave device connections and replace if necessary.
Magnetic switch operated number x loop y	Device	The reset switch in the base has closed for 3400 devices only.	Check base and proximity of magnetic field.
Main program not running at card x	Any Card	Problem with software. The watchdog will operate the MCC / MCB (main control card/board).	Software errors will activate a system reset. Ignore single occurrence. Record event in the log book.
Mains failed / restored	Power supply	Failure is detected when the ADC output from the rectifier is equal to or less than 105.	Check mains and fuse. Replace PSU if necessary. This should normally equal to 60V to get a restored message.
Mains too high	Power supply	The ADC value is too high.	Check the PSU values in the [test eng] menu. Replace PSU if necessary.
Mains too low	Power supply	Mains supply is too low.	
Master alarm oc or sc Restored	Power supply	No end-of-line unit seen. ADC value is greater than 92 or ADC value is less than 37 ADC value is 74	Check the wiring to the master alarm circuits. Ensure EOLs are connected inside the control panel if master alarm circuits are not used.

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Message	associated with..	meaning..	..possible action
Master alarm enabled/disabled	MCC / MCB (main control card/board)	The master alarm has been enabled or disabled automatically or manually.	If necessary, manually enable or disable the master alarm using the [control] menu.
Master alarm restored	Power supply	Master alarm fault cleared. The ADC value should be 74.	
Master polling/not polling at card x	MCC / MCB	PC or SRI (service request interface) is not handshaking with the control panel	Check the wiring
Master Sector Actioned	MCC / MCB (main control card/board)	Master sector has been switched ON/OFF.	-
Master Sector Enabled/Disabled	MCC / MCB (main control card/board)	Master sector has been enabled or disabled manually or automatically.	If necessary manually enable or disable the master sector.
Memory corrupt number x loop y	Device	FLASH checksum failure	Replace the device
Message discarded	Network Card LPC (loop processor card)	Message received that couldn't be routed or understood.	Check the location displayed, try resetting card.
Neighbour ACK Failed at card n	-	Repeated address	Check domain addresses are unique
Neighbour ACK Invalid at card n	TBA	TBA	TBA
Network initialised at card x	Network Card	Successful network map with polled panels.	-
Network Insecure / Secure at card x	Network Card	There are 3 missing replies in a row on End-2 of the controller.	Check the wiring. Check the error rate.
Network starting at card x	Network Card	Communication starting with other connected panels.	-
Network wiring fault at card x	Network Card	Wiring is presumed to be OC between panel n and panel n.	Check the wiring.
New address n at card x	MCC / MCB Network Card	The new address set has been read.	-

Message	associated with..	meaning..	..possible action
New domain address n at card x	Network Card	The new address set been read at the particular card.	-
NMI's missed at card x	LPC (loop processor card)	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
No periodic interrupts at card x	Any Card	Problem with software, the watchdog will operate the MCC / MCB (main control card/board).	Software error will activate a system reset. Ignore single occurrence. Record event in the log book.
No response number x loop y	LPC (loop processor card)	There is no communication with repeat or mimic panel software.	Power-up the repeat or mimic panel.
No room on NVM card	NVM (non volatile memory) on backplane or MCB	User attempts to backup or save to an NVM that does not have enough free space	If there is no memory left then the card 14 located on the Backplane or on MCB (Card 0) needs to be initialised
Node double allocated at card x	Network Card	Two network cards have same address.	Keep address unique.
Node found at card x	Network Card	Another panel on the network acknowledged.	-
Node lost/restored at card x	Network Card	There have been 15 missing replies from another panel.	Check the wiring or check and reset the panel Network card to restore.
Not enough RAM at card x	Any Card	Not enough memory.	Reset the card and ignore single occurrence. If necessary replace the card or add RAM, where applicable.
Nothing found at card x	LPC (loop processor card)	No devices found on End 1 or End 2.	Check wiring of loop circuit.
NVM initialised	NVM (non volatile memory)on backplane or MCB	Memory available verified.	-
NVM is not write protected	NVM (non volatile memory)on backplane or MCB	Read and write is possible to the memory card.	Enable protect if required using the [set up] menu.

Commissioning instructions

Message	associated with..	meaning..	..possible action
NVM is read only	NVM (non volatile memory)on backplane or MCB	User attempts to write to a hardware write protected NVM	Remove the hardware write protect before writing
Panel in/out of commissioning mode	MCC / MCB (main control card/board)	Commissioning mode switched ON/OFF.	To manually switch commissioning mode use the [test/eng] menu.
Panel Label Set up	MCC / MCB (main control card/board)	Panel label has been set from the menu.	-
Panel powerup	MCC / MCB (main control card/board)	Software reset of system or there has been a manual reset of card 0.	If an automatic reset has occurred then check for system errors.
Pattern number x on loop y channel x	LPC (loop processor card)	Sensor has detected an interesting event eg fire/fault.	-
Pointer corrupted at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Pointer missing at card x	Any Card	Software error	Software errors will cause a system reset. Card fault should be ignored for single occurrence. Record the event in the log book.
Prefire number x on loop y channel z	LPC (loop processor card)	Pattern match less than a Fire.	-
Printer lost/fitted at card x number y on loop z	MCC / MCB / DAC / MRC	The external printer has not been seen and is 'lost'.	Check the connections to the printer. Also check for paper jam in the printer mechanism.
Program memory corrupted at card x	Any Card	Checksum failure on FLASH at midnight.	Clear and ignore single occurrence. Record in the log book.

Message	associated with..	meaning..	..possible action
Program not running at card x	Any Card	Problem with software, the watchdog will operate the MCC / MCB (main control card/board).	Software errors will activate an automatic system reset. Ignore for single occurrence. Record event in the log book.
Program runaway at card x	Any Card	CPU has crashed.	Clear and ignore single occurrence. Record in the log book.
Q buffer full at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Q buffer number Failed at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Radio devices recovered	LRT	Radio LRT has completed finding all of the radio devices	-
Recover failed at card x	LPC (loop processor card)	Loop does not match with what is in NVM.	Check and confirm/correct the difference and then backup.
Recover succeeded at card x	LPC (loop processor card)	Successful recovery after a previous failure.	-
Replaced Device Wrong type number x on loop y	LPC (loop processor card)	Device is a different type.	Check and confirm installation of correct type and reallocate. Back up to memory.
SAFE Address not Set up number x on loop y	LPC (loop processor card)	The replaced device is not SAFE addressed. The device previously installed in the location was SAFE addressed.	Set up the SAFE address.
Scan error number x at card y	LPC (loop processor card)	Invalid universal scan reply.	Check loop length and screening

Commissioning instructions

Message	associated with..	meaning..	..possible action
Sector actioned	LPC (loop processor card)	Sector is switched ON/OFF.	-
Sector enabled/ disabled at card x	LPC (loop processor card)	The sector(s) has been enabled or disabled automatically or manually.	If necessary, manually enable/disable the sector using the [control] menu.
Seeking radio devices	LRT	Radio LRT is finding all of the radio devices	-
Sensor faulty	LPC (loop processor card)	Wireless system - sensor head needs cleaning or sensor head lost from a wireless base (wireless adaptor) Also mean - Fault pattern has been matched.	Check and replace the sensor if necessary.
Sensor out of specification number x on Loop y channel z	LPC (loop processor card)	Fault pattern has been matched.	Check and replace the sensor if necessary.
Sensor restored	Radio (Plexus) Device	A Radio (Plexus) Device head has been restored.	-
Shared memory hardware is faulty at card x	Any Card	There is MCC / MCB problem.	Replace all other cards first other than the MCC/MCB to see if the problem is resolved. If it persists then replace MCC / MCB.
Slave Device Replaced number x on loop y	LPC (loop processor card)	Slave type device has been replaced.	-
Slave micro failed at card x	LPC (loop processor card)	Loop driver is watchdogged by loop processor.	Reset and ignore single occurrence. Record in the log book.
Software errors	Software errors will activate a system reset	Software error	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Sounder circuit failed	Device	There is a fault with the sounder hardware on the S-Cubed or S-Quad.	Replace the device
Sounder circuit Restored	Device	The fault is on the sounder circuit associated with S-Quad or S-Cubed has been restored.	-

Message	associated with..	meaning..	..possible action
Speech circuit has failed	Device	There is a hardware fault possibly associated with the voice chip on the speech S-Cubed or S-Quad	Replace the device
Speech circuit Restored	Device	The fault on the speech S-Quad and S-Cubed has been restored.	-
Spurious acknowledgment at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Spurious FAB number x on loop y	LPC (loop processor card)	Device has FAB but shouldn't have any.	Ignore single occurrence as the panel will remove the indication.
Stack overflow at card x	Any Card	Software error	Software errors will activate a system reset Ignore single occurrence. Record the event in the log book
Stack too deep at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Strobe circuit has failed	Device	The strobe circuit in the S-Quad or S-Cubed device has failed.	Replace the device
Strobe circuit restored	Device	The strobe hardware fault, such as internal connection, in the S-Cubed or S-Quad has been restored	-
Sub Faults			See exceptions
Superfire number x on loop y channel z	LPC (loop processor card)	Pattern match is equal to or greater than a Fire.	-
Supply totally Failed	MCC / MCB (main control card/board)	The message does not normally appear.	-
Switching inhibited at end 1 at card x	Network	Automatic end swapping on the network controller has been disabled for test purpose.	-

Commissioning instructions

Message	associated with..	meaning..	..possible action
System Printer fitted at card x number y loop z	MCC / MCB (main control card/board)	A 80 column serial printer is fitted to the panel.	-
System Printer Lost at card x number y on loop z	MCC / MCB (Printer option)	TBA	TBA
System Restarted /Started	MCC / MCB (main control card/board)	Link on back plane restored.	-
System stopped	MCC / MCB (main control card/board)	Backplane link moved causing the system to stop.	The link should not be normally accessed .
Task stuck at card x	Any Card	Software error.	Software errors will activate a system reset. Ignore single occurrence. Record the event in the log book.
Test set found / removed at card x	LPC (loop processor card)	Remote allocation unit has been found connected to the system.	Disconnection of the test set will bring about a removed message.
Too many errors number x on loop y	Device	More than 15 errors from a device in 34 hours.	Check the loop wiring to the device.
Too many errors x at Card y	Network Card	Up to 6 consecutive communication errors.	Check wiring and cable screen. There may be faults with the installation.
Two controller at card x	Network	There are two network controllers.	One network controller will automatically shut down.
Two master has the same address at card x	MCC / MCB / DAC	Two PCs have been given the same address.	Each PC must have a unique address.
Unable to claim buffers at card x	Any Card	System too busy	Wait or reset card if stuck, record in the log book
Unrecoverable Tx fault number x loop y	LPC (loop processor card)	More than 3 consecutive reply errors	Check the device connections and replace if necessary.

Message	associated with..	meaning..	..possible action
Voice Alarm comms failed Port 1 / Node x	Panel's Serial Link	There is a serial link wiring fault between Vigilon Plus panel 'Port 1' and DOM/Comprio. Other Vigilon Plus panels in the connected network will indicate the 'Node' number of the panel connected to DOM /Comprio.	Check the serial link is working, ensure: ☐ Serial link cable is connected to DOM to socket TWI ☐ Serial link cables are connected to the two 9 way D Type connectors on the Adapter TWI-RS232 and ☐ Serial link cable is wired correctly at a Vigilon plus/Compact plus panel to terminals Tx, Rx and 0V.
Voice Alarm comms failed Port 1 / Node x	Panel's Serial Link	There is a serial link wiring fault has been restored between Vigilon Plus panel 'Port 1' and DOM/Comprio. Other Vigilon Plus panels in the connected network will indicate the 'Node' number of the panel connected to DOM /Comprio.	-
Voice Alarm common fault Node x	Connected D1 Voice Alarm system	There is a common fault, that is any fault in D1 Voice Alarm system, for example Speaker line fault, Amplifier fault, Mains power fault, etc. The ERROR LED is lit at the DOM/Comprio connected via serial link to Vigilon Plus panel.	Find the cause of common fault at a D1 Voice Alarm system: ☐ Refer to user guides for D1 Voice Alarm system modules to locate the fault. ☐ Once the fault is rectified the fault message and notice fault indications are removed at both Vigilon Plus and D1 Voice Alarm system.
Voice Alarm configuration fault Nos. 17	Configuration fault at a D1 Voice Alarm System	This message and indication is given at Vigilon Plus panel of a configuration fault in a D1 Voice Alarm system. The affected output (VOP) in this case No 17 is not recognised.	Test for a configuration error which occurs when a non existent or not configured properly Virtual Output Point is made active. ☐ Confirm the VOP displayed as being faulty is non existent. ☐ Rectify the configuration at Vigilon Plus and/or at the D1 Voice Alarm system. ☐ Repeat this test to ensure the revised configuration works. Check to ensure the fault has cleared.

Commissioning instructions

Message	associated with..	meaning..	..possible action
Warm restart	MCC / MCB (main control card)	The 5V rail has seen a dip activating the watchdog or the button has been pressed on MCC / MCB or panel power reconnected.	-
Warning cleared	MCC / MCB (main control card/board)	Warning has been removed.	-
Wiring changed - ground break at card x number y on loop z	LPC (loop processor card)	A test has shown a break on the 0V line. The test is conducted every minute.	Check wiring of 0V around loop.
Wiring changed - short at card x number y loop z	LPC (loop processor card)	Loop short circuit after the particular device.	Check wiring and device after indicated device.
Wiring changed - split/closed at card x number y on loop z	LPC (loop processor card)	No reply received from a device at End-2 of loop.	Check wiring.
Wiring changed part short at card x number y on loop z	LPC (loop processor card)	Less than 80 ohms between the +ve and 0V of loop wiring.	Check wiring and the device after the partially shorted device.
Wrong card type at Card x	Any Card	Card in wrong slot.	Move the cards to the right location.
Zone enabled / disabled at card x	MCC / MCB (main control card/board)	The zone has been enabled or disabled automatically or manually.	If necessary, manually enable or disable the zone using the [control] menu.

Appendix C - Guidelines for standalone system commands

This appendix provides guidance on factors that must be taken into consideration when applying *standalone system functions*, such as:

- ☐ labels
- ☐ sectors (including flag set or integral sectors)
- ☐ delay blocks
- ☐ time slots and time blocks
- ☐ zones
- ☐ groups
- ☐ voice alarm zones
- ☐ command builds
- ☐ sounder configuration
- ☐ auxiliary relays.

Labels

- ☐ Labels are given to identify location of areas on a site where products are installed

- ☐ A previously created label can be modified



Where manual call points are being used then each can only have a label of up to 28 characters.

- ☐ a 32 character label can be assigned to a:
 - device
(must have a label having 28 characters if it is an MCP)
 - each Input/Output line
(must have a label having 28 characters maximum if one or more MCPs are installed on the input line)
 - zone
 - group
 - control panel.

Input channel label

- ☐ A **supervisory input channel label** of an *interface unit* will only be displayed at the *control panel* of the standalone system.
 - By adding an * in front of the *channel label* it will allow the label to appear on all the networked panels

The networked panels must however be setup to display **all events**. In this configuration the label will also be displayed at a *network node and Central monitoring system*

- When an interface input is operated the standalone system's (local) *control panel buzzer* can only be activated by configuring the input to trigger a **Command build**. In this case the Command build should not have actions
- To activate buzzers of control panels in a network an * must be inserted before the **command build** label.

Commissioning instructions

Long labels

Long labels having up to 64 characters can be given to devices and command builds. This is achieved by replacing commonly used words with token values (special codes). **Tokenised labels** :When tokens are decoded, a trailing space is automatically added to the decoded word. Likewise when a label is being tokenised, the word must have a space following it, or be at the end of a line. Therefore a token word cannot be followed with a comma, dash or other such character.

To switch ON long labels: Press the **Menu on/Off** button -> **[Test/Eng]** -> **[UserCode]** -> enter your PIN (where necessary) -> **<etc>** -> **[Config]** -> **<etc>** -> **[Label]** -> **[On]** -> **[Enter]**.



All tokenised labels are displayed in Upper case.

Banks 0, 1 and 2 show the list of tokens used by the software to reduce the character label to 32 characters or less.

Bank 0

General Building Features

AREA	ANNEXE	BLOCK	BUILDING
CORE	CORRIDOR	DOOR	DUCT
ENTRANCE	ESCALATOR	EXIT	EXTENSION
FLAT	HALL	HOUSE	LIFT
LOBBY	RISER	ROOM	ROUTE
STAIRS	TOWER	UNIT	VENTILATION
VOID	ZONE		

Vertical Location

LEVEL	MEZZANINE	FLOOR	BASEMENT
GROUND	FIRST	SECOND	THIRD
FOURTH	CEILING	ROOF	ATRIUM

Geographic Location

NORTH	SOUTH	EAST	WEST
NEAR	OUTSIDE	ABOVE	UPPER
CENTRE	LOWER	LEFT	RIGHT
FRONT	REAR		

Building Usage

ACCOMMODATION	ADMIN	BEDROOM	BOILER
CENTRAL	CHANGING	COMPUTER	CONTROL
CUPBOARD	ELECTRICAL	ENGINEERING	EXTERNAL

GALLERY	GARAGE	GENERAL	HOUSING
KITCHEN	LAUNDRY	LOUNGE	OFFICE
PLANT	RECEPTION	RESTAURANT	SERVICE
SERVICES	SHOP	STAFF	STORE
STORES	SWITCH	SYSTEM	TOILET
WORKSHOP	WARD	WAREHOUSE	

Airport Tokens

AIR SIDE	ARRIVALS	BAGGAGE	BRIDGE
BUREAU DE CHANGE	CUSTOMS	CONCOURSE	DEPARTURES
DUTY FREE	EXCHANGE	GATE	HANDLING
IMMIGRATION	LANDSIDE	LUGGAGE	MALL
PASSENGER	RETAIL	SECURITY	STATION
TERMINAL	TRANSFER		

Medical Tokens

MEDICAL	THEATRE	X-RAY	CLINIC
PATIENTS	PHYSIOTHERAPY	GERIATRICS	PAEDIATRICS
RECORDS	HEALTH		

Components of the Fire System

SOUNDER	DETECTOR	OPTICAL	IONISATION
HEAT	INTERFACE	BEAM	REPEAT
SPRINKLER			

Bank 1**General Building Features**

CHAMBER	FOYER	SHAFT	STAIRCASE
STAIRWELL			

Vertical Location

FIFTH	SIXTH	ATTIC	BALCONY
LANDING	PASSAGE	SUBWAY	TUNNEL

Positions

ADJACENT	BOTTOM		
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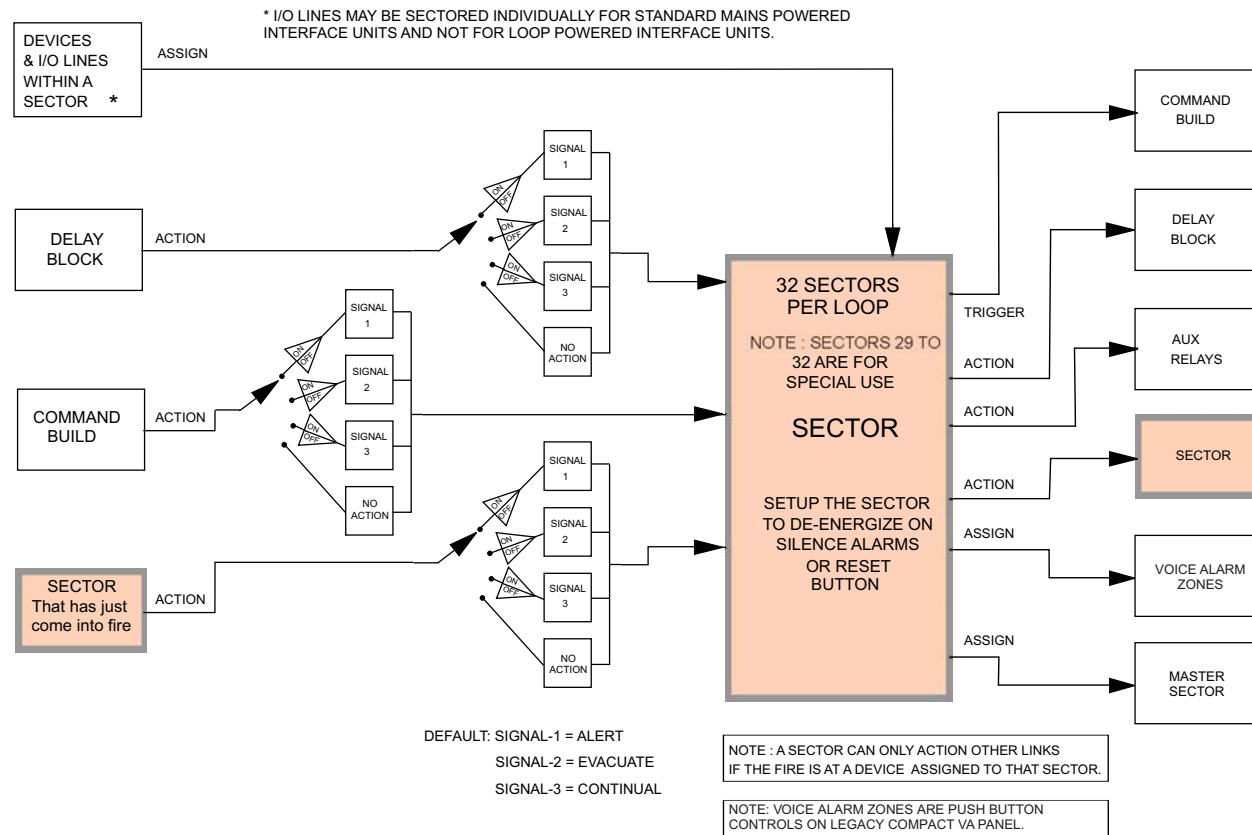
Building Usage

CATERING	COLLEGE	CONFERENCE	DEPARTMENT
DISPATCH	EMERGENCY	EQUIPMENT	ESCAPE
MANAGER	MEETING	PACKING	PHYSICS

Commissioning instructions

POINT	PREPARATION	SHOPPING	SITTING
STATIONERY	SUITE	SUPPLY	TELEPHONE
TRANSFORMER			
Airport Tokens			
AIRLINES	BONDED	CHECK-IN	CLORIFIER
CONVEYOR	CUL-DE-SAC	DOMESTIC	FORECOURT
INFORMATION	INTERCONNECTOR	INTERNATIONAL	RECLAIM
SCREENING	SECRET SIGN	TRAVOLATOR	TRUCKING
Medical Tokens			
DISPENSARY			
Components of the Fire System			
ASPIRATING	INPUT	OUTPUT	PRESSURE
SHUTTER			
Bank 2			
General Building Features			
BLOWER	CHUTE	DOORS	HOIST
ROLLER	SLUICE		
Position			
BEHIND	BELOW	UNDER	MIDDLE
Building usage			
BISTRO	BUSINESS	CANTEEN	CHILLER
CLASS	CLASSROOM	CLEANERS	DISABLED
ELECTRONICS	FACTORY	FEMALE	LIBRARY
LOCKER	MAINTENANCE	MOTOR	MUSIC
NURSERY	PHASE	QUIET	SEATING
SHOWER	STORAGE	TENANT	VEHICLE
VOLTAGE	WALKWAY		
Airport token			
APRON	AUTOWALK	BOARDING	CABIN
CONNECTOR	CRAFT	DEPARTURE	FLIGHT
STAND	TOUCHDOWN		

Sectors



- ☐ a Sector that is actioned ON will activate its alarm devices, such as interface outputs, S Cubed and S-Quad devices.
a sector can be configured to give one of three signals
 - Signal 1 Alert (priority 3)
 - Signal 2 Evacuate (priority 2)
 - Signal 3 Continuous (priority 1).
- ☐ Sectors 1-28 operate automatically with 'Sound Alarms' and 'Silence Alarms' buttons
- ☐ a Sector (any one of 1-28) that is configured for NO action on 'Sound Alarms' button being pressed will be silenced on operation of the 'Reset' button
- ☐ the Sectors 29-32 are independent of the 'Sound Alarms' button and are primarily used for the control of plant equipment. However, the Sectors 29-30 can be configured to normal sector operation
- ☐ sensors, call points and interface inputs can be assigned to a sector and not normally advised with caution may be assigned to more than one Sector
- ☐ a loop powered interface is assigned to a Sector and consequently the inputs and outputs lines of that interface will be assigned to the same Sector
- ☐ S4 devices having alarm sounder and interface outputs are restricted to one Sector only

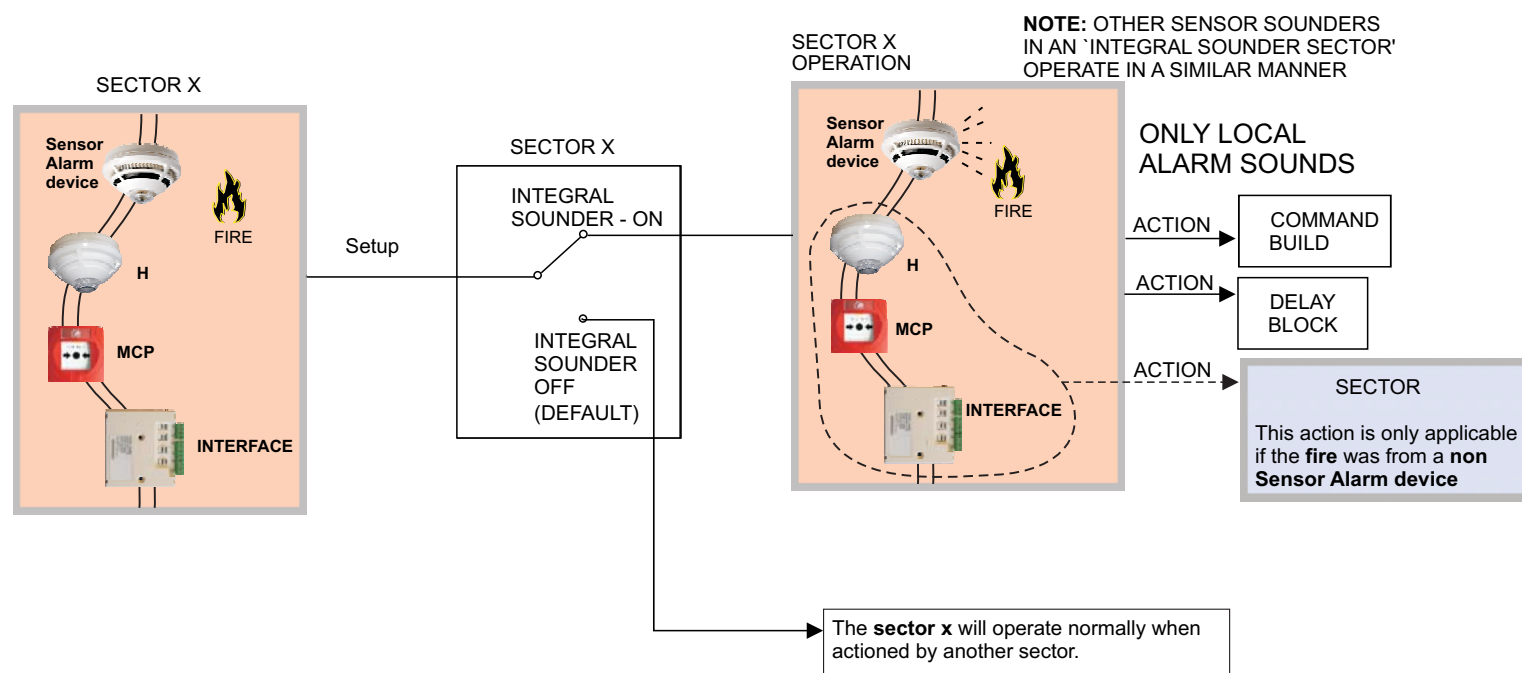
- ☐ There can be up to 32 sectors per loop
- ☐ a Sector is a collection of devices on the same loop having a common action

Commissioning instructions

Integral sounder operation

A sensor sounder device can have its integral sounder only actioned in the event of a local fire event being detected at the device.

- ☐ Integral sounder Sector operation can be configured using the panel menu commands [SetUp]-> [Sector] -> [Sounder] -> [On]
- ☐ when a Sector is configured for 'integral sounder mode' then a fire event from a sensor sounder in the Sector will cause only the integral sounder to operate
- ☐ linked Sectors are not affected when sensor/sounder type device is triggered from Sector configured for 'integral operation mode', only its local sounder operates and fire is reported at the panel
- ☐ Delay Blocks can be configured to action other Sectors from sector operating in an integral sounder mode
- ☐ Sectors configured for integral sounder mode operation can be setup to action command builds.



Default fire plan

The fire plan on power up is known as 'one out all out'. This is when Sector 1 is assigned with all devices of the system. A fire will activate evacuate (signal 2) on all interface outputs and alarm sounders of S-Quads, S-Cubed and DAU devices.

- ☐ Fixed extinguishant inputs / outputs are automatically assigned to Sector 29. Note the Fixed extinguishing interface unit is a legacy product and is no longer a part of the product range.

To assign a default fire plan

- ☐ To manually configure the default fire plan using the set up menu:
 - Assign all devices and IO lines to Sector 1
 - and then action all sectors (except Sectors 29-32) on all sectors to signal 2.

To remove the default fire plan.

- ☐ If the system is not intended to operate with the default 'Fire Plan', then do exactly the same as making the default fire plan, but this time replace the **[Assign]** with **[Remove]**. Also remove IO lines.

All sensors, call points, sounders and interface lines are now completely separate from each other.



The use of monitored line off a sensor device having MCP is not allowed in the UK.

- ☐ also **[Deaction]** all sectors on all Sectors. This will remove any configured Sector actions
- ☐ a site specific 'Fire Plan' can now be set up by assigning devices within common initiation areas to Sectors and then actioning them together to form the required 'Fire Plan'.

Fail safe fire plan

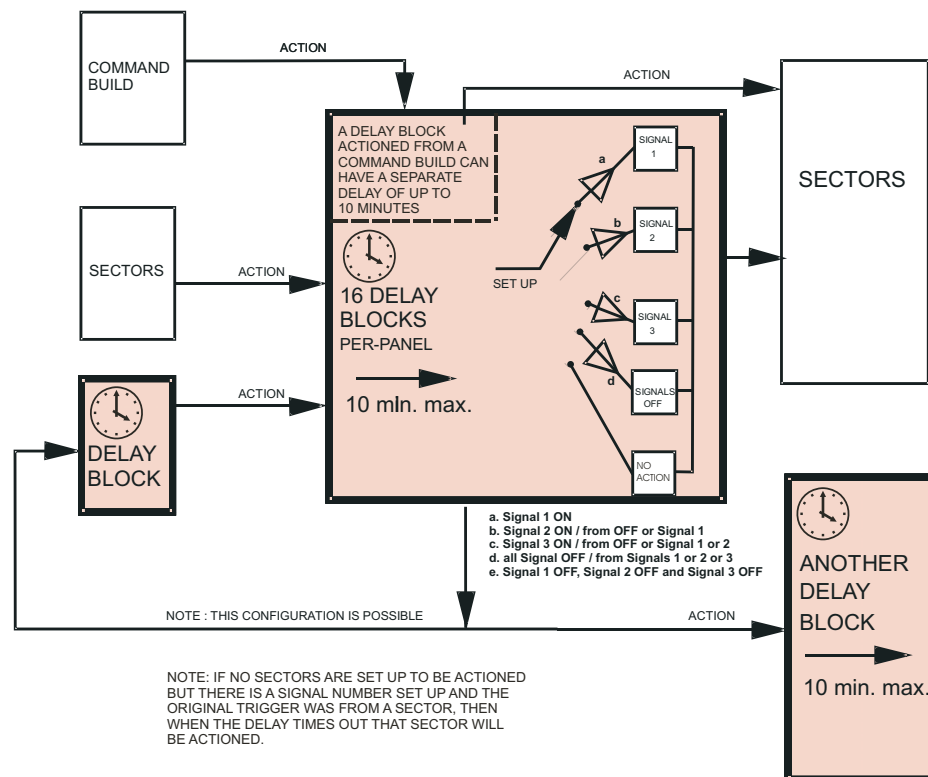
- ☐ It is advisable to build a fail safe mechanism when Sectoring. If for any reason certain Sector assignments have been lost due to system power down and subsequent NVM recovery conflict on system power up, the sounders will still operate in the event of a fire.
- ☐ the fail safe mechanism will involve leaving Sector 1 on each loop free from any device assignment, as any devices that lose their sector assignment will default to Sector 1
- ☐ the following action will give an evacuate tone to any alarm sounders that have defaulted to Sector 1:
 - Action all Sectors (except Sectors 29-32) on Sector 1 to signal 2
 - Action Sector 1 on all sectors to signal 2.

Site specific fire plan

- ☐ A site specific fire plan can be created by configuring Sectors to site requirement.

Commissioning instructions

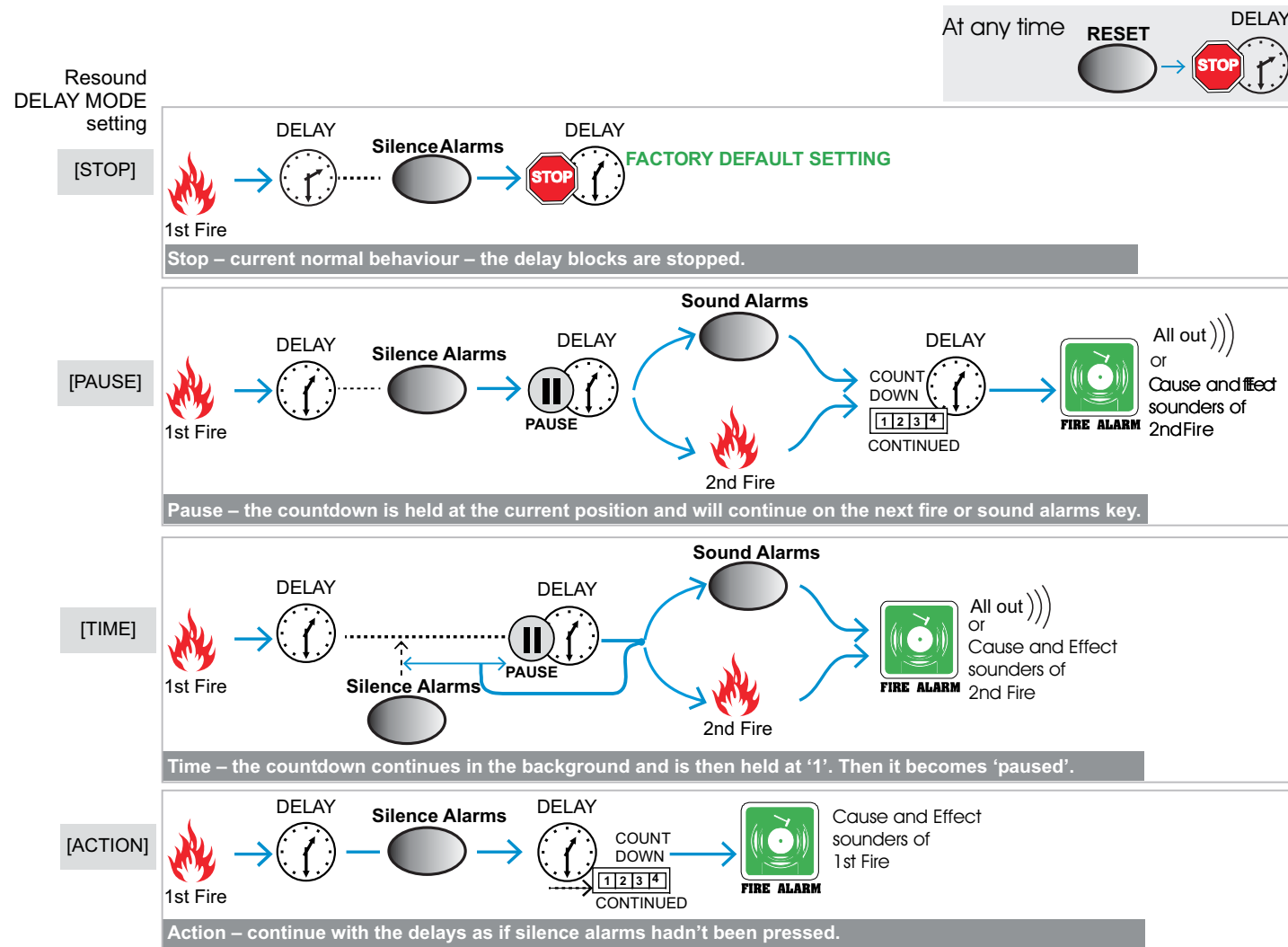
Delay Blocks



- ☐ There can be up to 16 Delay blocks per panel
 - ☐ each Delay block can provide a delay of between 4 seconds to 10 minutes (in 4 second increments) before starting a desired action
 - ☐ a Delay block can action:
 - Sectors
 - Delay blocks
 - ☐ a Delay block may be actioned by:
 - Sector fire event
 - Command build
 - Delay blocks
 - ☐ if the required delay is beyond 10 minutes. This can be achieved by a Delay block actioning another Delay block before the required action is taken
- If Delay Blocks are linked to produce a delay of longer than 10 minutes, then the panel will no longer meet the requirements of EN54: Part 2:1997.**
- ☐ where a Delay block is actioned by a command build, a different delay time can be set up (within the Command build entry line).

Resound Delay mode functions

There are 4 control settings for Delay mode under [Resound]. These settings control what happens to the delay blocks when the Silence Alarms button is pressed.

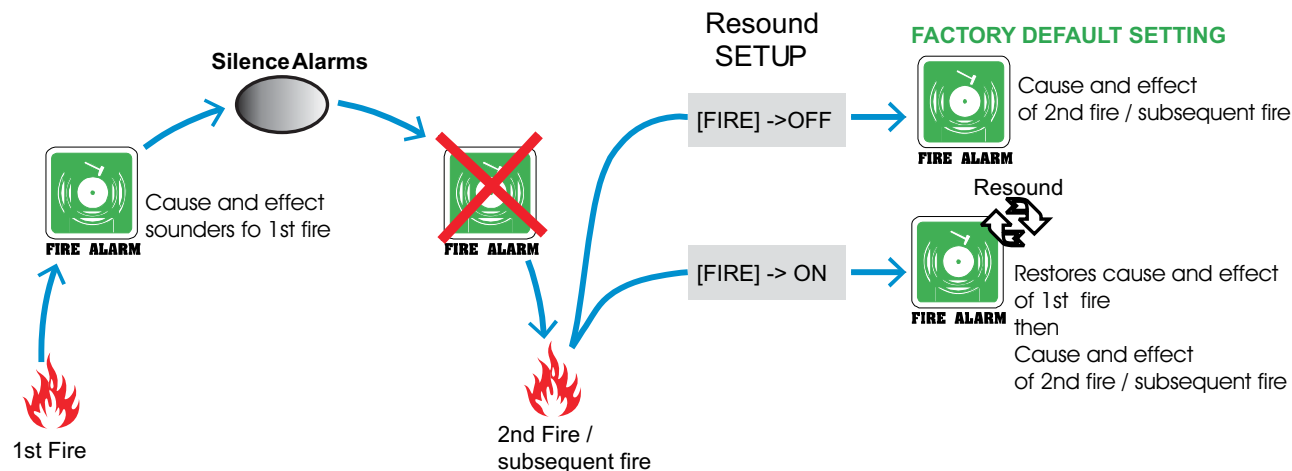


Commissioning instructions

Sound and Resound alarms options

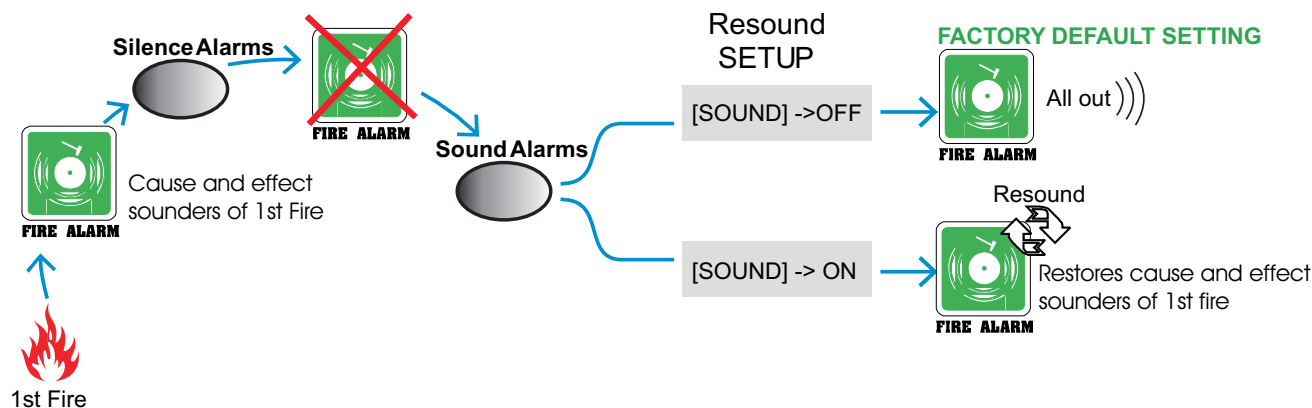
Resound [FIRE] -> ON / OFF setting

The FIRE ON/OFF setting under [Resound] controls the way sounders operate when a 2nd or subsequent fire occurs after the alarms have been silenced following 1st fire.



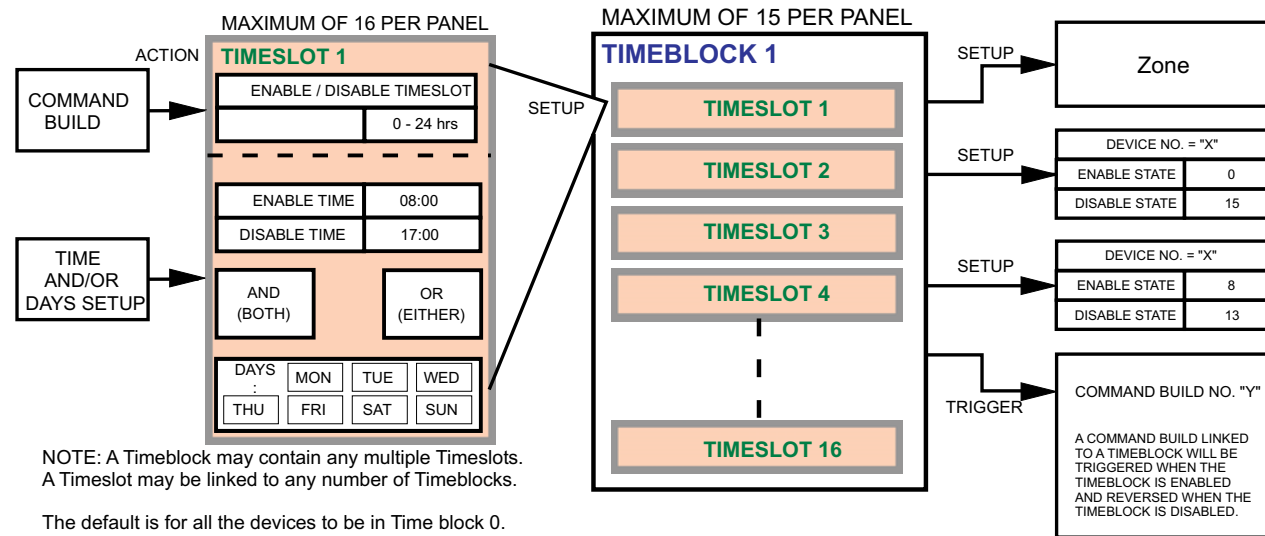
Resound [SOUND] -> ON / OFF setting

The SOUND ON/OFF setting located under [Resound] controls the way sounders operate on 1st press of the Sound Alarms button during a fire condition after the alarms have been silenced. When set to ON the sounders will resound and when set to OFF the sounders will perform the standard action of all-out alarms.



See also resound delay mode functions.

Time slots and Time blocks



NOTE: A Timeblock may contain any multiple Timeslots. A Timeslot may be linked to any number of Timeblocks.

The default is for all the devices to be in Time block 0. A Timeblock is enabled if any one of the Timeslots within it is in the enable state.

Time slots

A Time slot must be linked with a Time block. It is the Time block that perform actions on the system.



Timeslots do not process when the panel is in a fire condition.

- ☐ a Time slot can be:
 - triggered automatically
 - switched ON / OFF manually
 - or switched ON / OFF by Command build
- ☐ it performs an action immediately or after a delay of 0 to 24 hours.
- ☐ There can be up to 16 Time slots per panel

- ☐ each Time slot has to be given an Enable time and a Disable time
- ☐ each Time slot must be ANDed / ORed to the days of week, (Mon, Tue, Wed, Thu, Fri, Sat, Sun).
- ☐ when a Time slot is ANDed with selected days, it will only operate on the selected days
- ☐ when a Time slot is ORed with selected days, it will be Enabled during selected times and will remain ON throughout the days selected.

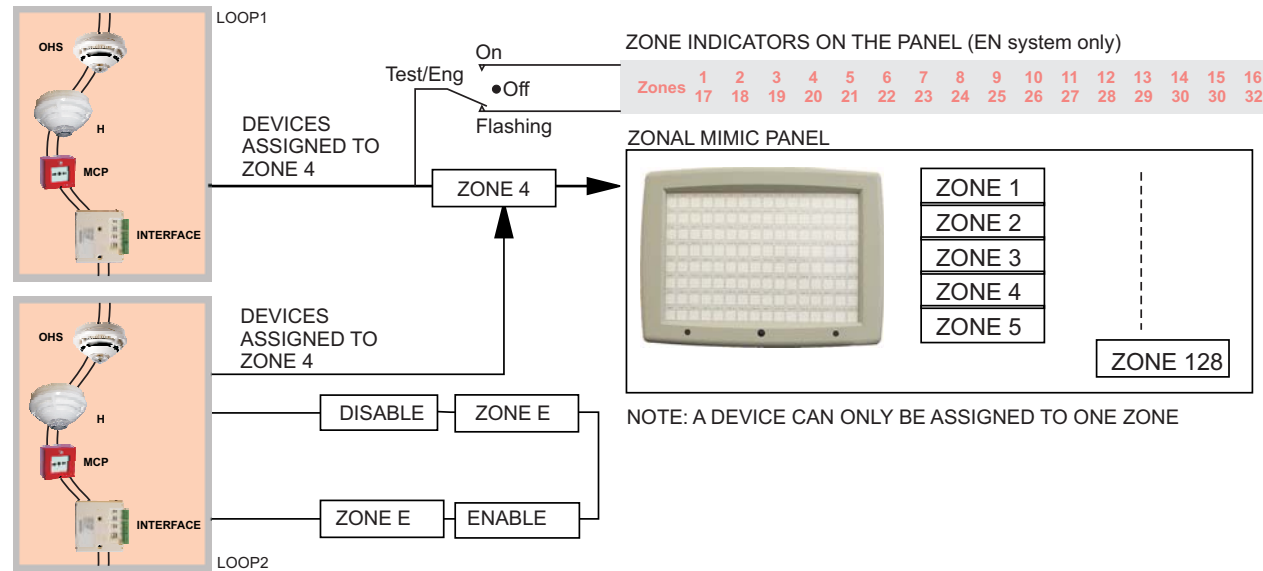
Time blocks

A Time block is used to perform periodic enable disable action on the system.

- ☐ There are 0 - 15 Time blocks per panel
- ☐ the Time block 0 is always in the Enable state and is the default for all sensors
- ☐ a Time block can be linked to one or more Time slots
- ☐ when a Time slot is enabled the time block it is linked to is also Enabled
- ☐ a sensor or interface input can only be assigned to one Time block
- ☐ when a Time block is in an Enable state:
 - any sensor(s) linked to it will operate in the Enable state that has been set up
 - any Command build linked to it is triggered.
- ☐ when a Time block is in a Disabled state any Zone tasks linked to it are also Disabled.

Commissioning instructions

Zones

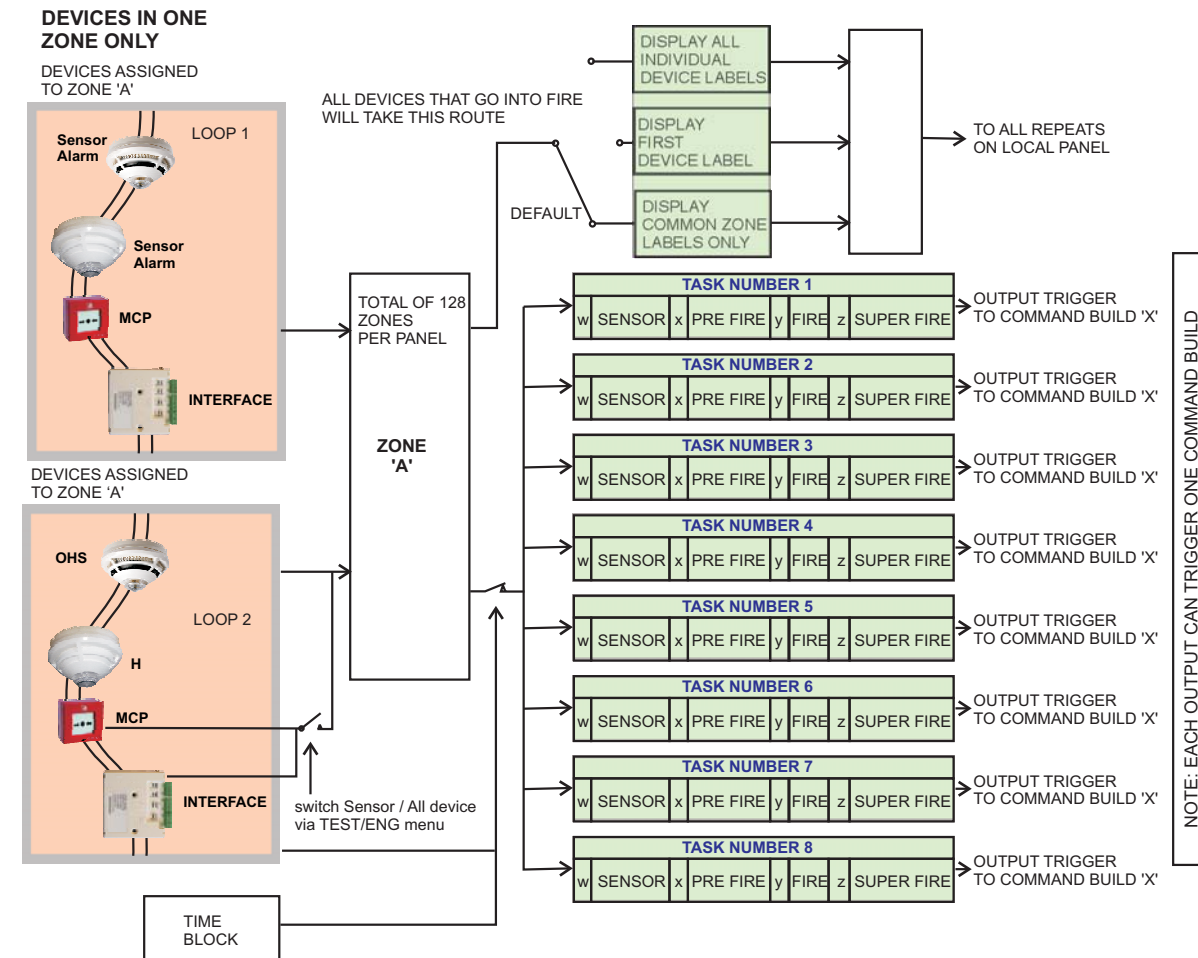


- ☐ A Zone has a collection of trigger devices that are located in the same general area. It may be used for one or all of the following:
 - for illumination of zonal fire lights on Main panel or Zonal panel or Mimic panel
 - are also used for common disablement purposes.
- ☐ There can be up to 128 Zones per A3 or on A2 zonal mimic panel / 64 Zones per A4 zonal mimic panel, the latter two panel are legacy products
- ☐ Zones are used to provide local indication of fire at the main panel and to send fire messages to Zonal mimic panel, Loop Repeat panels and Repeat indicator panels off main panel.
 - integral zonal indicators on a local panel can be configured to give steady / flashing indication of 1st zone in fire.
- ☐ Devices from any loop may be assigned to the same Zone
- ☐ A device may only be assigned to one Zone
- ☐ Input/output lines of an interface can not be assigned individually to Zones. The interface lines can only be assigned collectively to a Zone.

Groups

- ☐ There can be up to 128 Groups
- ☐ all devices default to Group 1
- ☐ all local controller events are in Group 0, ie faults, warning, silence alarms and reset.

Zone Tasks



- ☐ Zone mode may be used to control the display of fire messages, to provide:

- Common zone label only (this is the default).
- label of first device in fire
- or all individual device labels.

- ☐ a Zone can provide 3-level sensing:

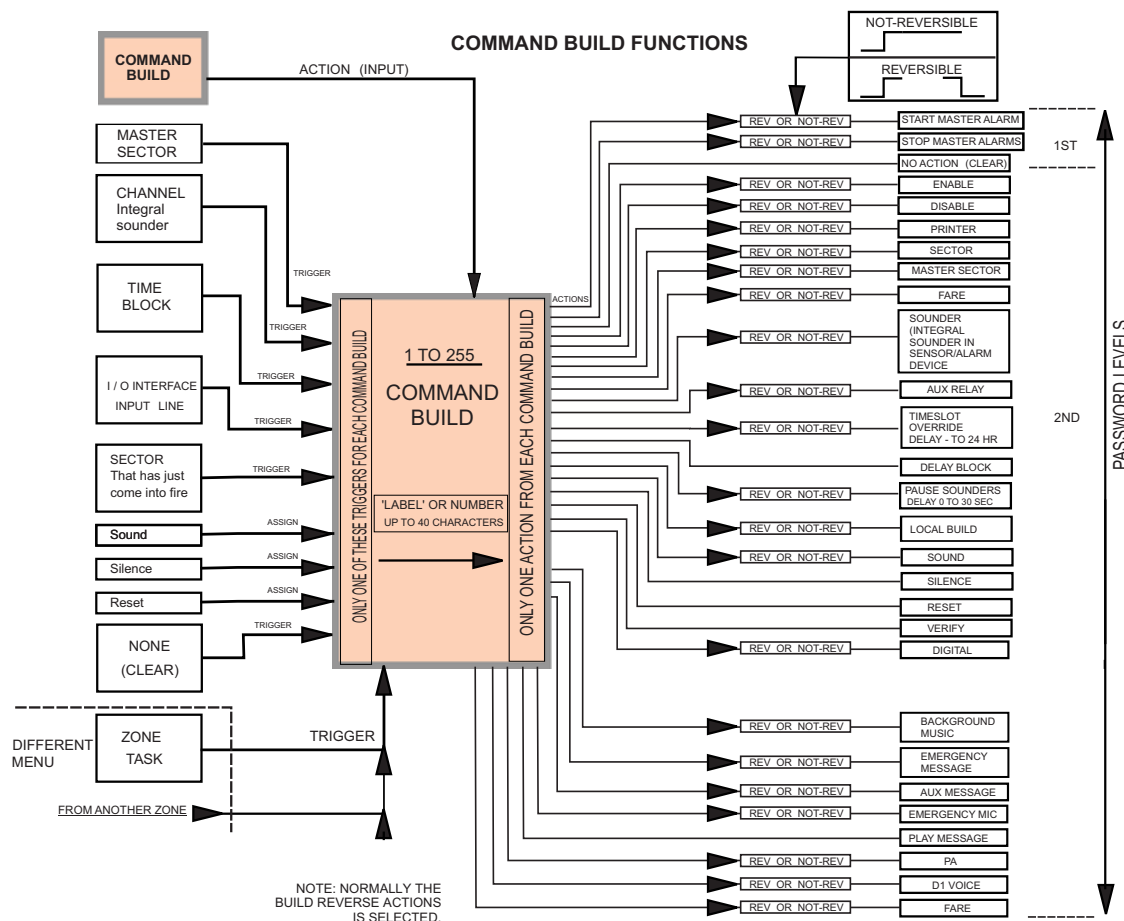
Condition	meaning..	..and for State 0 it implies
PreFire	Fire detection is at a higher sensitivity than the selected STATE .	Fire detection that will easily pass the respective EN54 test .
Fire	Fire detection at the sensitivity of the selected STATE .	Fire detection that is referenced to the respective EN54 test .
SuperFire	Fire detection at lower sensitivity than the selected STATE .	Fire detection that will not pass the Standards test.

- ☐ sensors can be configured to provide coincidence operation using a combination of all 3 levels if required
- ☐ up to 8 permutations of fire detection (Zone tasks) may be configured per zone
- ☐ a Zone task can be made to trigger a command build, this is done in the zone set up menu
- ☐ the same Command build can be triggered by different tasks within a Zone
- ☐ disabling a Zone will disable the sensors within that Zone
- ☐ disabling a Zone via a Time block disables the tasks performed by the Zone
- ☐ all devices default to Zone 1.

- ☐ There can be up to 128 Zones of which the first 64 are for standalone system and remaining zones are for network use. The number of Zones for standalone system can be re-configured via the commissioning tool
- ☐ a Zone is a collection of devices, sharing a common label
- ☐ a device can only be assigned to one Zone

Commissioning instructions

Command Builds



Where possible use sectored action instead of command build action.

- ☐ There can be up to 255 Command builds per panel
- ☐ a Command build can only be set up with one trigger (can be a range)
- ☐ a Command build can be triggered:
 - for reversible action (automatic deaction on removal of trigger)

- or non-reversible action, (deactioned by another Command build for a non-fire trigger or reset for a fire trigger).

- ☐ the Command build trigger for Zone tasks reside under the zone menu/function
- ☐ Command builds that are triggered through a Sector fire event will deaction on fire reset. This is independent of whether the Sector is configured to deaction on silence alarms
- ☐ Command builds that are configured for turn Off actions have no reverse action to turn On again, another Command build or manual action is required
- ☐ Command builds can be used under fire and non fire conditions to carry out pre-configured actions built up from the Control menu command
- ☐ a Command build can only be set up to perform one action (can be a range), any split range would require a second Command build
- ☐ each Command build can be given a 40 character label to be displayed and/or printed when triggered. If no label is given then nothing will be displayed
- ☐ if a label is given, the local buzzer will operate for 1 minute when the Command build is active
- ☐ Command builds are logged with an On/Off time, this information can be viewed by selecting [info] -> <etc> -> [Event]-> <etc> -> [Supervis]-> [Log] -> [Enter] and if still On via [Info]-> [Event]-> <etc> -> [Supervis]-> [Active] -> [Enter], this is only possible if a label has been given to the Command Build
- ☐ when speed of operation is required the lowest number Command builds should be used as these are processed first
- ☐ if a digital output line is to be actioned by a Command build, the I/O line should not be assigned to a Sector as it may deaction when the silence alarms is pressed regardless of the Command build still being active
- ☐ a fire condition takes priority over a Command build



Take great care not to action a digital output to an input line, as this will effectively disable the input.

Sounders Configuration

- ☐ The standard alarm devices of each standalone system can be configured for three sound output signals
- ☐ every 250mS over a 2 second time the sound output can be altered to high, low or off tone
- ☐ sound signal configuration can be altered by modifying the FAB-SAB pairs by pressing the button **Menu On/Off** to select **[Set-Up] -> <etc-> -> [Modify]-> <etc-> -> [Signal n] -> [Sounder] -> [Enter]** and enter the FAB/SAB for *Signal n* and press **enter**.

FABs & SABs

The FAB (first action byte) determines tone and On/Off control while the SAB (second action byte) determines the action to be performed on the output.

Factory Default FABs SABs	Time in seconds							
	0	0.25	0.5	0.75	1.0	1.25	1.5	1.75
Signal 3 (Continuous) Output turns on Low frequency sound continuous every 0.5 seconds.	4F	00	4F	00	4F	00	4F	00
Signal 2 (Evacuate) First output turns on Low frequency sound continuously, then every half second the High frequency output is pulsed for 250mS.	4F	25	4F	25	4F	25	4F	25
Signal 1 (Alert) Pulses high frequency for 250mS every half second.	65	00	65	00	65	00	65	00

Number	FAB action	SAB action
0	no output	Turns Off selected Outputs
1	-	10mS pulse on selected Outputs
2	frequency tone	40mS pulse on selected Outputs
3	-	90mS pulse on selected Outputs
4	On/Off control.(On being low frequency)	160mS pulse on selected Outputs
5	-	250mS pulse on selected Outputs
6	high frequency tone	350mS pulse on selected Outputs
7	-	480mS pulse on selected Outputs
8	-	630mS pulse on selected Outputs
9	-	800mS pulse on selected Outputs
A	-	980mS pulse on selected Outputs
B	-	1.2S pulse on selected Outputs
C	-	1.4S pulse on selected Outputs
D	-	1.6S pulse on selected Outputs
E		1.9S pulse on selected Outputs
F		Turn on selected Outputs

Commissioning instructions

Auxiliary Relays

The two Auxiliary relays in the control panel are defaulted to operate on:

- a Sector fire event the Auxiliary relay 1 and
- a fault event the Auxiliary relay 2.

☐ the Auxiliary relays can be configured for normally-open or normally-closed operation

☐ the Auxiliary relays may be configured to be operated with:

- a Fault
- a Warning
- any combination of Sector Fire events
- or any of the above combination

☐ It is possible to delay the operation of each relay by up to 10 minutes after the first **FIRE** event. The accuracy of the delay is 0 - 4 seconds

☐ The relay will operate immediately for fault and warning events.

Radio Device

A Plexus radio device is bound to a LRT and can be configured using the Plexus Commissioning tool in a similar manner to any other system device, however see the following note.



It is important to note that a Radio sensor device do not have multiple STATES and therefore a radio sensor device cannot operate with Timeslot, like for example a S-Quad sensor where the sensor STATE can be switched at pre defined times of a day.

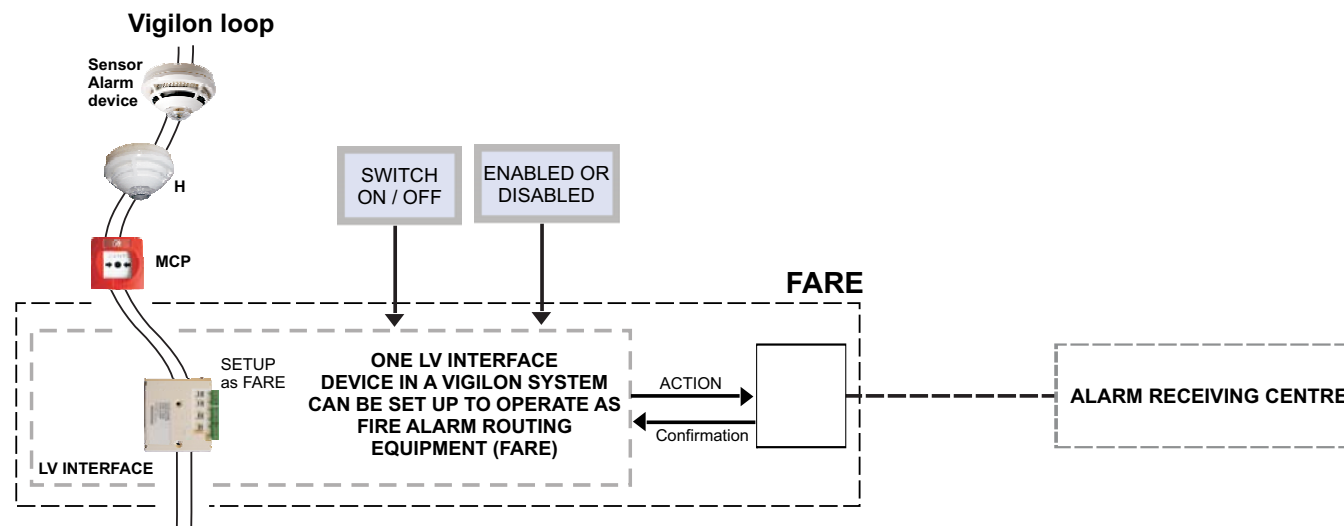
Prior to commissioning radio devices using the commissioning tool ensure they are all pre-commissioned using the Plexus Pre-commissioning tool.

Fire Alarm Routing Equipment (FARE)

- ☐ There can be one FARE Interface per Vigilon Plus System.
FARE Interface can be a LV Interface unit S4-34420, but can also be a S4-34450.
Other interface units can be configured FARE, these include S4-34440-02, S4-34440-12, S4-34401 and S4-34404.
- ☐ a FARE Interface device is automatically placed in plant Sector 32 (as default) and is unaffected by operation of SOUND ALARMS control
- ☐ a FARE Interface device may be switched ON or OFF from the panel menus select:
Menu On/Off button to select [Control] -> <etc> -> [FARE] -> [ON] or [Off] -> [E]
- ☐ a FARE device may be Enabled or Disabled from the panel menus, select:
Menu On/Off button to select: [Control] -> [Enable] / [Disable] -> <etc> -> <etc> -> <etc> -[FARE] -> [E]



Where a FARE interface is required to be disabled then use the FARE device disablement option.



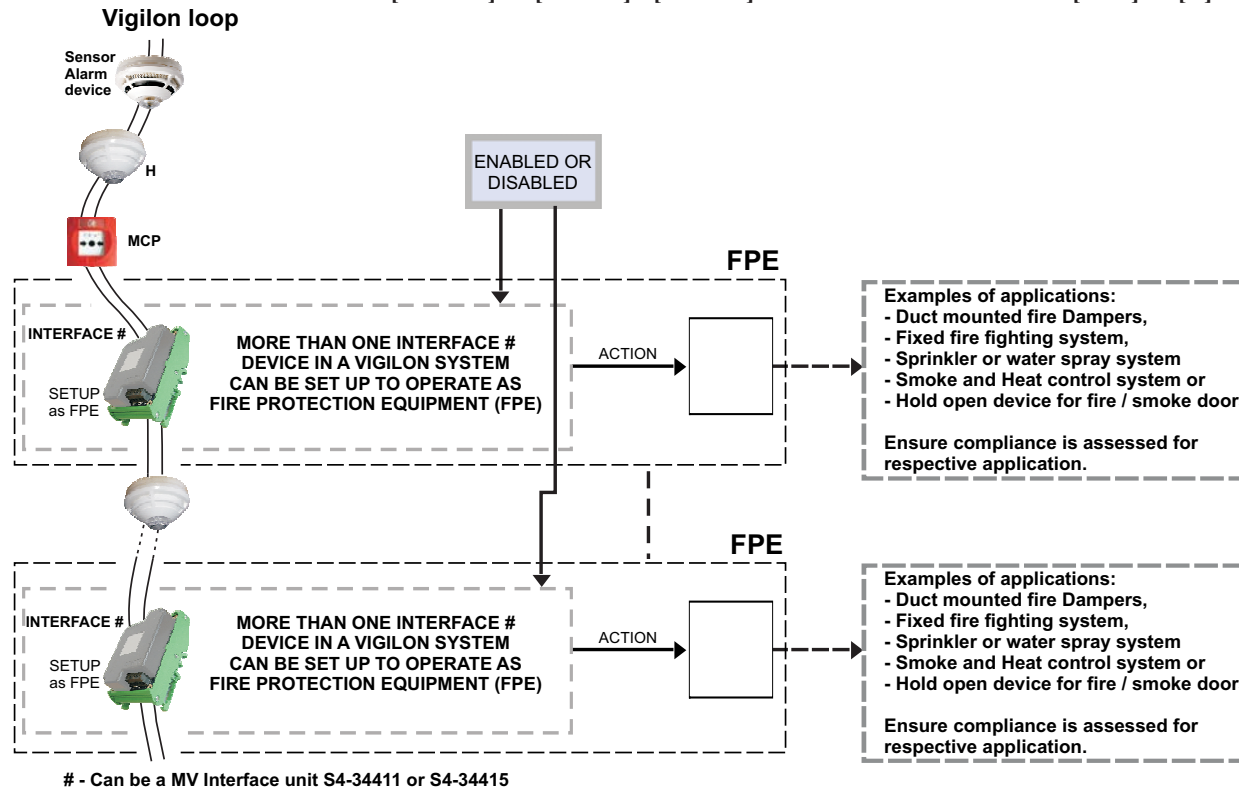
Commissioning instructions

Fire Protection Equipment (FPE)

- ☐ There can more than one Interface configured FPE application in a Vigilon Plus System.
FPE Interface can be a MV Interface unit S4-34411, but can be also be a S4-34415 or S4-34420.
Other interface units can also be configured as FPE, these include S4-34440-02, S4-34440-12, S4-34401 and S4-34404.
- ☐ a FPE Interface device is automatically placed in plant Sector 32 (as default) and is unaffected by operation of SOUND ALARMS control
- ☐ a FPE Interface device may be Enabled or Disabled from the panel menus select:
Menu On/Off button to select: **[Control] -> [Enable] / [Disable] -> <etc> -> <etc> -> <etc> -> [FPE] -> [E]**.



Where a FPE interface is required to be disabled then use the FPE device disablement option.

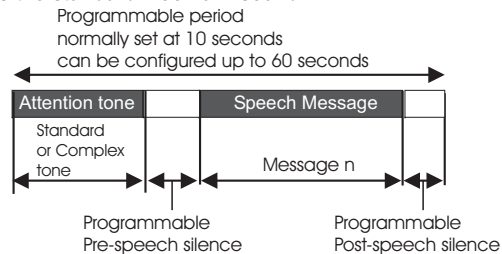


S-Cubed Mark III or II and S-Quad

Speech function

The Mark II or III S-Cubed or S-Quad Speech sounder function is provided by stored messages on a flash memory chip within the assembly. The standard flash memory can hold up to 20 seconds of audio and additionally it holds local complex attention tones, such as the bell and DIN signals.

Each signal output from a S Cubed Mark III or II, or S-Quad Speech device consists of an attention tone followed by a message. The attention tone can be a local complex tone or a panel tone, such as the standard 'nee naw' sound.



It is possible to reprogram factory set speech messages

Visual Alarm function - Part 23 complaint

It is possible to set the flash rate of devices like S-Cubed Mark III (including S-Quad with VAD) using the commissioning tool. The flash rate can be set to 1s, 2s or 4s, but only the 2s rate is compliant with EN54-23.



The Equality Act 2010 recommends visual alarms like those offered by S-Cubed and S-Quad devices, are installed in protected premises to warn occupants who are hard of hearing.

Strobe function (Legacy)

In the event of a fire the appropriate legacy S-Cubed Mark II or I and legacy S-Quad devices when installed in the system will output alarm signals according to the site specific needs and these can be Signal 1, 2 and 3. The strobe alarm flash rate is every 2s for signal 1, every 1s for signal 2 and every 1s for signal 3. The output from these Strobes are not Part 23 compliant.

The strobe flash rate for signals 1, 2 and 3 can be changed using the commissioning tool. The output flash rates are synchronised with S-Cubed Mark II or I strobe and legacy S-Quad strobe devices installed in the same system.

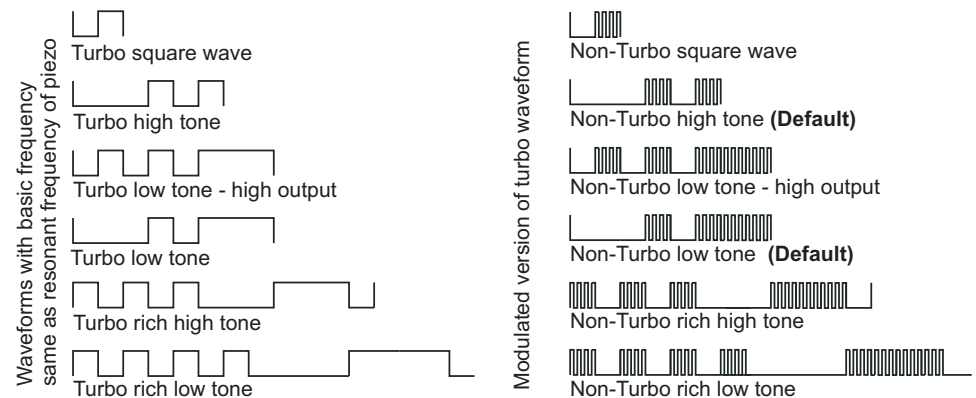
Sounder function

A S-Cubed Mark III and II and S-Quad sounder element can operate in turbo mode if configured during commissioning to provide further 3dB output. The sound outputs are based on the settings of the FABs and SABs at the panel that gives changing levels over 2 seconds duration in 8 time slots. The standard outputs are synchronised with the S-Quad devices, S-Cubed device and 34000 sounder devices installed in the same system.

The volume of the sound output can be individually set at the device, note that the sound level should not be set lower than 65dBA at 1m for standards compliance. The system prevents adjustment of volume down to zero. Another feature that can be configured is the soft start that ramps the sound volume gradually to the maximum level set at the device.

Turbo and Non Turbo Modes

The S-Cubed or S-Quad sounder can output in turbo and/or in non turbo modes. The high and low FAB mappings are used to change the tone output of the S-Cubed or S-Quad sounder. The turbo mode outputs a waveform with a basic frequency that is the same as the resonant frequency of the piezo inside each device. The non turbo mode outputs modulated turbo waveform.



Commissioning instructions

S-Cubed Mark I

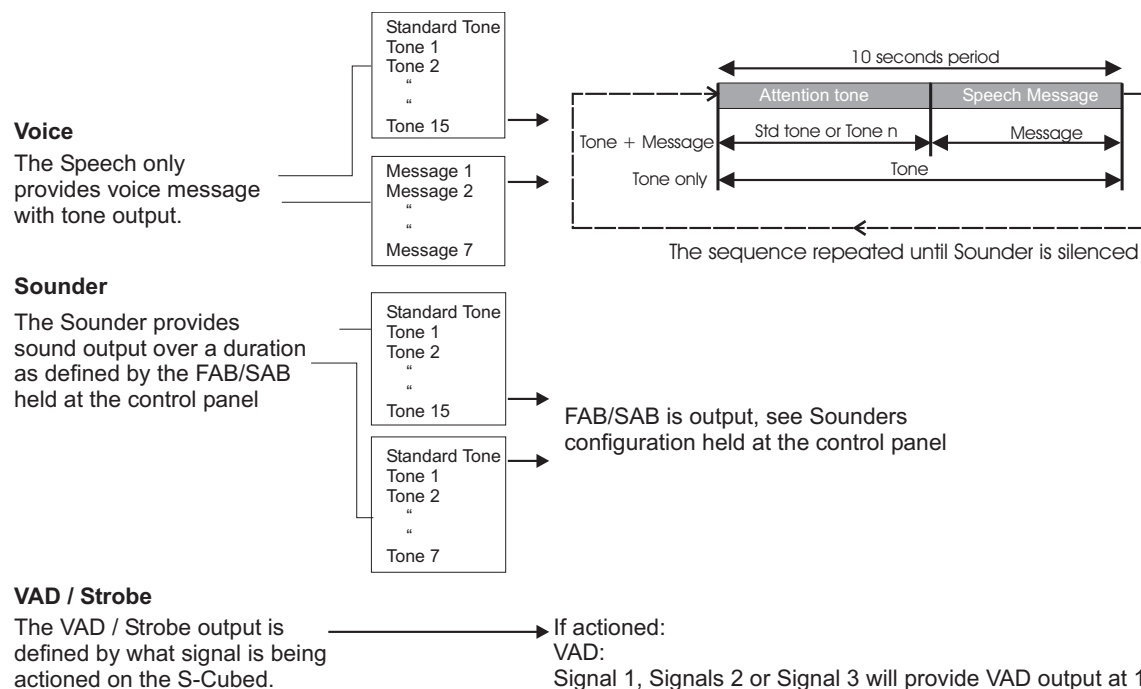


The S-Cubed Mark 1 strobe outputs is not compliant to EN54 part 23. The S-Cubed Mark 3 Visual Alarm Devices are compliant to EN54 part 23.

- ☐ An S-Cubed device can operate in Tone or Voice mode dependent on type.
- ☐ All S-Cubed in a system can share a common start volume setting from a range between 16% to 100% sound output.
- ☐ All S-Cubed in a system can share a common soft start facility, if set the sound output will increment the volume every 0.25s from a defined start level to maximum output.

Voice and Sounder mode pulse VAD/Strobe action

- ☐ S-Cubed/S-Quad operating in tone mode allows selection from a range of low and high frequency tones that are outputted according to FAB/SAB settings, see sounders configurations section.
- ☐ S-Cubed/S-Quad set for Voice mode operation allows selection from a range of tones and messages that span over 10 seconds duration. The tones and messages are those held in respective S-Cubed/S-Quad device.



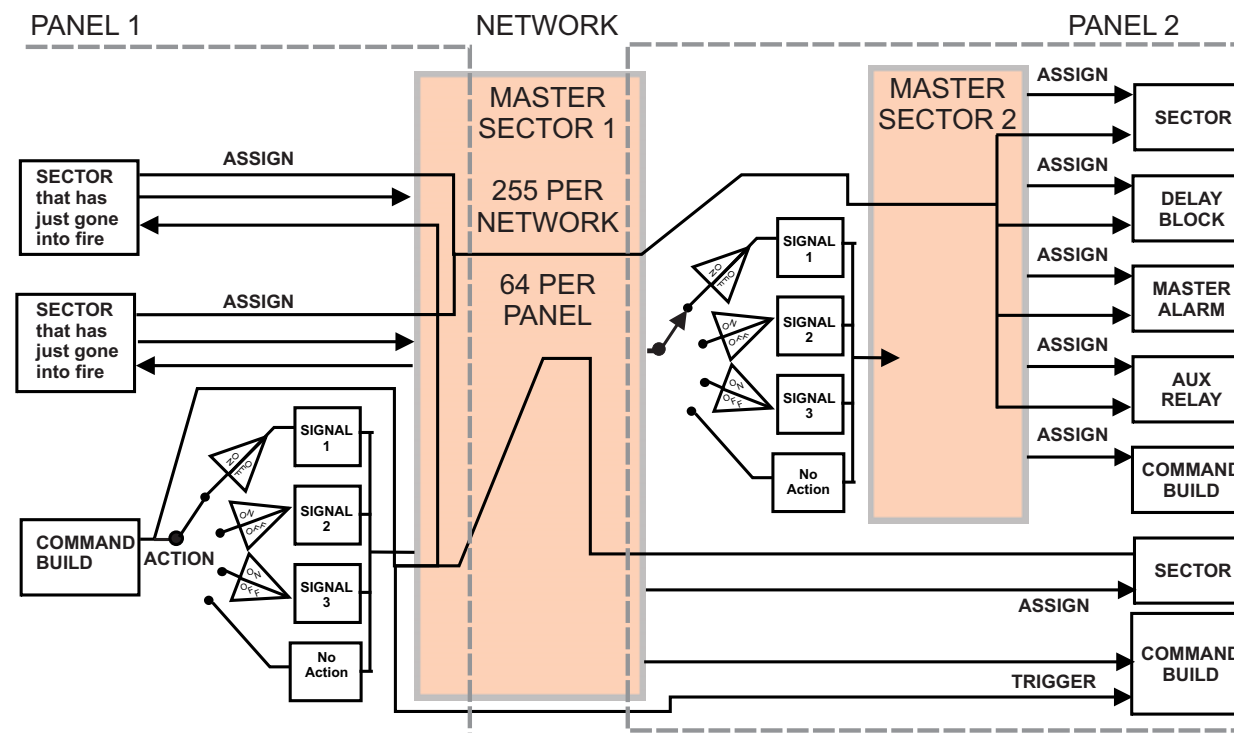
Appendix D - Guidelines for Networked system commands

The functions explained in this section apply to networked systems. See also guidelines for standalone system. For information on Command build actions across a network, see Appendix C.



When commissioning a new network of control panels, ensure the software version is the same across all the panels. This is recommendation may also be applied where an existing network is to be installed with additional panels.

Master Sectors



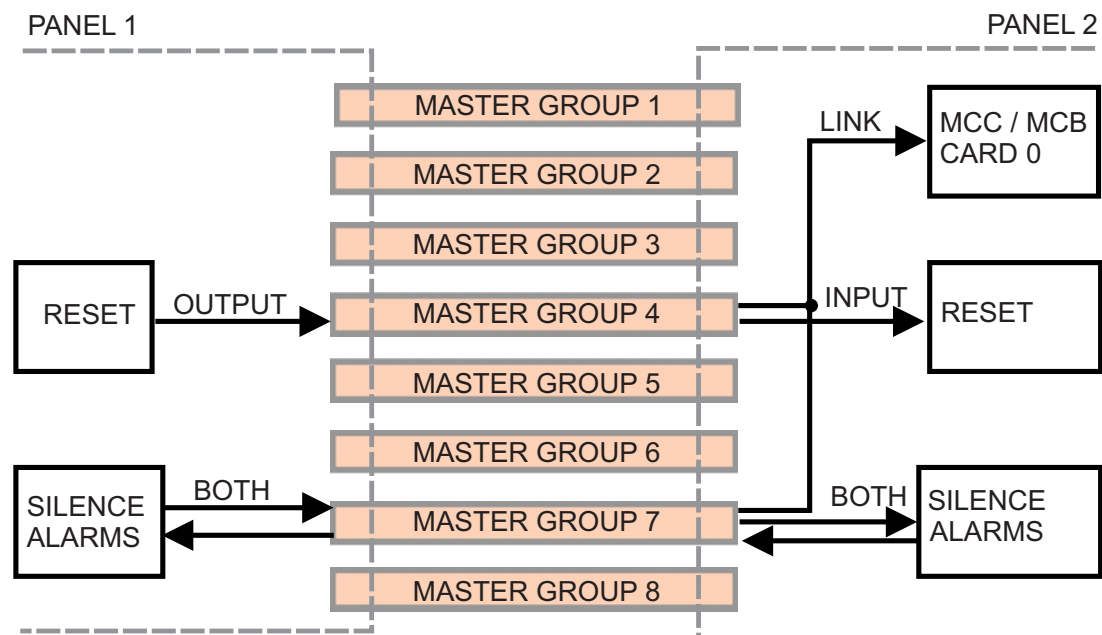
- ☐ There can be up to a maximum of 64 Master Sectors per panel
- ☐ a total of 255 Master sectors per network system
- ☐ they are therefore a collection of:
 - Sectors
 - Delayblocks
 - Master Alarms
 - and Aux Relays from one or more Systems.
- ☐ they provide global alarm actions across a network of control panels

Guidelines

- ☐ Fire Input Sectors must only be assigned to one Master Sector.
- ☐ Output Sectors can be assigned to one Master Sector.
- ☐ Master Sectors can be actioned by a Sector Fire Event.
- ☐ a triggered Master Sector can action one or any combination of:
 - Sector
 - another Master Sector
 - Command Build (not locally)
 - Delay Blocks
 - Master Alarms
 - or Auxiliary Relays.
- ☐ if a Sector assigned to a Master Sector has a Fire Event, the Master Sector will not perform any local actions on the Panel in alarm.
- ☐ if a Command Build actions a Master Sector, the actions will be performed on other panels, but Command Builds will not be triggered.
- ☐ a Master Sector must have something assigned to it before any Master Sector action links can be made to it.
- ☐ Master Sector Fire Event actions on other Panels can be stopped by global Silence Alarms.

Commissioning instructions

Master Groups



- ☐ There can be up to a maximum of 8 Master Groups per network.
- ☐ These are a collection of Groups used for:
 - selective fire and non fire event message routing within a Control Panel or around a Network of Panels
 - and for routing Global Controls 'Silence Alarms, Sound Alarms and/or Reset' to designated Control Panels in a Network.
- ☐ Messages can be displayed or printed on Control Panels and Repeat Panels. Messages are routed on a Card by Card basis, ie:
 - Card 0 = Panel LCD
 - Card 15 on Vig Plus = Remote RS232 Printer
 - Cards 1-6 = Repeats and Mimics panels on Loops 1-6
 - Each Card can be configured to display/print **All Events** (default) or just **Fire Events**. Repeat Panels on the same loop will always give the same information.



Once message passing has been setup for global repeat or mimic indication, the local panel will also have to route reset messages to other groups, besides Group 0.

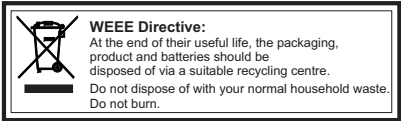
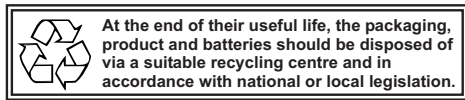
- ☐ Groups within different Control Panels can be assigned to the same Master Group
- ☐ Global controls can be configured to output/input or both.
- ☐ Messages and controls within a standalone system do not require any setting up
- ☐ the Group 0 contains all panel events and fire resets and therefore must always be assigned to a Master Group.



The Verify control button is applicable for EN panels only can operate globally, like the Sound alarms, Silence alarms and Reset buttons.

Appendix E - Cards

Refer to the instruction leaflet supplied with each card.



Honeywell Gent reserves the right to revise this publication from time to time and make changes to the content hereof without obligation to notify any person of such revisions of changes.

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